Topological Methods in Condensed-Matter Physics

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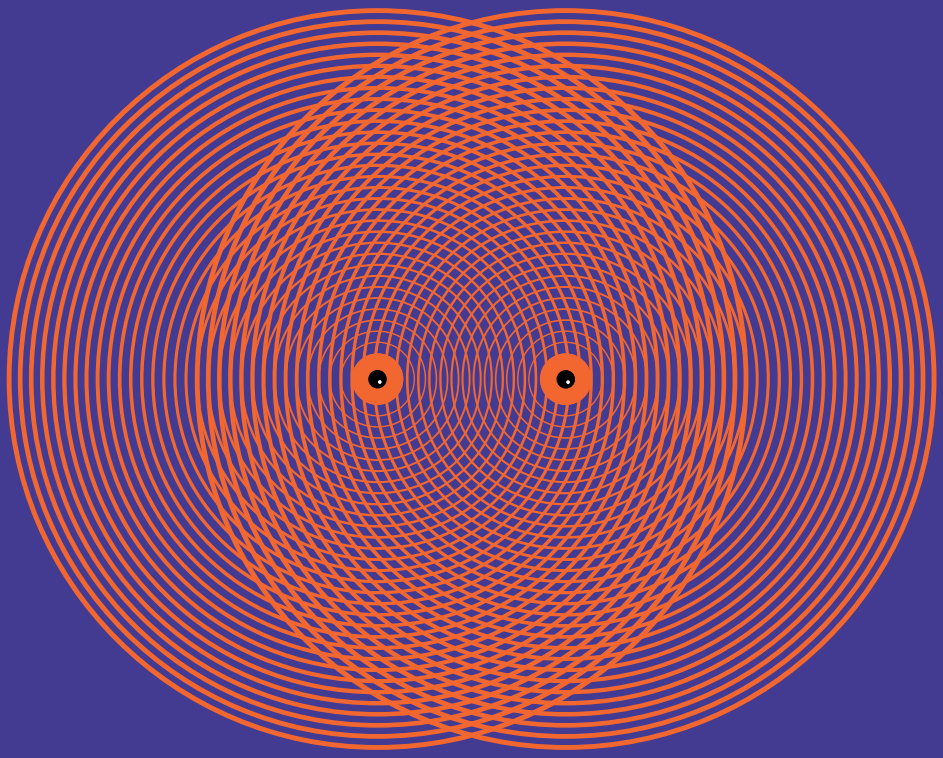
Topological methods in condensed matter physics

The basic mathematical concepts behind the physical phenomena



Berry phase – Quantum Hall effect ∗ Chern number – Topological insulators
Vector bundles – Field theory ∗ Hopf fibration – Quantumbit

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1 Introduction

This document is a lecture note for the course Topological methods in condensed-matter physics in BME (BMETEEFMSFTMSZ-00 in Neptun). The lecture notes are not ready, they are being prepared continuously during the semester. The mathematical parts are presented by Gergő Pintér, while the physical parts are presented by Levente Rózsa.

Topological concepts provide the key to understanding various phenomena in contemporary condensed-matter physics research. This course gives an introduction to the mathematical basics of these concepts located at the intersection of algebraic topology, differential topology and differential geometry, as well as to their physical applications. It will cover the following topics:

- Topological spaces and smooth manifolds
- Real and complex projective spaces
- Homotopy theory, fundamental group
- Degree of a mapping, Hopf theorem
- Tangent vector fields, Poincaré-Hopf theorem
- Vector bundles, Euler number, Chern number
- Connection and curvature of vector bundles
- Localized field configurations characterized by topological invariants (domain walls, skyrmions, hopfions)
- Topology of the band structure in reciprocal space (integer quantum Hall effect, Chern insulators, topological insulators, Weyl semimetals)
- Berry connection, Berry curvature and their connection to topological invariants and observable phenomena
- Connection between topology and quantum statistics, fractional statistics, Majorana bound states, braiding

Actually, this is a huge material, the topological part usually takes 3-4 semesters by itself in mathematics courses. Therefore, this one-semester course is definitely not a full topology course. Instead, we want the present an intuitive overall picture, avoiding the precise proofs and details. We introduce only the minimally necessary theoretical background, with many examples and intuition instead of precise proofs. Many details are collected in the exercises, which serve to form a kind of 'peripheral knowledge'. We recommend to read all the exercises and solve your favorite ones.

Based on this topological introduction we focus on the physical applications. They require the three components

- theoretical background,
- intuition, and
- computational methods

of topology and differential geometry. We only sketch the theoretical background, while giving a strong intuition through many examples. Most of the computational methods are left to the end of the semester, because they are based on differential forms, connections and curvatures from differential geometry. These tools themselves are not intuitive, however, very intuitive notions are expressed with them, which will be introduced earlier during the semester.

Prerequirement: We will frequently use multivariable analysis, we refer to the continuity and partial derivatives of multivariable functions as well-known notions, and actually we generalize these notions for topological spaces and manifolds. We also use linear algebra and group theory – although, we will shortly recall the required notions all the time. In more detail:

- *Complex numbers* and the basic operations with them.
- *Linear algebra:* Real and complex vector spaces, matrices, linear maps, linear transformations, rank, determinant, eigenvalues and eigenvectors, orthogonal, unitary, symmetric, self-adjoin (hermitian) matrices.
- *Sets:* basic operations, relations, equivalence relations and partitions, maps (functions), image and preimage of subsets.
- *Groups:* finite groups, Abelian groups, subgroup, normal subgroup, cosets, quotient group, free groups, groups presented by generators and relations.
- *Muti-variable calculus, vector calculus:* continuity, partial derivatives, total derivative (Jacobian matrix), multi-variable integration, line integral, surface integral, gradient, curl, divergence.

2 How does topology come into quantum physics?

2.1 Physical examples

Ez a te bevezetod Levente.

2.2 One qubit states and Hopf fibration

By Wikipedia: 'A qubit is a two-state (or two-level) quantum-mechanical system, one of the simplest quantum systems displaying the peculiarity of quantum mechanics. Examples include the spin of the electron in which the two levels can be taken as spin up and spin down; or the polarization of a single photon in which the two spin states (left-handed and the right-handed circular polarization) can also be measured as horizontal and vertical linear polarization. In a classical system, a bit would have to be in one state or the other. However, quantum mechanics allows the qubit to be in a coherent superposition of multiple states simultaneously, a property that is fundamental to quantum mechanics and quantum computing.' – *But what does this have to do with topology?*

The states of a qubit are superpositions, i.e., complex linear combinations of the two basis states $|0\rangle$ and $|1\rangle$,

$$\Psi = z|0\rangle + w|1\rangle = \begin{pmatrix} z \\ w \end{pmatrix} \in \mathbb{C}^2. \tag{2.2.1}$$

The probability of a measurement resulting in $|0\rangle$ is $|z|^2$, while the probability of $|1\rangle$ is $|w|^2$. This implies that $|z|^2 + |w|^2 = 1$, that is, Ψ is a unit vector (normalized). Ψ can be represented by the two vectors $z = a + bi$ and $w = c + di$ on the place $\mathbb{C} \cong \mathbb{R}^2$, hence, by identifying $\mathbb{C}^2$ with the 4D space $\mathbb{R}^4$, we have

$$\Psi = \begin{pmatrix} z \\ w \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} \in \mathbb{C}^2 \cong \mathbb{R}^4, \tag{2.2.2}$$

such that $a^2 + b^2 + c^2 + d^2 = 1$. This means that Ψ is an element of the unit sphere

$$S^3 = \{v \in \mathbb{R}^4 \mid \|v\| = 1\}. \tag{2.2.3}$$

We conclude that space of one qubit states is the 3 dimensional unit sphere S^3 in the 4D space $\mathbb{R}^4$. *Is it possible to imagine and visualize this sphere?*

However, from physical aspects the global phase is usually irrelevant. This means that Ψ cannot be distinguished physically from another state

$$\Psi' = u \cdot \Psi = \begin{pmatrix} uz \\ uw \end{pmatrix} \tag{2.2.4}$$

where $u \in \mathbb{C}$, $|u| = 1$. More precisely, we say that Ψ and $\Psi' = u \cdot \Psi$ determine the same *projective state* $[\Psi] = [\Psi']$, and this projective state can be identified with the set

$$[\Psi] = \{u \cdot \Psi \mid u \in \mathbb{C},\ |u| = 1\} \tag{2.2.5}$$

of states determining the same projective state[1]. *Yes, this is a kind of tautology, but it makes sense!*

What 'shape' does the set of all projective states have? To answer this question we provide here an intuitive argument, which will be made precise in one way in the Exercises 2.3 and in another way later. Multiplication by u rotates the vectors z and w simultaneously, and in this way z can be rotated to the positive part of the real axes. More precisely, for every (z, w) there is a u such that $uz \in \mathbb{R}^+$ (or 0). Then uw completely determines the projective state $[\Psi]$, and uw can be every complex number with $|uw| \leq 1$. Hence, one might guess that the space of projective states can be identified with the closed unit disc

$$D^2 = \{(c, d) \in \mathbb{R}^2 \mid c^2 + d^2 \leq 1\}, \tag{2.2.6}$$

however, this claim is not correct. A vector (complex number) $w = c + di$ with $|w| < 1$ really determines a unique projective state by choosing $z = \sqrt{1 - |w|^2}$. But for $|w| = 1$, $z = 0$, all of these states Ψ determine the same projective state $[\Psi]$, indeed, $u \cdot \Psi$ can be everything with first coordinate 0. In other words: if z is the zero vector, then the simultaneous rotation of z and w by u is the same as rotating w alone. At this point we established that

- The *interior points* of the disc D^2 are one-to-one correspondence with projective states,
- The *boundary points* of D^2 determine the same projective state. These points form a circle S^1, the boundary of D^2.

Thus the space of all projective states can be obtained from D^2 by contracting its boundary S^1 to one point. It is intuitively clear that the resulted shape is a sphere of dimension 2, denoted by S^2, see Figure 1, 2. We claim that the space of all projective states can be identified with a sphere S^2. In the physicist terminology this is the *Bloch sphere*, while the mathematician call it *Riemann sphere* or *complex projective line*. The following remarks anticipate the role of the topology:

1. The sphere S^3 of all states is the unit sphere in $\mathbb{R}^4$. In contrast, the sphere S^2 of all projective states is not contained in the 3-space $\mathbb{R}^3$ by default. This is an abstract sphere obtained from a disc by contracting its boundary circle to a point.
2. The main point of identifying the space of all projective states with S^2 would be the *continuity*, in sense that two projective states are 'close' to each other if and only if the corresponding points of S^2 are 'close' to each other. However, we haven't studied this property, this will be done by using topology.
3. The above discussion can be made precise in two different ways. One can identify S^2 with the unit sphere in $\mathbb{R}^3$, and give the identification with the projective states by formulas. The details of this process are collected in Exercisees 2.3. Another way is using the framework of topology to give a precise meaning of the concepts like 'shape of the space of all projective states' or 'contacting the boundary to one point'. In more complicated situations the second way is more convenient, as it will be demonstrated in this course.

[1] $\Psi' = u \cdot \Psi$ is an equivalence relation on the set S^3 of states, and $[\Psi]$ is the equivalence class of Ψ. One can also consider the group action of the unitary group $U(1) \cong S^1 = \{u \in \mathbb{C} \mid |u| = 1\}$ on S^3 defined by $(u, \Psi) \mapsto u \cdot \Psi$, then $[\Psi]$ is the orbit of Ψ under this action.

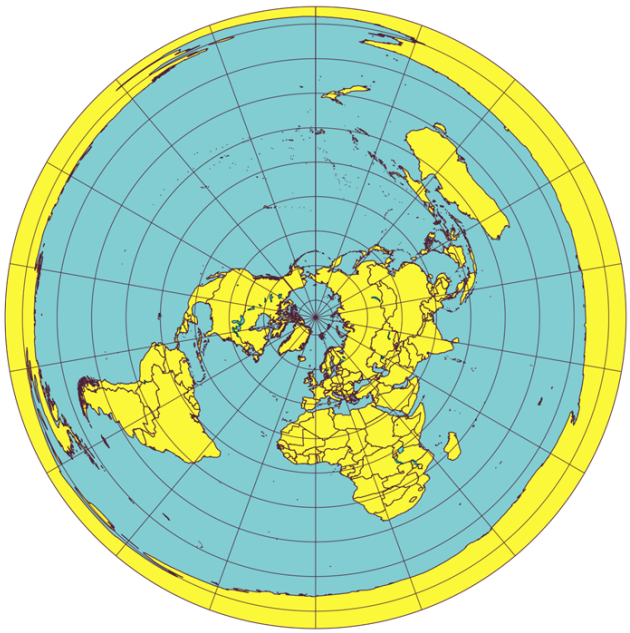


Figure 1: The 2D sphere S^2 obtained by contracting the boundary circle S^1 of a disc D^2.

The space S^3 of all states decomposes to the disjoint union of the projective states. In other words, there is a (continuous, moreover, smooth) map

$$h : S^3 \to S^2 \tag{2.2.7}$$
$$\Psi \mapsto [\Psi]. \tag{2.2.8}$$

Importantly – and confusingly –, $[\Phi]$ denotes two things: (1) $[\Psi] \subset S^3$, i.e. the subset of all states S^3 determining the same projective state, and (2) $[\Psi] \in S^2$, i.e. a point of the Bloch sphere S^2 corresponding the projective state. The map h is 'tautological' in the sense that

$$h^{-1}([\Psi]) = [\Psi], \tag{2.2.9}$$

i.e. the preimage of a point of S^2 is the corresponding subset of S^3. By Equation (2.2.5), each projective state $[\Psi]$ has a shape of circle S^1, as a subset of S^3, since it is parametrised by the unit complex numbers u. Again, S^3 decomposes as the disjoint union of circles S^1, and the set of circles is parametrised by S^2 'continuously'.

The above correspondence between S^3, S^1 and S^2 is called *Hopf fibration*. *Fibrations*, also called *bundles* or *fiber bundles* play central role in topology and also in its physical applications. In this particular case, S^3 is the *total space* of the fibration defined by the *projection map* h with S^1 *fibers* over the *base space* S^2. The S^1 fibers are called Hopf circles.

Is it possible to visualize the Hopf circles, i.e. the sets of states determining the same projective state? Yes, it can be done by using stereographic projection. Choose $N = (0, 0, 0, 1) \in S^3$ ('north pole') and $\mathbb{R}^3 = \{(a, b, c, 0)\} \subset \mathbb{R}^4$. The *stereographic projection* is the map

$$S^3 \setminus \{N\} \to \mathbb{R}^3 \tag{2.2.10}$$
$$P \mapsto P', \tag{2.2.11}$$

where P' is defined to be the unique intersection point of $\mathbb{R}^3$ and the line joining N and P, see Figure 3.

The image of each Hopf circle $[\Psi] \subset S^3$ is a circle in $\mathbb{R}^3$, except the circle corresponding to $z = 0$, $|w| = 1$, i.e., the projective state of $\Psi = |1\rangle$. Indeed, since this is the Hopf circle containing the point N, its image is a straight line in $\mathbb{R}^3$. We can observe the following properties from Figure 4:

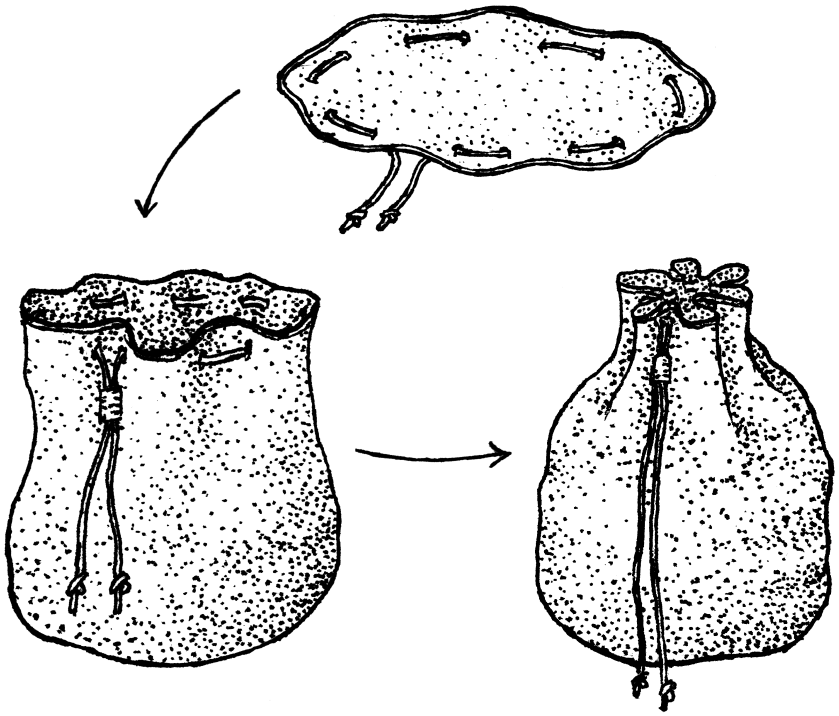


Figure 2: The 2D sphere S^2 obtained by contracting the boundary circle S^1 of a disc D^2. Graphics by Mátyás Szabari, Új világok teremtése.

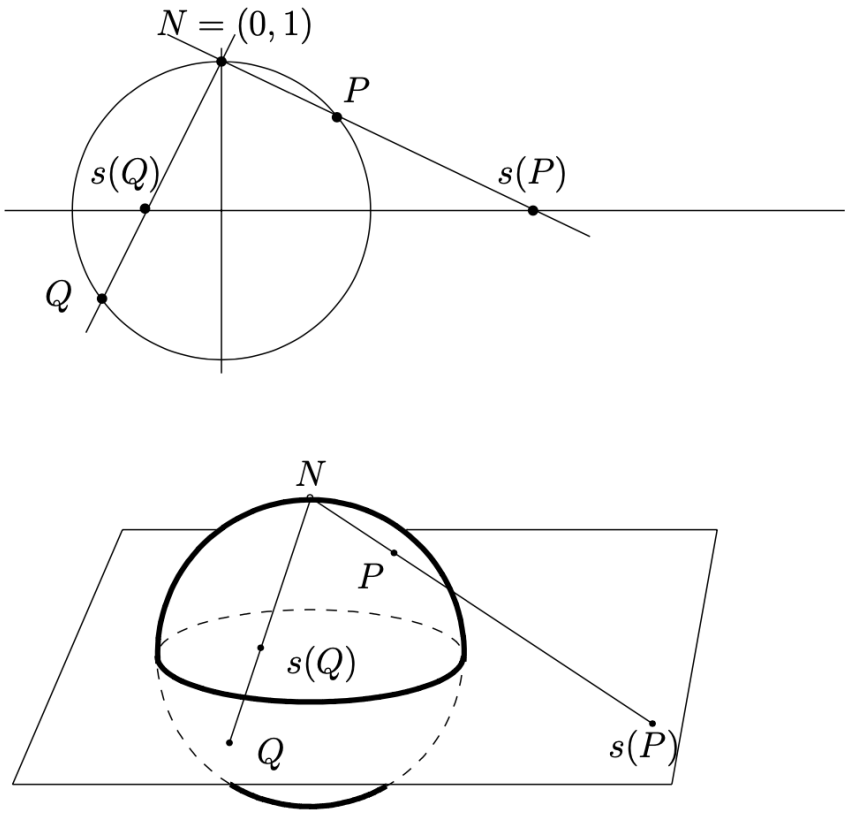


Figure 3: Stereographic proejctions $S^1 \setminus \{N\} \to \mathbb{R}^1$ and $S^2 \setminus \{N\} \to \mathbb{R}^2$.

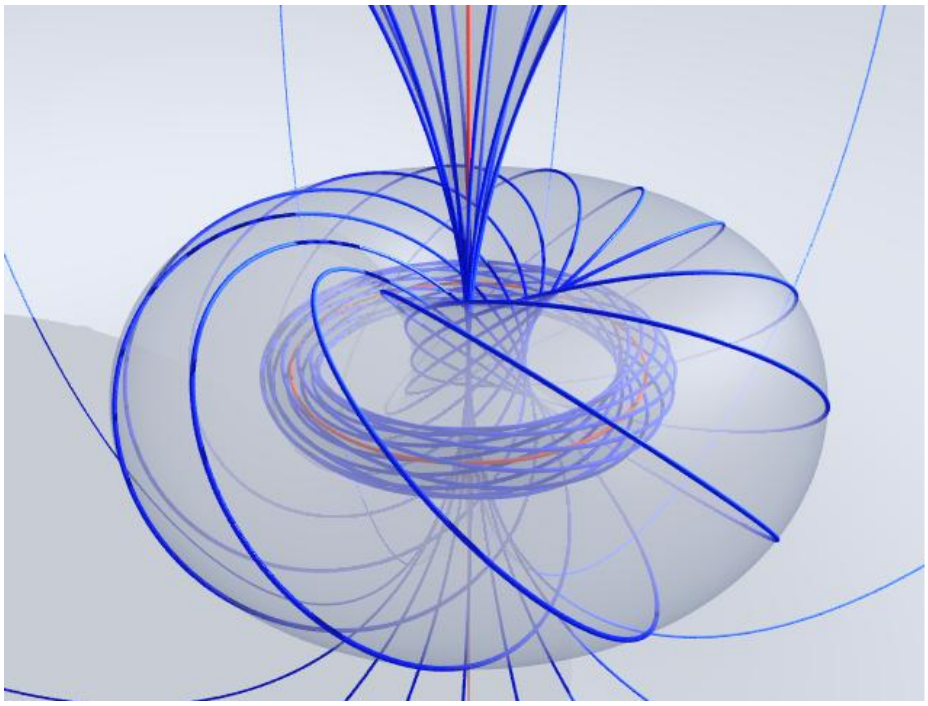


Figure 4: The images of the Hopf circles and the tori in $\mathbb{R}^3$ via stereographic projection. The tori correspond to fixed lengths $|z| = r$ and $|w| = \sqrt{1-r^2}$.

1. There is a fundamental configuration in $\mathbb{R}^3$ consisting of a circle and a straight line through the center of the circle, which is orthogonal to the plane of the circle (red circle and red line on Figure 4). These two objects correspond to the projective states of $|0\rangle$ and $|1\rangle$, respectively.

2. Every other state lies on the surface of a torus. The system of the tori can be obtained by 'thickening' the circle corresponding to $|0\rangle$. More precisely, these tori are the surfaces of revolution obtained by rotating a hyperbolic pencil of circles around the central line, see Figure 5.

3. The images of the Hopf circles are 'slant' circles on the tori surfaces in sense that they go around the meridian and the longitude once. *What is the reason of this property?*

Now we have three pictures about the quantum bit states: (1) Ψ can be visualized as two vectors (complex numbers) z and w on the plane, (2) the corresponding projective state $[\Psi]$ is a point on the Bloch sphere S^2, (3) every state $\Psi \neq N$ can be considered as a point of $\mathbb{R}^3$ via the stereographic projection. To compare these pictures, let's see what happens if we modify a state Ψ. See Figure 6 or this video of Niles Johnson.

1. *Varying the length of w between 0 and 1:* If $w = 0$, $|z| = 1$, the corresponding projective state is $[|0\rangle]$, this is the south pole on the Bloch sphere, 'spin down', the image of Ψ in $\mathbb{R}^3$ is a point of the fundamental circle. Increasing $|w|$, $|z|$ decreases, the point on the Bloch sphere moves along a meridian of the sphere, the torus containing the image of Ψ in $\mathbb{R}^3$ gets bigger (thicker). Reaching $|w| = 1$ and $z = 0$, the point on the Bloch sphere moves to the north pole ('spin up'), and the torus in $\mathbb{R}^3$ narrows down to the fundamental straight line, this is the projective state of $|1\rangle$.

2. *Varying the relative phase, i.e. the angle between z and w, while keeping the lengths $|z|$ and $|w|$ constant:* If $|z| = 0$ or $|w| = 0$, then the projective state stays in the south or the north pole of the Bloch sphere, the point in $\mathbb{R}^3$ moves along the fundamental circle or straight line.

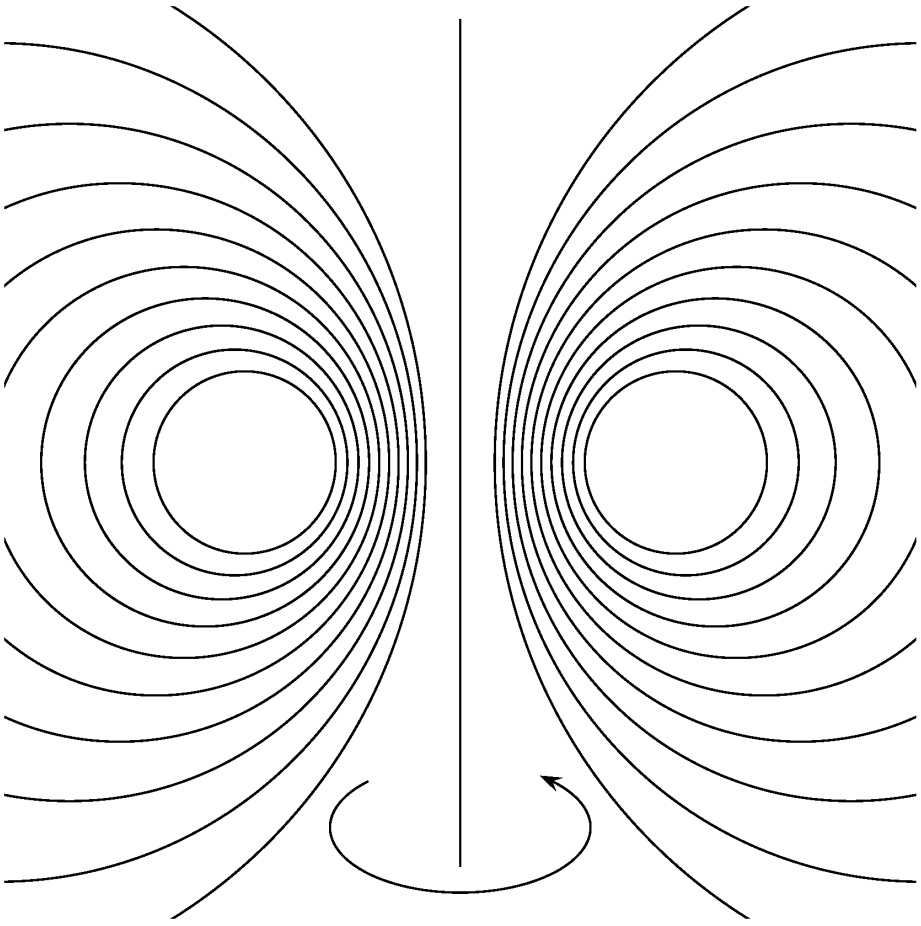


Figure 5: The tori in Figure 4 are the surfaces of revolution obtained by rotating a hyperbolic pencil of circles around the central line.

If $|z| \neq 0 \neq |w|$, then the point on the Bloch sphere moves along a longitude of the sphere. The image of Ψ in $\mathbb{R}^3$ moves on a fixed torus, following a longitude of the torus if z is rotated, and following a meridian of the torus if w is rotated. Importantly, each torus surface in $\mathbb{R}^3$ corresponds to a longitude of the Bloch sphere, which corresponds to fixing the lengths $|z| = r$ and $|w| = \sqrt{1-r^2}$ (where $0 < r < 1$), or equivalently, fixing the probabilities $\mathcal{P}(|0\rangle) = r^2$ and $\mathcal{P}(|1\rangle) = 1 - r^2$.

3. *Rotating z and w simultaneously:* The angle between z and w does not change. The point of the Bloch sphere does not move, indicating that the projective state does not change. The image of Ψ in $\mathbb{R}^3$ moves along a Hopf circle. Moving through the state N, i.e., $z = 0$ and $w = i$, the image of Φ in $\mathbb{R}^3$ escapes to the infinity.

2.3 Exercises

Exercise 2.3.1 (Hopf map)**.** Define the map $f : \mathbb{C}^2 \supset S^3 \to \mathbb{R}^3$

$$f(\Psi) = \begin{pmatrix} \mathrm{Re}(2\overline{z}w) \\ \mathrm{Im}(2\overline{z}w) \\ |z|^2 - |w|^2 \end{pmatrix}, \tag{2.3.2}$$

where

$$\Psi = \begin{pmatrix} z \\ w \end{pmatrix} \in S^3 \subset \mathbb{C}^2, \text{ i.e. } |z|^2 + |w|^2 = 1. \tag{2.3.3}$$

(a) Show that $|f(\Psi)| = 1$, that is, $f(\Psi) \in S^2 \subset \mathbb{R}^3$.

(b) Show that $f : S^3 \to S^2$ is surjective.

(c) Express $f(\Psi)$ in terms of the quotient w/z, if $z \neq 0$.

(d) In polar coordinates let $z = r \cdot e^{i\alpha}$, $w = s \cdot e^{i\beta}$; $r^2 + s^2 = 1$. Find $0 \leq \theta \leq \pi$ such that $r = \cos(\theta/2)$ and $s = \sin(\theta/2)$. Let $\phi := \beta - \alpha$ (the relative phase). Show that

$$f(\Psi) = \begin{pmatrix} \sin\theta \cdot \cos\phi \\ \sin\theta \cdot \sin\phi \\ \cos\theta \end{pmatrix}.$$

(e) Verify that f is essentially the same map as h we introduced in Equation (2.2.7) via the topological construction of the Bloch sphere. Explain the relationship of f and h in your own words.

Exercise 2.3.4 (Visualization of the Hopf circles)**.** (a) Express the stereographic projection (Equation (2.2.10)) by an explicit formula.

(b) Find a 'nice' parametrization of the tori

$$T_r = \{(z, w) \in \mathbb{C}^2 \mid |z| = r,\ |w| = \sqrt{1-r^2}\} \tag{2.3.5}$$

and the Hopf circles in S^3.

(c) By composing the paramterizations with the stereographic projection, find parametrizations for the images of the tori and Hopf circles in $\mathbb{R}^3$.

(d) Use Geogebra, Mathematica or other software to visualize the above objects in $\mathbb{R}^3$. Depict the three pictures of the state Ψ together with variable parameters, according to the description at the end of Section 2.2.

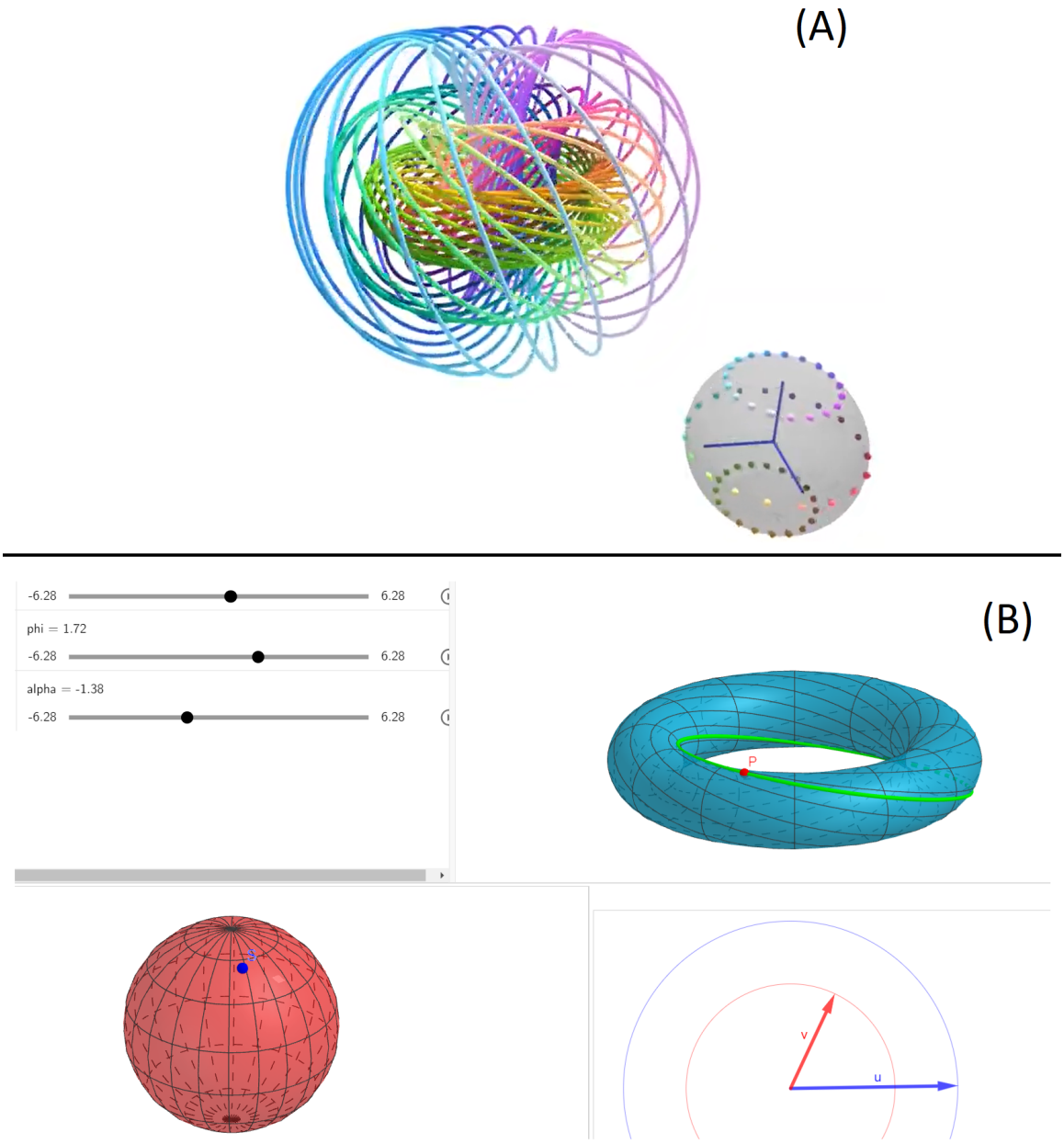


Figure 6: Panel (A) is a snapshot of the video of Niles Johnson, showing the tori corresponding to three longitudes of the Bloch sphere with the Hopf circles on them. Panel (B) is a snapshot of an interactive GeoGebra animation made by Jozef Bodnar showing all three visualizations of the states Ψ simultaneously.

Exercise 2.3.6 (Hermitian matrices)**.** Recall that a complex square matrix H is hermitian (or self-adjoint) if $H^\dagger = H$, where $H^\dagger = \overline{H^T}$ denotes the conjugate transpose of H. Every 2×2 hermitian matrix H can be uniquely written as a linear combination

$$H = a\sigma_0 + x\sigma_x + y\sigma_y + z\sigma_z \tag{2.3.7}$$

with real coefficients $a, x, y, z \in \mathbb{R}$ of the Pauli matrices

$$\sigma_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{2.3.8}$$

(a) Find the eigenvalues of $H = a\sigma_0 + x\sigma_x + y\sigma_y + z\sigma_z$.

(b) Find an eigenvector Ψ of unit length ($|\Psi| = 1$) for both eigenvalue of H.

(c) Consider an H with $a = 0$ and $x^2 + y^2 + z^2 = 1$. Apply the Hopf map f (see Equation (2.3.2)) to the eigenvectors of H. What is the result?

(d) The above matrix H is the operator of the spin measured in the (x, y, z) direction. What can we learn from the above calculation about this quantity?

Exercise 2.3.9 (Hopf map from expectation values)**.** Recall that the *expectation value* of an operator (now, a 2×2 hermitian matrix) H on the state Ψ is defined as $\Psi^\dagger \cdot H \cdot \Psi$. Show that the Hopf map f (see Equation (2.3.2)) can be expressed as

$$f(\Psi) = \begin{pmatrix} \Psi^\dagger \cdot \sigma_x \cdot \Psi \\ \Psi^\dagger \cdot \sigma_y \cdot \Psi \\ \Psi^\dagger \cdot \sigma_z \cdot \Psi \end{pmatrix}. \tag{2.3.10}$$

3 Topological spaces

3.1 Why?

Very shortly:

- Topology is the theory of continuous deformations.
- Topological classifications in physics aims to find the properties of the systems which remain unchanged during continuous deformations.

We have to introduce the basic terminology of the topology, like *topological spaces* and *continuous functions (maps)* between them. However, it is important to emphasize that we do not need topological spaces in their whole generality. We will only use the basic constructions, for example, gluing geometric pieces together to obtain abstract topological spaces. From this reason we only sketch the abstract terminology of general topology in this section, and many examples are coming in the next sections.

The classical analysis (or calculus) in $\mathbb{R}^n$ is based on two essential properties of functions: continuity and differentiability. We want to generalize these notions to our more general spaces. We will see that topological spaces are the objects for which the continuity of functions makes sense, but there is no differential calculus on the topological spaces, by default. Later we will introduce the *manifolds*, which are special type of topological spaces for which the differentiability of functions is meaningful.

3.2 An intermediate step: metric spaces

Recall the usual definition of the continuity of functions, formulating in a way that makes it meaningful in $\mathbb{R}$ and in $\mathbb{R}^n$ as well. For it we introduce the notation $d(x, y) = \|x - y\|$, where in $\|v\|$ denotes the absolute value, if $v \in \mathbb{R}$, and it denotes the norm of v in $\mathbb{R}^n$ (using any type of norm, see Exercise 3.4.2, but you can think on the 2-norm $\|v\| = \sqrt{\sum_{i=1}^n x_i^2}$, where $v = (x_1, \ldots, x_n)$).

Definition 3.2.1. Let $f : \mathbb{R}^n \supset D(f) \to \mathbb{R}^m$ be a map, defined on its domain $D(f)$.

(a) Let $a \in D(f) \subset \mathbb{R}^n$ be a point of the domain of f. Then, f is *continuous at the point* a, if

$$\forall \epsilon > 0 \;\; \exists \delta > 0 \;\; \forall x \in D(f) :$$
$$d(x, a) < \delta \;\Rightarrow\; d(f(x), f(a)) < \epsilon. \tag{3.2.2}$$

(b) f is *continuous* if it is continuous at every point of its domain.

The definition formulates the intuitive description of continuity: 'f maps nearby points to nearby points'. Observe that the definition does not use that we work in $\mathbb{R}^n$, it is enough that the distance $d(x, y)$ of two elements are defined. This observation leads to the notion of *metric spaces*, which are just sets endowed with a reasonable notion of distance:

Definition 3.2.3. A *metric space* is a set X together with a function $d : X \times X \to \mathbb{R}$ (called *metric*) satisfying the following conditions (for every $x, y, z \in X$):

1. $d(x, y) = d(y, x)$,
2. $d(x, y) \geq 0$, and $d(x, y) = 0$ if and only if $x = y$,
3. $d(x, y) + d(y, z) \geq d(x, z)$.

The third condition is called *triangle inequality*, since it expresses that any side of a 'triangle' (like xyz) is shorter or equal than the sum of the lengths of the other two sides. d can be an arbitrary function satisfying conditions 1-3., it does not necessarily come from a 'real distance' (actually, what does it mean?). The metric space is an abstract concept inspired by the geometry. The notion of continuity makes sense for functions between metric spaces:

Definition 3.2.4. Let X be a metric space with metric d_X and let Y be a metric space with metric d_Y. Let $f : X \to Y$ be a map (defined on the whole X).

(a) Let $a \in X$ be a point of X. Then, f is *continuous at the point* a, if

$$\forall \epsilon > 0 \;\; \exists \delta > 0 \;\; \forall x \in X :$$
$$d_X(x, a) < \delta \;\Rightarrow\; d_Y(f(x), f(a)) < \epsilon. \tag{3.2.5}$$

(b) f is *continuous* if it is continuous at every point of X.

Example 3.2.6 (Metric spaces)**.** Every subset of a metric space X inherits the metric of X, hence the basic examples for metric spaces are the subsets of $\mathbb{R}^n$ with the inherited metric. A more abstract example is any set X endowed with the metric defined to be $d(x, y) = 1$ if $x \neq y$, and $d(x, x) = 0$. One can check that this function d satisfies the properties 1-3., hence it defines a metric on an arbitrary set X, called *discrete metric*. Other metric spaces come from various areas of the mathematics. For example, many function spaces, like L^2 spaces or Sobolev spaces are normed spaces, and the norm defines a metric $d(x, y) = \|x - y\|$. Another surprising example is the set X consists of all closed and bounded (i.e., *compact*) sets of $\mathbb{R}^2$. There is a natural metric on X which is a key ingredient of the fractal geometry.

However, the metric spaces are definitely not suitable enough for us, because they are not 'flexible'. In general, two metric spaces (or two parts of one metric space) cannot be glued together to create a new metric space. Before the next step of abstraction we recall some basic notions for metric spaces.

Definition 3.2.7. Let X be a metric space with metric d.

(a) The *open ball* centered at a point $a \in X$ with radius $r > 0$ is defined as

$$B_r(a) = \{x \in X \mid d(x, a) < r\}. \tag{3.2.8}$$

(b) A subset $A \subset X$ is *open* if for every point $a \in A$ there is a radius $r > 0$ such that $B_r(a) \subset A$.

(c) A subset $A \subset X$ is *closed* if its complement $X \setminus A$ is open.

3.3 Topological spaces

The topological spaces are generalizations of the metric spaces, but without metric. Instead, the family of open sets is given.

Definition 3.3.1. A *topological space* is a set X together with a family of subsets of X denoted by τ (and called a *topology* on X), satisfying the following conditions:

1. $\emptyset \in \tau$ and $X \in \tau$.
2. If $A_1, \dots, A_n \in \tau$, then $\bigcap_{i=1}^{n} A_i \in \tau$.
3. If $A_j \in \tau$ for $j \in J$, then $\bigcup_{j \in J} A_\alpha \in \tau$ (J is an arbitrary index set).

Shortly we denote the above topological space by (X, τ), but we omit τ, if the topology on X is clear from the context, writing like 'a topological space X'. The next definition helps to understand the previous very abstract definition.

Definition 3.3.2. In a topological space (X, τ), a subset $A \subset X$ is called *open* if $A \in \tau$.

In other words, for topological spaces the 'primitive concept' is the notion of open sets. The elements of any family τ satisfying the conditions 1-3. in Definition 3.3.1 can be declared to be open sets, defining a topology on X. The three conditions require that

1. the empty set and the whole space are open,
2. the intersection of finitely many open sets is also open,
3. the union of arbitrary many open sets is also open.

One can show that the open sets of a metric space satisfy these conditions, hence every metric space is a topological space, see Exercise 3.4.1. The converse is not true, not every topology comes from a metric. The must extreme example is the *anti-discrete topology* $\tau = \{\emptyset, X\}$ on an arbitrary set X. In this case the only open sets are the empty set and the whole space, and the three conditions obviously hold for this τ. The fact that this topology cannot come from any metric can be deduced from Exercise 3.4.9.

Every concepts in topological spaces are defined via the open sets.

Definition 3.3.3. Let X be a topological space with topology τ.

(a) A subset $A \subset X$ is *closed* if its complement $X \setminus A$ is open, i.e., if $X \setminus A \in \tau$. Let τ' denote the family of closed sets.

(b) A subset $A \subset X$ is a *neighborhood* of a point $a \in X$ if A contains an open set B containing a, i.e., $a \in B \subset A$ and $B \in \tau$.

(c) An *open neighborhood* of $a \in X$ is an open subset B containing a, i.e., $a \in B \subset X$, $B \in \tau$.

(d) The *interior* of a subset $A \subset X$, denoted by $\mathrm{int}(A)$, is the biggest open set contained in A, i.e.,

$$\mathrm{int}(A) = \bigcup_{B \in \tau,\ B \subset A} B. \tag{3.3.4}$$

The elements of $\mathrm{int}(A)$ are called *interior points* of $A \subset X$.

(e) The *closure* of a subset $A \subset X$, denoted by $\mathrm{cl}(A)$, is the smallest closed set containing A, i.e.,

$$\mathrm{cl}(A) = \bigcap_{C \in \tau',\ A \subset C} C. \tag{3.3.5}$$

(f) The set

$$\partial A = \mathrm{cl}(A) \setminus \mathrm{int}(A) \tag{3.3.6}$$

is called the *boundary* of $A \subset X$, i.e., it is the difference between the closure and the interior.[2] The elements of ∂A are called the *boundary points* of $A \subset X$.

Importantly, all the above properties of subsets $A \subset X$ depend on the ambient space X and the topology τ on X, see Exercise 3.4.7. (A more precise notation for the interior, closure and boundary would be $\mathrm{int}_X(A)$, $\mathrm{cl}_X(A)$, $\partial_X(A)$.)

The definition of the continuity becomes extremely simple for topological spaces:

Definition 3.3.7. Let X and Y be topological spaces with topology τ and σ, respectively. A map $f : X \to Y$ is *continuous* if the preimage[3] of every open set is open, i.e., if

$$B \in \sigma \;\Rightarrow\; f^{-1}(B) \in \tau. \tag{3.3.8}$$

For metric spaces this definition agrees with the original Definition 3.2.4, see Exercise 3.4.5.

Definition 3.3.9. Let X and Y be topological spaces with topology τ and σ, respectively.

(a) A map $f : X \to Y$ is called *homeomorphism* if it is continuous and bijective and its inverse is also continuous.

(b) Two topological spaces (X, τ) and (Y, σ) are *homeomorphic* if there is a homeomorphism $f : X \to Y$ between them.

We denote by $X \cong Y$ that X and Y are homeomorphic. However, if the homeomorphism is very natural, we simply write $X = Y$, that is, X and Y are 'the same' topological spaces.

Homeomorphism is the natural equivalence of topological spaces. Two spaces being homeomorphic means that they are 'the same' from topological point of view. Intuitively this means that they can be continuously deformed to each other. If both shapes are 'made of gum', then one can 'stretch them to each other without cutting or gluing together faraway (non-nearby) pieces'. This imprecise description gives a good intuition for topology, but it does not replace the exact formulations, based on the precise definitions.

From topological point of view, the traditional geometry collapses. There are no distances, angles, the straight lines are meaningless, and the usual shapes represent very few homeomorphism classes. For example, every convex polygon on the plane (without its boundary) is homeomorphic to an open disc, which is also homeomorphic to the whole plane. On the other hand, we can introduce and investigate more interesting shapes (Möbius band, projective spaces etc.) focusing on their most essential properties, which do not change for continuous deformation.

3.4 Exercises

Exercise 3.4.1 (Metric and topology)**.** (a) Show that every metric induces a topology. More precisely, show that the open sets of a metric space (Definition 3.2.7) satisfy the conditions 1-3. of Definition 3.3.1. Conclude that every metric space is a topological space. Did you use all the three properties of the metric listed in Definition 3.2.3?

(b) Show that in a metric space every open ball is an open set (see Definition 3.2.7).

[2] Later, for manifolds we will have a more convenient notion of boundary ∂A, which does not depend on the ambient space X, it only depends on the manifold A. In general, the boundary of a manifold is not equal to its boundary as a subset of a topologycal space X. However, we take care to avoid confusion. In our examples in Section 4 the two notions of the boundary coincide. In Hungarian the terminology distinguishes the two notions, we say 'az $A \subset X$ részhalmaz *határa*' and 'az A sokaság *pereme*'.

[3] Reacall that for a map $f : X \to Y$ between two sets, the *image* of a subset $A \subset X$ is $f(A) = \{f(a) \mid a \in A\} \subset Y$, and the *preimage* of a subset $B \subset Y$ is $f^{-1}(B) = \{a \in X \mid f(a) \in B\} \subset X$. Images and the preimages always exist, while the inverse map f^{-1} of f exists if and only if f is bijective. The coincident notations f^{-1} for the inverse map and $f^{-1}(B)$ for the preimage of a subset is a bit confusing.

Exercise 3.4.2 (Equivalent norms)**.** Recall that a *norm* on a (real) vector space V is a function $\|\cdot\| : V \to \mathbb{R}$ satisfying the conditions (for every $v, w \in V$ and $r \in \mathbb{R}$)

1. $\|v\| \geq 0$, and $\|v\| = 0 \Leftrightarrow v = 0$,
2. $\|rv\| = |r| \cdot \|v\|$,
3. $\|v\| + \|w\| \geq \|v + w\|$.

Recall that two norms $\|\cdot\|_{(1)}$ and $\|\cdot\|_{(2)}$ are *equivalent* if there are real numbers $c, C > 0$ such that for every $v \in V$,

$$c\|v\|_{(1)} < \|v\|_{(2)} < C\|v\|_{(1)}. \tag{3.4.3}$$

(a) Given a norm on V, show that $d(v, w) := \|v - w\|$ defines a metric on V, i.e., every normed vector space is a metric space.

(b) A well-know theorem states that on a finite dimensional vector space V every norm is equivalent. Verify the statement for the most common norms on $\mathbb{R}^n$, i.e., show that the following three norms are equivalent:

$$\|v\|_1 = \sum_{i=1}^{n} |x_i|, \quad \|v\|_2 = \sqrt{\sum_{i=1}^{n} x_i^2}, \quad \|v\|_\infty = \max_{i=1}^{n}\{|x_i|\}, \tag{3.4.4}$$

where $v = (x_1, \ldots, x_n)$.

(c) Show that equivalent norms induce the same topology on V.

Exercise 3.4.5 (Continuity)**.** (a) Show that the definition of the continuity for topological spaces generalizes the definition of the continuity for metric spaces. More precisely, let X and Y be metric spaces, and let $f : X \to Y$ be a map. Show that f is continuous according to Definition 3.2.4 if and only if it is continuous according to Definition 3.3.7.

(b) Formally, a sequence of real numbers is a function $f : \mathbb{N} \to \mathbb{R}$. What is the condition for a sequence, as a function, to be continuous?

(c) What are the continuous functions $f : X \to Y$ if X or Y is a discrete or anti-discrete topological space?

Exercise 3.4.6 (Interior, closure and boundary)**.** (a) By condition 2. in Definition 3.3.1, the union of finitely many open sets is open. Show that it is not true for arbitrary many open sets, find a counterexample.

(b) Reformulate the conditions 1-3. in Definition 3.3.1 for the closed sets τ' (instead of the open sets τ).

(c) Let X be a topological space, $A \subset X$ and $a \in X$. Show that $a \in \mathrm{int}(A)$ if and only if there is a neighbourhood B of a with $B \subset A$.

(d) Let X be a topological space, $A \subset X$ and $a \in X$. Show that $a \in \mathrm{cl}(A)$ if and only if for every neighbourhood B of a, $B \cap A \neq \emptyset$.

(e) Let X be a topological space, $A \subset X$ and $a \in X$. Show that $a \in \partial A$ if and only if for every neighbourhood B of a, $B \cap A \neq \emptyset$ and $B \cap (X \setminus A) \neq \emptyset$.

(f) Show that $A \cup \partial A = \mathrm{cl}(A)$ and $A \setminus \partial A = \mathrm{int}(A)$.

(g) Show that $\partial A = \partial(X \setminus A)$.

(h) Show that $\mathrm{int}(A)$, ∂A and $\mathrm{int}(X \setminus A)$ are pairwise disjoint subsets of X, and their union is X.

(i) Show that $\partial A \subset X$ is a closed subset.

Exercise 3.4.7 (Open and closed subsets)**.** For the following topological spaces (X, τ) and their subsets $A \subset X$ decide, whether is A an open or closed subset, and give $\mathrm{cl}(A)$ and $\mathrm{int}(A)$ and ∂A.

(a) $X = \mathbb{R}$, τ is its usual metric topology, $A = [0; 1)$ is an interval.

(b) $X = [0; 5]$ is an interval, τ is its usual metric topology, $A = [0; 1)$ is an interval.

(c) $X = \mathbb{R}$, τ is the discrete topology, $A = [0; 1)$.

(d) $X = \mathbb{R}$, τ is the anti-discrete topology, $A = [0; 1)$.

(e) $X = A = [0; 1)$ with the metric topology.

(f) $X = \mathbb{R}$ with the usual metric topology, $A = (0; 1)$.

(g) $X = \mathbb{R}^2$ with the usual metric topology, $A = (0; 1)$.

(h) $X = \mathbb{R}$ with the usual metric topology, $A = \mathbb{Q}$ is the set of rational numbers.

Exercise 3.4.8 (Homeomorphic spaces)**.** Show that the topological spaces X and Y are homeomorphic. Find an explicit homoemorphism. (X and Y, as subsets of the plane $\mathbb{R}^2$, are endowed with metric topology.)

(a) $X = [a, b]$, $Y = [c, d]$ (where $a < b$ and $c < d$ are real numbers).

(b) $X = (a, b)$ (where $a < b$), $Y = \mathbb{R}$.

(c) $X = S^1 \setminus \{\text{point}\}$ (a circle minus one point), $Y = \mathbb{R}$.

(d) $X = S^1$, Y is the boundary line of a square.

(e) $X = D^2$ is a closed disc, Y is a square (as a closed region of the plane).

Exercise 3.4.9 (Convergence of sequences)**.** Let $(a_n) = (a_1, a_2, \dots)$ be an infinite sequence of elements of a topological space (X, τ), i.e., $a_n \in X$ for every n. We say that (a_n) *converges* to $b \in X$ if for every neighborhood A of b there is an integer N such that if $n > N$, then $a_n \in A$.

(a) Show that in $\mathbb{R}^n$ this agrees with the classical definition of convergence!

(b) Let (X, τ) be a discrete or an anti-discrete topological space. What are the convergent sequences in these spaces and what are their limits, where they converge?

(c) Show that in a metric space X every sequence converges to at most one point, i.e., if (a_n) converges to b and to c, then $b = c$. Show that this is not true in general for topological spaces, find a counterexample![4]

4 Basic constructions and examples for topological spaces

Again, we emphasize that we do not need the topology in its whole generality. Our topological spaces have geometric nature: we construct them from subspaces of $\mathbb{R}^n$.

So, the basic example is $\mathbb{R}^n$, endowed with the topology induced by the metric. Note that the metric of $\mathbb{R}^n$ comes from the norm i.e., $d(v, w) = \|v - w\|$, where is $\|\cdot\|$ is the the 2-norm by default, but any other norm induces the same topology on $\mathbb{R}^n$, see Exercise 3.4.2.

We note that the topological spaces do not have a well-defined dimension by default, although, most of our examples have an obvious dimension. It's reason is that these examples are special types of topological spaces (cell complexes and manifolds), for which the dimension is defined. The dimension is usually labeled in upper-right index in the notation.

[4]This exercise illustrates which kind of 'patology' can appear in general topological spaces. However, this will be irrelevant for us, because every topological space appearing in our context has geometric nature, in particular, every sequence in this space converges to at most one point.

4.1 Subspace topology, balls and spheres

Let X be a topologycal space with topology τ. Every subset $A \subset X$ inherits a topology $\tau|_A$, called *subspace topology*, defined by

$$\tau|_A = \{B \subset A \mid \exists C \subset X,\ C \in \tau \text{ such that } B = A \cap C\}. \tag{4.1.1}$$

That is, a subset $B \subset A$ is declared to be open if and only if it is the intersection of an open subset C of X with A. One can verify that $\tau|_A$ defines a topology on A, i.e., it satisfies conditions 1-3. of Definition 3.3.1. Hence, by the subspace topology, *every subset of a topological space is a topological space.*

Example 4.1.2 (Spheres and balls)**.** The topological space consists of the subset

$$S^n = \{(x_1, \dots, x_{n+1}) \in \mathbb{R}^{n+1} \mid \sum_{i=1}^{n+1} x_i^2 = 1\} = \{v \in \mathbb{R}^{n+1} \mid \|v\| = 1\} \tag{4.1.3}$$

endowed with the subspace topology from $\mathbb{R}^{n+1}$ is the *sphere of dimension* n.

The *closed ball of dimension* $n+1$ is the topological space consists of the subset

$$B^{n+1} = \{(x_1, \dots, x_{n+1}) \in \mathbb{R}^{n+1} \mid \sum_{i=1}^{n+1} x_i^2 \leq 1\} = \{v \in \mathbb{R}^{n+1} \mid \|v\| \leq 1\} \tag{4.1.4}$$

endowed with the subspace topology from $\mathbb{R}^{n+1}$.

The *open ball of dimension* $n+1$ is the topological space consists of the subset

$$\breve{B}^{n+1} = \{(x_1, \dots, x_{n+1}) \in \mathbb{R}^{n+1} \mid \sum_{i=1}^{n+1} x_i^2 < 1\} = \{v \in \mathbb{R}^{n+1} \mid \|v\| < 1\} \tag{4.1.5}$$

endowed with the subspace topology.

The names and the symbols label the homeomorphism type of the corresponding topological spaces. That is, every topological space which is homeomorphic to S^n (B^{n+1}, $\breve{B}^{n+1}$) is also denoted by S^n (B^{n+1}, $\breve{B}^{n+1}$), and called n dimensional sphere (closed ball, open ball).

In dimension $n = 1$, the balls are the intervals $B^1 = I = [-1; 1]$, $\breve{B}^1 = (-1; 1)$, the sphere contains only two points: $S^0 = \{-1; 1\}$. For $n = 2$, the ball B^2 is the unit disc in the plane $\mathbb{R}^2$, also denoted by $D^2 = B^2$. S^1 is the unit circle (and everything which is homeomorphic to it).

S^n and B^{n+1} are closed subsets of $\mathbb{R}^{n+1}$ with interior $\mathrm{int}(S^n) = \emptyset$ and $\mathrm{int}(B^{n+1}) = \breve{B}^{n+1}$. $\breve{B}^{n+1}$ is an open subset of $\mathbb{R}^{n+1}$, its closure is $\mathrm{cl}(\breve{B}^{n+1}) = B^{n+1}$. S^n is the boundary of B^{n+1} and also $\breve{B}^{n+1}$, i.e., $S^n = \partial B^{n+1} = \partial \breve{B}^{n+1}$. However, we emphasise that all these properties are valid *only if* S^n, B^{n+1}, $\breve{B}^{n+1}$ *are considered as subsets of* $\mathbb{R}^{n+1}$. For example, B^3, considered as its own subset, is open and closed and its boundary is empty[5]. The open interval $\breve{B}^1$, considered as a subset of $\mathbb{R}^2$, is not an open subset.

Proposition 4.1.6. *The open ball* $\breve{B}^n$ *is homeomorphic to the space* $\mathbb{R}^n$*.*

Sketch of the proof. Consider $\mathbb{R}^{n+1}$ with coordinates $x := (x_1, \dots, x_n)$ and $y := x_{n+1}$, and consider the southern open hemiphere with center $N = (0, 1)$, i.e.,

$$H := \{(x, y) \in \mathbb{R}^{n+1} \mid \|x\|^2 + (y-1)^2 = 1 \text{ and } y < 1\} \tag{4.1.7}$$

We define the desired homeomorphism in two steps: $\breve{B}^n \to H \to \mathbb{R}^n$.

First step. H is the graph of the continuous function $h : \breve{B}^n \to \mathbb{R}$,

$$h(x) = \sqrt{1 - \|x\|^2} + 1, \tag{4.1.8}$$

[5]Later we will introduce the notion of *manifolds*, and the boundary will get another, more natural meaning. In this sense, B^{n+1} is a manifold with boundray, and its boundary is the manifold S^n, independently on the ambient space.

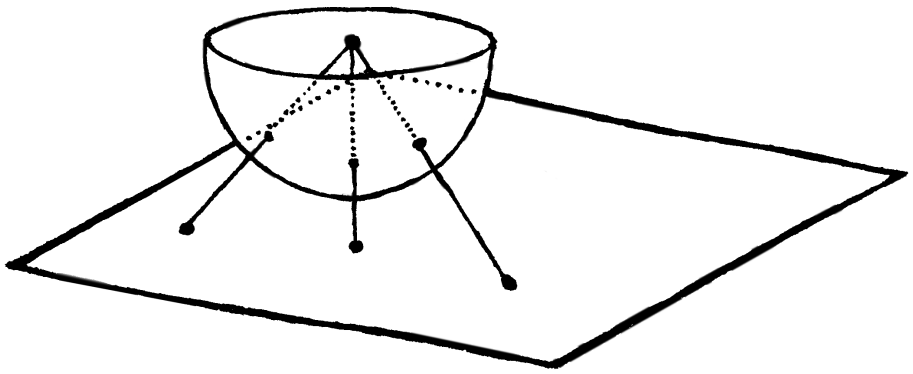


Figure 7: The central projection is a homeomorphism between the southern hemisphere H (homeomorphic to $\breve{B}^2$) and $\mathbb{R}^2$. Graphics by Mátyás Szabari, Új világok teremtése.

hence, the map $f : \breve{B}^n \to H$, $f(x) = (x, h(x))$ is a homeomorphism between the open ball and the hemisphere. See Exercise 4.8.1 for details.

Second step. We define a homeomorphism $g : H \to \mathbb{R}^n$ as the central projection from N. That is, for a point $p \in H$, define $g(p)$ as the unique intersection point of the line joining N and p with $\mathbb{R}^n = \{(x, 0)\} \subset \mathbb{R}^{n+1}$, see Figure 7. For details see Exercise 4.8.3. □

Proposition 4.1.9. *$S^n \setminus \{point\}$ is homeomorphic to $\mathbb{R}^n$.*

Sketch of the proof. Consider the unit sphere $S^n \subset \mathbb{R}^{n+1}$ with coordinates $x = (x_1, \ldots, x_n)$ and $y = x_{n+1}$, as above. The point $N = (0, 1) \in S^n$ is 'the north pole'. The *stereographic projection* $f : S^n \setminus \{N\} \to \mathbb{R}^n$ is defined as follows. For a point $N \neq p \in S^n$, let $f(p)$ be the unique intersection point of the line joining N and p with $\mathbb{R}^n = \{(x, 0)\} \subset \mathbb{R}^{n+1}$, see Figure 3. Then, f is a homeomorphism between $S^n \setminus \{N\}$ and $\mathbb{R}^n$. For details see Exercise 4.8.3. □

Since the image of N by the stereographic projection f escapes to the infinity, the above proposition is intuitively referred as 'S^n is obtained from $\mathbb{R}^n$ by extending it with a point at infinity', i.e., '$S^n = \mathbb{R}^n \cup \{\infty\}$'.

4.2 Product topology, cylinders and tori

Given two topological spaces, X with topology τ and Y with topology σ, the a Cartesian product (or direct product) of the sets, i.e.,

$$X \times Y = \{(x, y) \mid x \in X,\ y \in Y\} \tag{4.2.1}$$

becomes a topological space in a natural way. The *product topology* π on $X \times Y$ consists of the unions of subsets $A \times B$, where $A \in \tau$ and $B \in \sigma$. One can check that in this way we define a topology, i.e., π satisfies the conditions 1-3. of Definition 3.3.1.

Example 4.2.2 ($\mathbb{R}^n$)**.** The first example is $\mathbb{R}^n = \mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R}$. One can check that the topology induced by the metric agrees with the product topology. This example also shows that not every open subset $C \subset \mathbb{R}^2$ has the form $A \times B$, where A and B are open sets in $\mathbb{R}$, for example, the open disc $C = \breve{B}^2$ is not a product. Hence, we really need the unions of the sets $A \times B$ in the definition of the product topology.

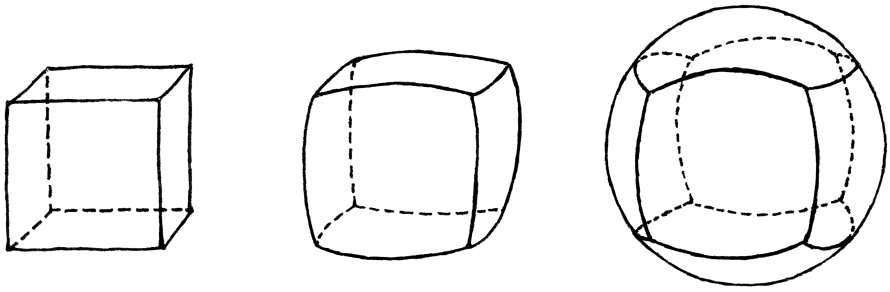


Figure 8: The cube I^3 is homeomorphic to the ball B^3, its boundary is homeomorphic to the sphere S^2. Graphics by Mátyás Szabari, Új világok teremtése.

Example 4.2.3 (Cubes and balls)**.** Consider $I = B^1 \subset \mathbb{R}$, its power $B^1 \times B^1 \in \mathbb{R}^2$ is a square. One can verify that $B^1 \times B^1$ is homoemorphic to the disc B^2, and its boundary is homoemorphic to S^1, see Exercise 3.4.8. More generally, the n-th Cartesian power $(B^1)^n = B^1 \times B^1 \times \ldots B^1 \subset \mathbb{R}^n$ is the (body of the) hypercube of dimension n. One can verify that $(B^1)^n$ is homeomorphic to B^n, and its boundary in $\mathbb{R}^n$ is homoemorphic to S^{n-1}. Much more generally, $B^n \times B^m$ is homeomorphic to B^{n+m}, shortly writing $B^n \times B^m \cong B^{n+m}$. See Figure 8.

Example 4.2.4 (Cylinders)**.** $S^1 \times \mathbb{R}$ is the *open cylinder* and $S^1 \times I$ is the *closed cylinder*. Since $S^1 \subset \mathbb{R}^2$ in a natural way, $S^1 \times \mathbb{R}$ and $S^1 \times I$ can be identified as subspaces of $\mathbb{R}^3 = \mathbb{R}^2 \times \mathbb{R}$, these are the usual cylinders in the 3-space. However, one can verify that they are also homeomorphic to certain subspaces of $\mathbb{R}^2$. $S^1 \times \mathbb{R}$ is homeomorphic to the complement of the origin, $\mathbb{R}^2 \setminus \{0\}$, and $S^1 \times I$ is homemorphic to an annulus

$$\{(x, y) \in \mathbb{R}^2 \mid 1 \leq x^2 + y^2 \leq 4\}, \tag{4.2.5}$$

i.e., the complement of an open disc in a closed disc.

Example 4.2.6 (Tori)**.** $\mathcal{T}^2 = S^1 \times S^1$ is the *torus* (of dimension 2), while $S^1 \times B^2$ is called the *solid torus*. Both are homeomorphic to certain subspaces of $\mathbb{R}^3$. Consider a circle S^1 on the xz plane which does not intersect the z axes, e.g.

$$\{(x, y) \in \mathbb{R}^2 \mid (x-2)^2 + z^2 = 1\}. \tag{4.2.7}$$

By rotating this circle around the z axes results in a surface of revolution $T \subset \mathbb{R}^3$. The surface T is homeomorphic to the torus $\mathcal{T}^2$. Indeed, one of the S^1 components can be identified with the rotated circle, and the other one is the rotation, hence, every point of T can be determined by two S^1 valued coordinates, see Exercise 4.8.5 for details.

In $\mathbb{R}^3$, the surface T is the boundary of a body obtained by the revolution of the disc

$$\{(x, y) \in \mathbb{R}^2 \mid (x-2)^2 + z^2 \leq 1\}. \tag{4.2.8}$$

This body is a subspace of $\mathbb{R}^3$ and it is homeomorphic to the solid torus $S^1 \times B^2$.

There are also many different looking tori in $\mathbb{R}^3$, e.g. the knotted tori. Consider a *knot* $K \subset \mathbb{R}^3$, i.e., a subspace of $\mathbb{R}^3$ homeomorphic to S^1, like the trefoil knot, see Figure 9. The 'thickening' of K provides a solid torus around K, whose boundary is a torus $\mathcal{T}^2$. More precisely, one can show that there is an $0 < \epsilon$ such that the set of points p whose distance from K is lower or equal then ϵ is homeomorphic to a solid torus (it is a so-called tubular neighborhood of K).

Example 4.2.9 (Tori in S^3)**.** We saw in Section 2.2 that the qubit states in S^3 determining the same probability distribution form a torus. To show it, recall that the space of 1-qubit states is

$$S^3 = \{(z, w) \in \mathbb{C}^2 \mid |z|^2 + |w|^2 = 1 \ \}, \tag{4.2.10}$$

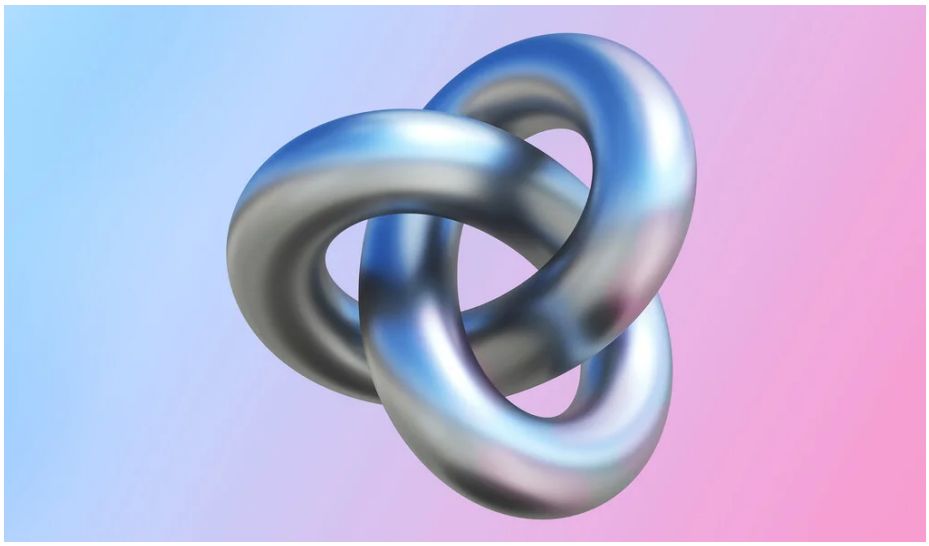


Figure 9: Knotted torus around a trefoil knot.

the probability of $|0\rangle$ is $|z|^2$, the probability of $|1\rangle$ is $|w|^2$. Hence, the set of states determining the same distribution is

$$T_r = \{(z, w) \in S^3 \mid |z|^2 = r\}, \tag{4.2.11}$$

which shows that z can move along a circle S^1 of radius $\sqrt{r}$, and w can move along a circle S^1 of radius $\sqrt{1-r}$. Thus, any point of T_r is identified by two circular coordinates, showing that T_r, as a subspace of S^3, is homeomorphic to a torus $\mathcal{T}^2$ for each r.

In S^3, T_r is the boundary of two closed subsets

$$P_1 = \{(z, w) \in S^3 \mid |z|^2 \leq r\}, \text{ and } P_2 = \{(z, w) \in S^3 \mid |z|^2 \geq r\} \tag{4.2.12}$$

One can verify that P_1 and P_2 are solid tori. Indeed, in case of P_1, z can be any point of a disc B^2 of radius $\sqrt{r}$, and for any fixed value of z one can chose w from a circle S^1 of radius $\sqrt{1-|z|^2}$ (and simaliarly for P_2). In particular, for $r = \frac{1}{2}$, the torus T_r divides S^3 into two congruent solid tori P_1 and P_2.

4.3 Quotient topology

In Section 2.2 we investigated the space of all projective states. Here the projective states are disjoint subsets of S^3 whose union is S^3, see Equation (2.2.5). The quotient topology gives explicit meaning to such constructions.

Recall the relationship between the equivalence relations and partitions, and the notion of the quotient set. Let X be a set, and let $Y = \{X_j\}_{j\in J}$ be a *partition* of X. That is, X decomposes as the disjoint union of the subsets $X_j \subset X$, i.e., $X = \bigcup_{j\in J} X_j$, where $X_i \cap X_j = \emptyset$ for all pairs $i \neq j \in J$. Here J is an arbitrary index set, but it can be identified with the set Y (self-indexing), hence, practically, $J = Y$.

An *equivalence relation* $\sim$ on X leads to the same setup, since an equivalence relation determines a partition and vice versa: two points p and q in X are in relation, $p \sim q$, if and only if they are in the same subset X_j. Therefore, the subsets X_j are the equivalence classes of the relation $\sim$, i.e., if $p \sim q$, then $[p] = [q] = X_j$, where $[p]$ denotes the equivalence class of p:

$$[p] = \{q \in X \mid p \sim q\}. \tag{4.3.1}$$

Y is the set of the equivalence classes called *quotient set*, denoted by $Y = X/\sim$. There is a natural *quotient map*

$$\begin{aligned} f : X &\to Y \qquad (4.3.2)\\ f(p) &= [p]. \qquad (4.3.3)\end{aligned}$$

Observe that – a bit confusingly – $[p]$ denotes a subset of X, $[p] \subset X$, and it also denotes an element of Y, $[p] \in Y$, with the tautological relationship

$$f^{-1}([p]) = [p]. \qquad (4.3.4)$$

Now, let X be a topological space with topology τ, let $\sim$ be an equivalence relation on the set X, and let $Y = X/\sim$ be the quotient set, let $f : X \to Y$ be the quotient map. Then there is a natural topology σ on Y, called *quotient topology*:

$$\sigma = \{A \subset Y \mid f^{-1}(A) \in \tau\}. \qquad (4.3.5)$$

In other words, a set of equivalence classes $A = \{X_j\} \subset Y$ is open if and only if the union of the sets X_j is open in X, i.e.,

$$f^{-1}(A) = \bigcup_{X_j \in A} X_j \in \tau. \qquad (4.3.6)$$

Intuitively, Y (endowed with the quotient topology σ) is the topological space obtained from X by identifying (gluing together) the equivalent points of X.

Example 4.3.7 (An illustrative example)**.** Consider $X = \mathbb{R}^2$ with coordinates (x, y) and consider the equivalence relation $(x_1, y_1) \sim (x_2, y_2)$ if $x_1 = x_2$. The equivalence classes are the vertical lines

$$X_r = \{(x, y) \in \ \mid x = r\}, \qquad (4.3.8)$$

for every $r \in \mathbb{R}$. The quotient set is $Y = \{X_r\}$, and the quotient map is $f(x, y) = X_x$, i.e., X_x is the equivalence class corresponding to $r = x$.

What is the topological space Y of the vertical lines of the plane? The indexing indicates that Y is homeomorphic to $\mathbb{R}$, indeed, $X_r \mapsto r$ is a homeomorphism. To see that this is really a homeomorphism (not only a bijection), one can analyse the meaning of the quotient topology on Y. A subset A of Y is open iff $f^{-1}(A) \subset X$ is open. $f^{-1}(A)$ is nothing but the union of the elements of A, which are vertical lines X_r on the plane:

$$f^{-1}(A) = \bigcup_{X_r \in A} X_r. \qquad (4.3.9)$$

Observe that the union of the lines $X_r \in A$ is an open subset of the plane $\mathbb{R}^2$ if and only if the set of the indices, i.e.,

$$\{r \in \mathbb{R} \mid X_r \in A\} \qquad (4.3.10)$$

is an open subset of $\mathbb{R}$, this establishes that the map $X_r \mapsto r$ is a homeomorphism between Y and $\mathbb{R}$.

Example 4.3.11 (Circle)**.** Let $X = [0; 1]$ be a closed interval, X is homeomorphic to B^1. Consider the equivalence relation $0 \sim 1$ and for r_1, r_2 not equal to 0 or 1, $r_1 \sim r_2$ if and only if $r_1 = r_2$. The equivalence classes are $\{0; 1\}$ and $\{r\}$ $(0 < r < 1)$. What is $Y = X/\sim$? Intuitively, the two end points of a line segment are glued together, hence, Y should be homeomorphic to S^1. Shortly we write

$$\frac{[0; 1]}{0 \sim 1} = \frac{B^1}{S^0} \cong S^1. \qquad (4.3.12)$$

How can we prove rigorously that Y is really homoemorphic to S^1?

Before the answer, let us analyse the open sets! Recall that the intervals $[0; r)$ and $(r, 1]$ $(0 < r < 1)$ are open subsets of X (although they are not open sets of $\mathbb{R}$). If we take image $f(A)$ of $A := [0, r)$

via the quotient map $f : X \to Y$, it contains the point $\{0;1\}$, hence its preimage $f^{-1}(f(A))$ is equal to $[0,r) \cup \{1\}$, which is not open in X. If we do the same with a subset $B = [0,r_1) \cup (r_2,1]$ (where $0 < r_1 < r_2 < 1$), then $f^{-1}(f(A)) = A$ is open subset of X, hence $f(A)$ is open in Y.

A precise and also intuitive way to see that Y is homeomorphic to S^1 is consider the parametrization

$$\begin{aligned} g : [0;1] &\to S^1 \subset \mathbb{R}^2 && (4.3.13)\\ g(t) &= (\cos 2\pi t,\ \sin 2\pi t). && (4.3.14)\end{aligned}$$

Observe that $g(0) = g(1)$, i.e., g glues together 0 and 1. Hence, g induces a map $\widetilde{g} : Y \to S^1$ with $\widetilde{g}(\{0;1\}) := g(0) = g(1)$, and $g = \widetilde{g} \circ f$ holds, see the diagram 4.3.15. Moreover, $\widetilde{g}$ is a bijection between Y and S^1. To show that $\widetilde{g}$ is a homeomorphism it is enough to check that $A \subset S^1$ *is an open subset if and only if its preimage* $g^{-1}(A) \subset X$ *is open.* In this way the rather abstract notion of quotient topology is reduced to the study of the map g between subspaces of $\mathbb{R}^2$.

$$\begin{array}{ccc} & Y & \\ {\scriptstyle f}\nearrow & & \searrow{\scriptstyle \widetilde{g}} \\ X & \xrightarrow{\quad g \quad} & S^1 \end{array} \qquad (4.3.15)$$

Example 4.3.16 (Balls and spheres)**.** More generally, contracting the boundary S^{n-1} of B^n to one point results in a quotient space homeomorphic to S^n. In this case, $X = B^n$, the equivalence relation is defined by $p \sim q$ if and only if $p = q$ or $p, q \in S^{n-1}$. The equivalence classes are S^{n-1} and the one point classes $\{p\}$ for $p \neq S^{n-1}$. The quotient space $Y = X/\sim$ consists of the points $\{p\}$ for $p \in B^n \setminus S^{n-1}$ and S^{n-1} itself. The quotient map $f : X \to Y$ is a bijection on $B^n \setminus S^{n-1}$, and it sends the points of S^{n-1} to one point, a bit confusingly denoted by $S^{n-1} \in Y$. Intuitively, f contracts S^{n-1} to one point. Shortly we write

$$\frac{B^n}{S^{n-1}} \cong S^n. \qquad (4.3.17)$$

As in the previous example, a precise and intuitive way to show that Y is homeomorphic to S^n is to find a parametrization $g : B^n \to S^n \subset \mathbb{R}^{n+1}$ which sends the boundary S^{n-1} to one point, e.g., to the north pole, see Exercise 4.8.3. Then g induces a homeomorphism $\widetilde{g} : Y \to S^n$, as above.

For the next examples see Figure 10, my popular science video Space creation (with English subtitles) and my book Új világok teremtése (in Hungarian).

Example 4.3.18 (Cylinder and torus)**.** The closed cylinder and the torus can be considered as quotient spaces of the square $X = [0;1] \times [0;1]$ (homeomorphic to B^2). Consider the equivalence relation $(0,y) \sim (1,y)$ for every $y \in [0;1]$, and $p \sim q \ \Leftrightarrow\ p = q$ for every other $p, q \in X$. In this way we glue the end points of the horizontal line segments together to make them a circle S^1. Hence, the quotient $X/\sim$ is homeomorphic to a cylinder $S^1 \times I$:

$$\frac{[0;1] \times [0;1]}{(0,y) \sim (1,y)} \cong S^1 \times I. \qquad (4.3.19)$$

To obtain a torus, we have to glue also the end points of the vertical line segments together:

$$\frac{[0;1] \times [0;1]}{(0,y) \sim (1,y),\ (x,0) \sim (x,1)} \cong S^1 \times S^1. \qquad (4.3.20)$$

These quotients are visualized on Figure 10, the equivalence relation is indicated by the arrows.

Example 4.3.21 (Möbius band and Klein bottle)**.** Encouraged by the previous example, one can define new spaces by gluing together the edges of the square in different ways.

$$\mathcal{M} := \frac{[0;1] \times [0;1]}{(0,y) \sim (1,1-y)} \qquad (4.3.22)$$

glues two opposite sides together with reversed orientation, the resulted space $\mathcal{M}$ is the *Möbius band*.

Combining the cylinder-type and the Möbius-type gluing for the two opposite pairs of edges results in a topological space

$$\mathcal{K} := \frac{[0;1]\times[0;1]}{(0,y)\sim(1,1-y),\ (x,0)\sim(x,1)}, \tag{4.3.23}$$

called *Klein bottle*. See Figure 10 and my popular science video Space creation (with English subtitles).

We have already seen that the cylinder and the torus can be realized as subspaces of $\mathbb{R}^3$, and one can verify the same for the Möbius band, indeed, it can be made from a piece of paper of rectangular shape. In contrast, the Klein bottle cannot be realized as a subspace of $\mathbb{R}^3$. Its usual visualization has a circle of double point (self-intersection), where the tube (a cylinder) is inserted through itself. Therefore, this visualization is not the Klein bottle, but a quotient of it (pairs of points along two circles are glued together). It can be proved that there is no subspace of $\mathbb{R}^3$ which is homeomorphic to the Klein bottle. See Exercise 4.8.6.

However, the Klein bottle can be realized as a subspace of $\mathbb{R}^4$, that is, the gluing can be done in $\mathbb{R}^4$ without creating self-intersection. Indeed, around the self-intersection circle in 3D, one of the branches can be endowed with a different fourth coordinate, in other words, in the first part of the gluing process the fourth coordinate increases to avoid the self-intersection, then decreases to 0 to realize the gluing.

Example 4.3.24 (Higher dimensional torus)**.** One can construct the n dimensional torus

$$\mathcal{T}^n := (S^1)^n = S^1\times S^1\times\cdots\times S^1 \tag{4.3.25}$$

as a quotient of the n dimensional cube $[0;1]^n$ (homeomorphic to B^n), for example,

$$\mathcal{T}^3 = S^1\times S^1\times S^1 \cong \frac{[0;1]\times[0;1]\times[0;1]}{(0,y,z)\sim(1,y,z),\ (x,0,z)\sim(x,1,z),\ (x,y,0)\sim(x,y,1)}, \tag{4.3.26}$$

i.e., the opposite faces of the cube are glued together with preserving the orientation. Learn more about these type of constructions in the book *Jeffrey R. Weeks: The shape of space*.

Example 4.3.27 (Circle and torus again)**.** Consider $X=\mathbb{R}$ with the equivalence relation $r\sim q \Leftrightarrow r-q\in\mathbb{Z}$. The equivalence classes are

$$X_r = r+\mathbb{Z} = \{\dots, r-1, r, r+1, r+2, \dots\} \tag{4.3.28}$$

for $0\le r<1$, i.e., each of them is represented by an element $r\in[0;1]$, but $0\sim 1$. The quotient is $Y=X/\sim = S^1$, we shortly write

$$\frac{\mathbb{R}}{\mathbb{Z}} \cong S^1. \tag{4.3.29}$$

The rigorous proof can be done by extending the parametrization in Example 4.3.11, obtaining

$$g:\mathbb{R} \ \to\ S^1\subset\mathbb{R}^2 \tag{4.3.30}$$

$$g(t) \ =\ (\cos 2\pi t,\ \sin 2\pi t). \tag{4.3.31}$$

The map g sends every equivalence class X_r to one point of S^1, hence it induces a bijection $\widetilde{g}: Y\to S^1$, which turns out to be a homeomorphism. In the same way we can see that

$$\frac{\mathbb{R}^n}{\mathbb{Z}^n} = (S^1)^n \cong \mathcal{T}^n. \tag{4.3.32}$$

See Figure 11.

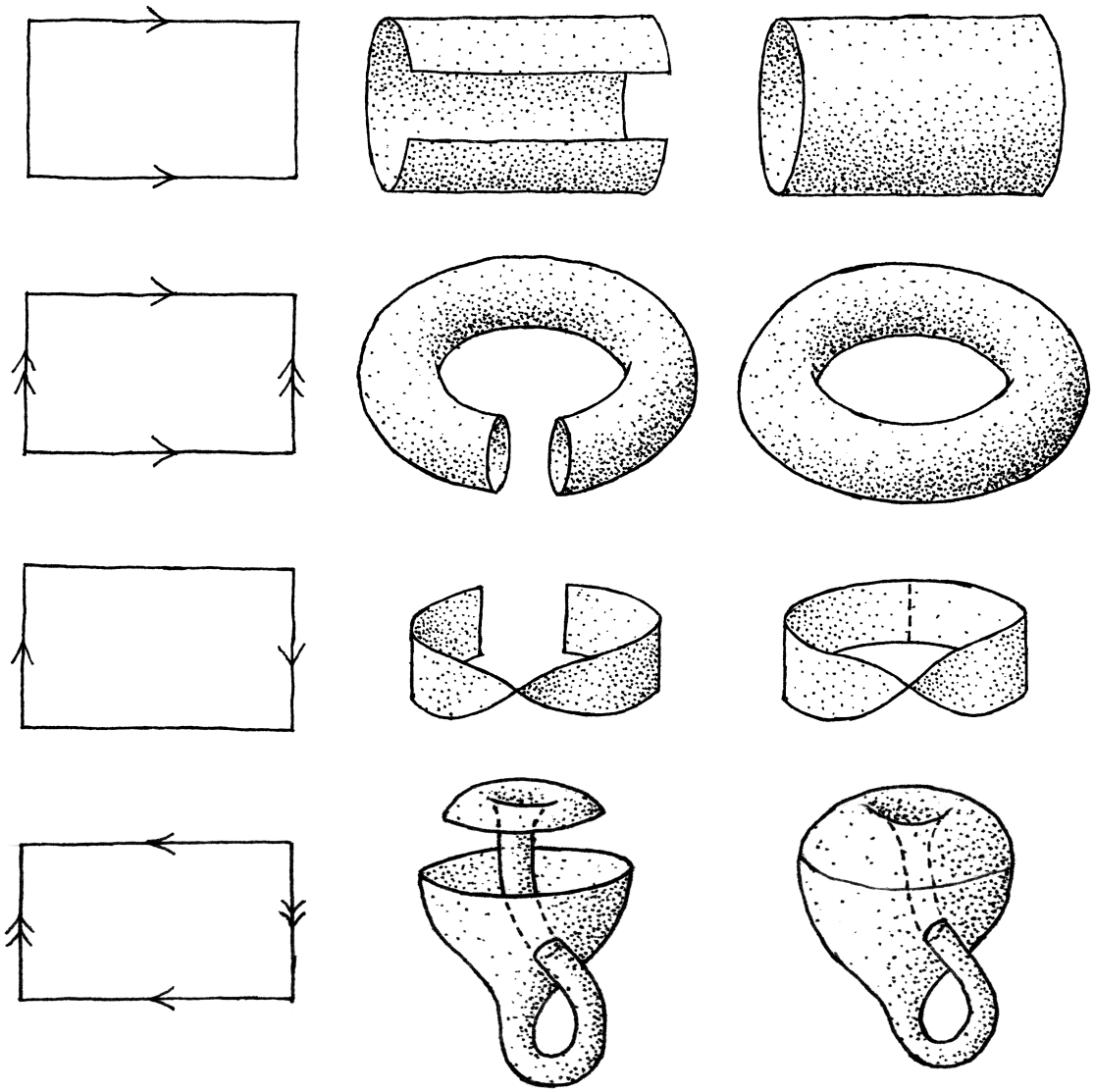


Figure 10: Cylinder, torus, Möbius, Klein, as quotient spaces of the square $[0;1] \times [0;1]$ and as subspaces of $\mathbb{R}^3$. The Klein bottle is not a subspace of $\mathbb{R}^3$, in this picture it has a circle of self-intersection points. Graphics by Mátyás Szabari, Új világok teremtése.

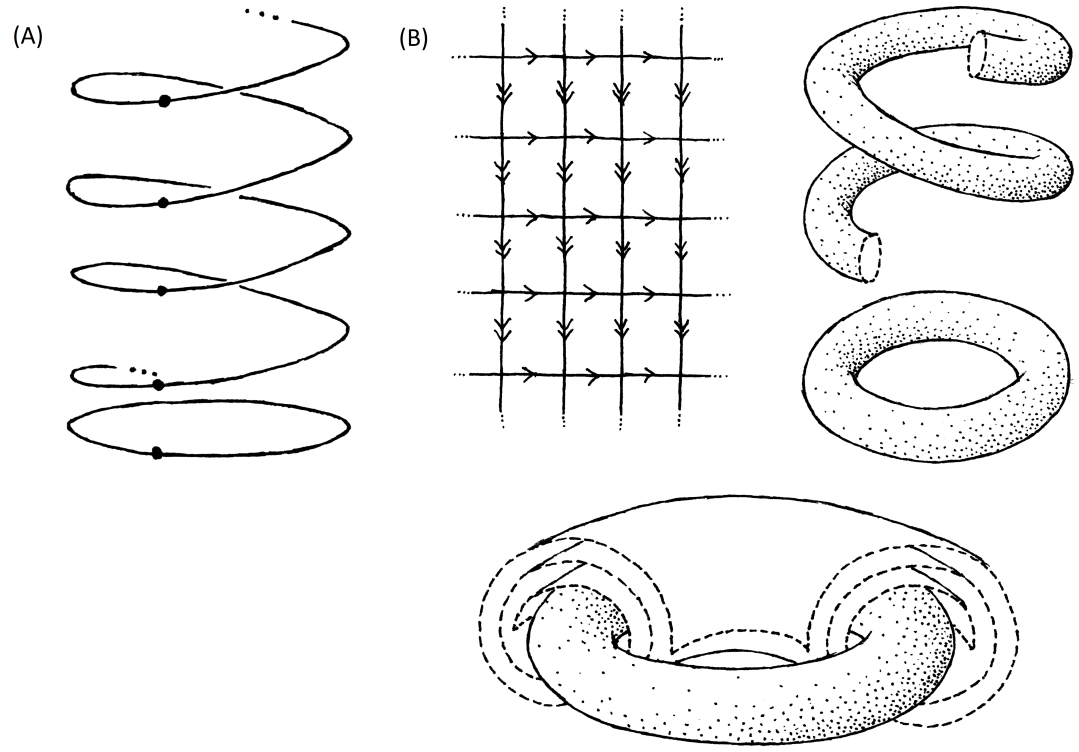
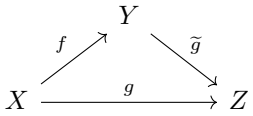


Figure 11: Visualization of the quotients $\mathbb{R}/\mathbb{Z} = S^1$ (panel (A)) and $\mathbb{R}^2/\mathbb{Z}^2 = \mathcal{T}^2$ (panel (B)). Graphics by Mátyás Szabari, Új világok teremtése.

The examples suggest a convenient way to identify a quotient space $Y = X/\sim$ with a more familiar topological space Z, according to Diagram (4.3.33). Find a continuous map $g : X \to Z$ which induces a bijection $\widetilde{g} : Y \to Z$ such that $g = \widetilde{g} \circ f$, where $f : X \to Y$ is the quotient map. To show that $\widetilde{g}$ is a homeomorphism it is enough to check that $A \subset S^1$ *is an open subset if and only if its preimage* $g^{-1}(A) \subset X$ *is open.* In this way the rather abstract notion of quotient topology is reduced to the study of the more familiar map g. Informally, we call such a g a parametrization, but it is also called quotient map or *identification map* in the literature.



(4.3.33)

4.4 Gluing topological spaces together

The quotient topology allows us to glue topological spaces together. Let X and Y be topological spaces (with topology τ and σ), let $A \subset X$ be a subspace of X and let $f : A \to Y$ be a continuous map. We *glue* X *and* Y *together along the map* f to obtain a new topological space

$$Z = X \cup_f Y = X \cup_A Y, \tag{4.4.1}$$

where the second notation is used if the map f is clear from the context. Z is defined as the quotient space of the disjoint union $X \cup Y$ with the equivalence relation $p \sim f(p)$ for $p \in A$, and $p \sim q \Leftrightarrow p = q$ for every other p and q. Then, $Z := (X \cup Y)/\sim$.

Example 4.4.2 (Balls and spheres)**.**

$$B^n \cup_{S^{n-1}} B^n \cong S^n, \tag{4.4.3}$$

more precisely, gluing two balls together along a homeomorphism between the boundaries

$$\partial B^n = S^{n-1} \to S^{n-1} = \partial B^n \tag{4.4.4}$$

results in a topological space homeomorphic to S^n. Indeed, the unit sphere $S^n \subset \mathbb{R}^{n+1}$ decomposes as the union of the northern and southern closed semispheres,

$$B_1 = \{(x, y) \in \mathbb{R}^{n+1} \mid \|x\|^2 + y^2 = 1,\ y \geq 0\} \text{ and } B_1 = \{(x, y) \in \mathbb{R}^{n+1} \mid \|x\|^2 + y^2 = 1,\ y \leq 0\},$$

where $x = (x_1, \ldots, x_n)$, $y = x_{n+1}$. Both B_1 and B_2 are homeomorphic to B^n, and their intersection is the equator $B_1 \cap B_2 = S^{n-1}$. (Note that this argument does not prove that using *any* homoemorhism $S^{n-1} \to S^{n-1}$ for the gluing results in a space S^n, but it also holds.)

Example 4.4.5 (Tori in S^3)**.** By Example 4.2.9, S^3 decomposes as the union of two solid tori P_1 and P_2. They are glued together along their common torus boundary $P_1 \cap P_2 = \mathcal{T}^2$. However, the identification is a bit tricky. Consider the solid tori $P_i = B^2 \times S^1$ with coordinates $x_i \in B^2$, $y_i \in S^1$ $(i = 1, 2)$, then, on the boundaries ∂P_i, x_i is the meridian coordinate, y_i is the longitude. One can conclude from the construction that the meridian of ∂P_1 is identified with the longitude of ∂P_2, hence, the gluing map is

$$f : S^1 \times S^1 \to S^1 \times S^1 \tag{4.4.6}$$

$$f(x_1, y_1) = (y_1, x_1), \tag{4.4.7}$$

and

$$S^3 \cong (B^2 \times S^1) \cup_f (B^2 \times S^1). \tag{4.4.8}$$

Example 4.4.9 (Möbius and Klein)**.**

$$\mathcal{M}_1 \cup_{S^1} \mathcal{M}_2 \cong \mathcal{K}, \tag{4.4.10}$$

that is, two Möbius bands glued together along their S^1 boundaries form a Klein bottle. It can be seen by cutting the Klein bottle in half, obtaining a Möbius band, see Figure 12. However, this visualization deals with a self-intersected Klein bottle in $\mathbb{R}^3$, and the result is a self-intersected Möbius band. A rigorous proof for Equation (4.4.10) is indicated by Figure 13.

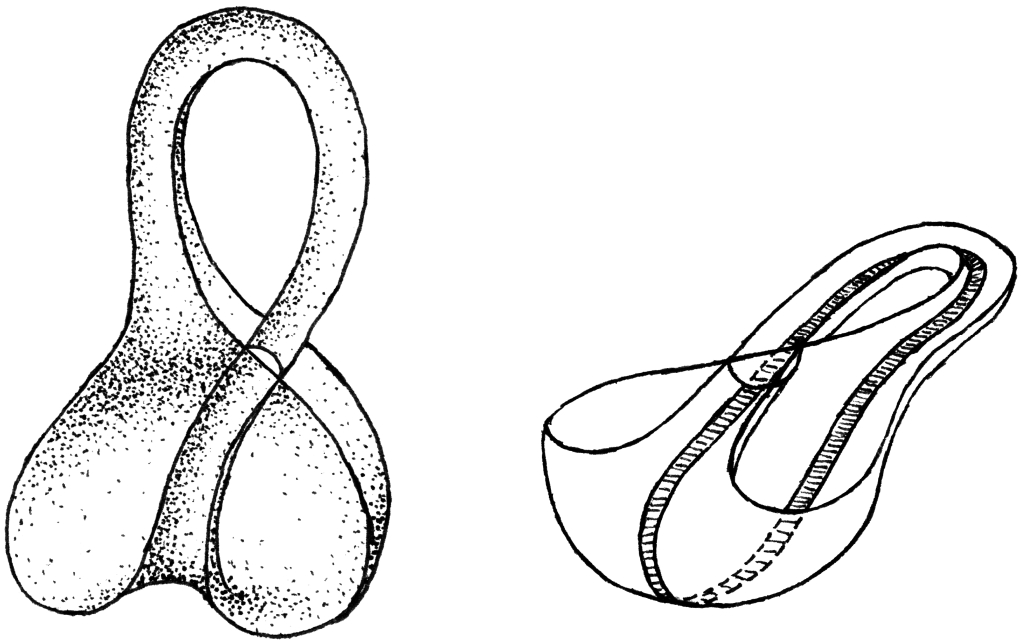


Figure 12: A 'half' of a Klein bottle is a Möbius band (with self-intersection in $\mathbb{R}^3$). Graphics by Mátyás Szabari, Új világok teremtése.

4.5 Cell complexes

Cell complexes (also called CW complexes) are special topological spaces assembled from building blocks (cells) B^n according to the following rules. The construction is recursive by the dimension. For $0 \le d \le n$, where n is the dimension of the cell complex X we want to construct, let $E_1^d, E_2^d, \dots$ [6] denote the d dimensional cells, each one is homeomorphic to B^d. Assuming that every cell of dimension up to d are already glued together forming the *d-skeleton* $\mathrm{sk}_d(X)$ of the cell complex X, then each $d+1$ cell E_j^{d+1} is glued to $\mathrm{sk}_d(X)$ along a continuous map

$$\phi_j^{d+1} : \partial E_j^{d+1} = S^d \to \mathrm{sk}_d(X). \tag{4.5.1}$$

In particular, the 0-cells $E_1^0, E_2^0, \dots$ are isolated points, hence, $\mathrm{sk}_0(X)$ is a discrete topological space consists of finitely many points. For a cell complex X of dimension $n = 0$ that's all, $X = \mathrm{sk}_0(X)$ is a discrete topological space consists of finitely many points.

A 1-cell $E_j^1 = B^1 = [-1, 1]$ is glued to $\mathrm{sk}_0(X)$ along its gluing map

$$\phi_j^1 : \partial E_j^1 = \{-1; 1\} \to \mathrm{sk}_0(X), \tag{4.5.2}$$

which is determined by $\phi_j^1(-1) = E_{j'}^0$ and $\phi_j^1(1) = E_{j''}^0$. Hence, $\mathrm{sk}_1(X)$ is a graph, whose vertexes are the 0-cells and the edges are the 1-cells. Since $\phi_j^1(-1) = \phi_j^1(1)$ is also possible, the graph possibly have loops, and also multiple edges. A cell complex X of dimension $n = 1$ is a graph $X = \mathrm{sk}_1(X)$.

Each 2-cell $E_j^2 = B^2$ is glued to $\mathrm{sk}_1(X)$ along a continuous map

$$\phi_j^2 : \partial E_j^2 = S^1 \to \mathrm{sk}_1(X), \tag{4.5.3}$$

etc.

[6] For simplicity, we use only finitely many cells. In general their number can be infinite, but this case requires to solve further technical difficulties.

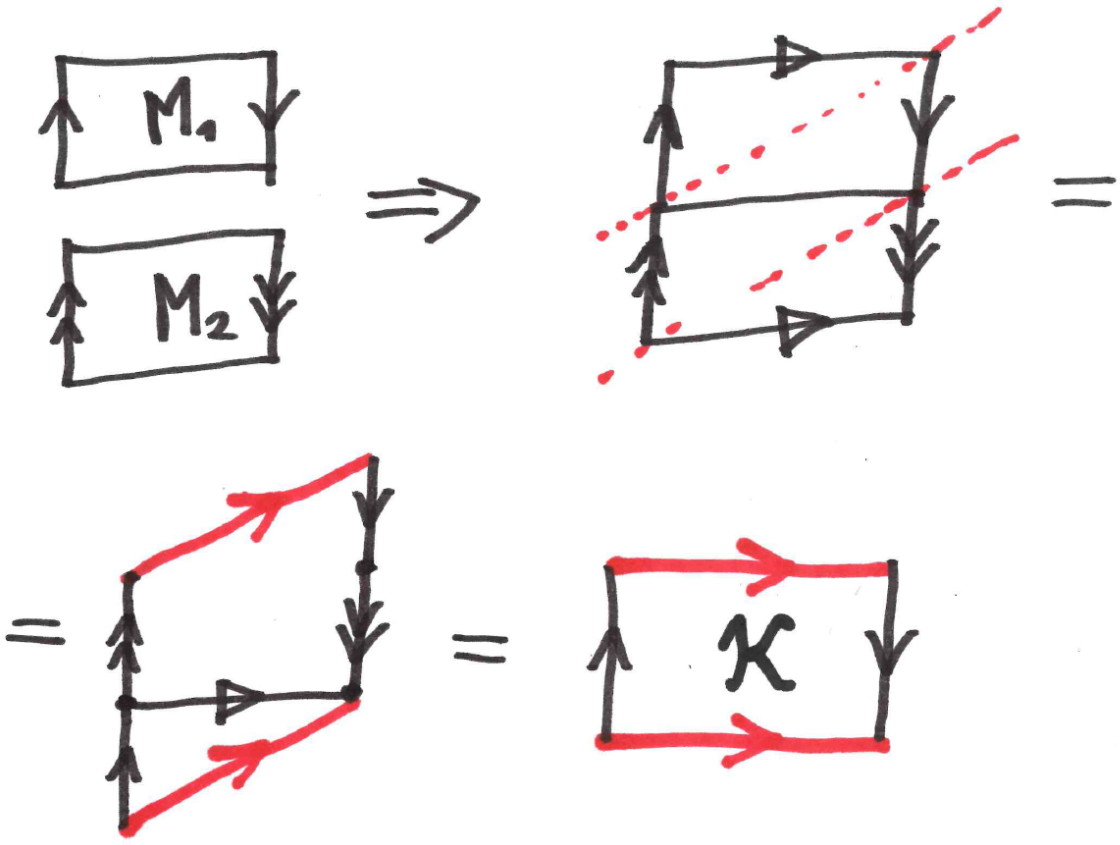


Figure 13: Two Möbius bands glued together along their S^1 boundaries form a Klein bottle.

Example 4.5.4 (Balls and spheres)**.** Take one 0-cell E^0 and one 1-cell E^1, then the only possibility for the gluing map $\phi^1 : \partial E^1 = \{-1; 1\} \to E^0$ is the constant map $\phi^1(-1) = \phi^1(1) = E^0$. We obtain a cell complex X of dimension 1, X is obviously homeomorphic to the circle S^1. The boundary line of any polygon is also a cell complex homeomorphic to S^1 with the same number of 0-cells (vertices) and 1-cells (edges).

Starting with $\mathrm{sk}_1(X) = S^1$ and taking a 2-cell E^2, one possible choice for to the gluing map $\phi^2 : \partial E^2 = S^1 \to S^1 = \mathrm{sk}_1(X)$ is a homeomorphism. The resulted cell complex of dimension 2 is a disc B^2. If we glue another 2-cell along a homeomorhism, we obtain a cell complex homeomorphic to S^2. This decomposition of S^2 has two 2-cells. The boundary surface of any polyhedron is also a cell complex homeomorphic to S^2, the 0-cells are the vertices, the 1-cells are the edges and the 2-cells are the faces of the polyhedron. Hence, the cell complexes are generalizations of the polyhedra.

One can build S^n starting from any cell complex homoemorphic to S^{n-1}, serves as the $n-1$ skeleton, and glue two n-cells E^n_1, E^n_2 to this along homeomorphisms $\phi^n_{1,2} : \partial B^n \to S^{n-1}$. The two n-cells can be considered as the northern and the southern hemispheres.

However, the simplest cell decomposition of S^n consists of only one 0-cell E^0 and one n-cell E^n. Since there are no other cells, $\mathrm{sk}_{n-1}(X) = E^0$ is only one point, and $\phi^n : \partial E^n = S^{n-1} \to E^0$ is the constant map, contracting S^{n-1} to a point. The resulted cell complex is $X = S^n$.

Example 4.5.5 (Torus)**.** A cell decomposition of the torus is indicated by its quotient construction, see Example 4.3.18 and Figure 10. The four vertices of the square correspond to the same point of the torus, let it be the 0-cell E^0. The two opposite pairs of edges define two 1-cells E^1_1 and E^1_2, all of their end points are glued to E^0. They form the 1-skeleton of the cell complex, which is homeomorphic to two circles glued together at one point. Then, we glue only one 2-cell, the square itself, to obtain the cell complex homeomorphic to $\mathcal{T}^2$.

4.6 Compactness

We have to mention two important properties of topological spaces.

Definition 4.6.1. A topological space X (with topology τ) is *compact* if every open cover of X has a finite subcover. That is, X is compact, if for every collection of open subsets $A_j \in \tau$, $j \in J$ such that

$$X = \bigcup_{i \in J} A_j, \tag{4.6.2}$$

there is a finite number of subsets, indexed by $j_1, \dots, j_n \in J$, such that

$$X = \bigcup_{k=1}^{n} A_{j_k}. \tag{4.6.3}$$

Instead of this abstract definition – which is, however, very useful in general proofs, see Exercise 4.8.8 – we have a more intuitive characterization in special cases, which is stated here without proof.

Theorem 4.6.4 (Heine–Borel)**.** *A subspace of $\mathbb{R}^n$ is compact if and only if it is closed and bounded. That is, if $X \subset \mathbb{R}^n$ is endowed with the subspace topology, then X is compact if and only if*

1. *X is a closed subset of $\mathbb{R}^n$, and*
2. *X is a bounded subset of $\mathbb{R}^n$, i.e., there is an $r \in \mathbb{R}^+$ such that $\|p\| < r$ holds for every $p \in X$ (in other words, $X \subset B_r(0)$, where $B_r(0)$ is the open ball of radius r centered at the origin).*

We also do not prove the following theorem.

Theorem 4.6.5 (Tychonoff)**.** *The product of compact topological spaces is compact.*

Theorem 4.6.6. *The image of a continuous map defined on a compact topological space is compact. That is, if X is compact, $f : X \to Y$ is continuous, then the image $f(X)$ is a compact subspace of Y.*

Proof. Exercise 4.8.8. □

Corollary 4.6.7. *Any quotient of a compact topological space is compact.*

Proof. Let X be a compact topological space, $\sim$ is an equivalence relation on X, $Y = X/\sim$ is the quotient space. The quotient map $f : X \to Y$ is continuous and surjective, hence, its image is Y, which is compact by the previous theorem. □

Corollary 4.6.8. *Let X be a compact topological space. Then any continuous function $f : X \to \mathbb{R}$ has a minimum and a maximum value.*

Proof. The image $A := f(X)$ is a compact subset of $\mathbb{R}$, hence it is closed and bounded. Because A is bounded, it has a finite infimum and supremum $\inf(A) = \inf(f)$, $\sup(A) = \sup(f) \in \mathbb{R}$. Since $A \subset \mathbb{R}$ is closed, $\inf(A), \sup(A) \in A$, hence, these are minimum and maximum values of f. □

Example 4.6.9 (Compact and not compact topological spaces)**.** The following topological spaces are compact: closed ball B^n, sphere S^n, closed cylinder $S^1 \times I$, torus $\mathcal{T}^2 = S^1 \times S^1$, solid torus $S^1 \times B^2$, Möbius band $\mathcal{M}$. This follows from the fact that all of these topological spaces can be realised as closed and bounded supspaces of $\mathbb{R}^3$. The Klein bottle $\mathcal{K}$ is also compact, since it can be realised as a closed and bounded subspace of $\mathbb{R}^4$.

The open ball $\breve{B}^n$ and the open cylinder $S^1 \times \mathbb{R}$ are not compact. Although we have seen several different realizations as subspaces of $\mathbb{R}^n$, each of them is either not closed or not bounded. For example, $\breve{B}^n \subset \mathbb{R}^n$, as originally defined, is not a closed subset of $\mathbb{R}^n$. But $\breve{B}^n$ is homeomorphic to $\mathbb{R}^n$, which is a closed subset of itself, but not bounded. The infinite cylinder $S^1 \times \mathbb{R} \subset \mathbb{R}^3 = \mathbb{R}^2 \times \mathbb{R}$ is closed, but not bounded. $S^1 \times \mathbb{R}$ is homeomorphic with $\mathbb{R}^2 \setminus \{0\}$, which is not closed and not bounded subset of $\mathbb{R}^2$. (One can also consider the open annulus, which is a not closed, but bounded realization of $S^1 \times \mathbb{R}$ in $\mathbb{R}^2$.)

What about the higher dimensional torus $\mathcal{T}^n = (S^1)^n$? Can it be realised as a closed and bounded supspace of $\mathbb{R}^N$, for some N? We don't need this information, cause we can apply that any product of compact spaces is compact, or, alternatively, $\mathcal{T}^n$ is a quotient of a compact space, namely, the cube $(B^1)^n = B^n$. Therefore, it is compact.

4.7 Connectedness

The other property we have to mention is whether we call a topological space connected.

Consider a topological space which is definitely not connected, e.g. $X = [0;1] \cup (3;5] \subset \mathbb{R}$, indeed, it is the disjoint union of two pieces, like a pair of shoe. Observe that $[0;1] \subset X$ is an open subset in the subspace topology, indeed, $[0;1] = X \cap (-2;2)$ is an open set of $\mathbb{R}$ intersected with X. But $[0;1]$ is also a closed subset of X, since its complement $(3;5] \subset X$ is open. Finding a subset $A \subset X$ which is open and closed at the same time is equivalent with finding a decomposition $X = A \cup B$ of X as the union of two disjoint open subsets. This decomposition always exists in a trivial way, namely, $A = X$ and $B = \emptyset$ are always open subsets (hence, they are also closed).

Definition 4.7.1. A topological space (X, τ) is *not connected* if there is a decomposition $X = A \cup B$ with $A \cap B = \emptyset$; $A, B \in \tau$; $A \neq \emptyset \neq B$. The topological space is *connected* if the only decomposition to the union of disjoint open subsets is $X = \emptyset \cup X$.

However, there is another related notion, expressing the connectedness in another way. Intuitively, a space is (path) connected if every pair of points can be joined by a (continuous) path. A *path* in a topological is (X, τ) is a continuous map $f : [0;1] \to X$.

Definition 4.7.2. A topological space (X, τ) is *path connected*, if for every pair of points $p, q \in X$ there is a continuous map (path) $f : [0;1] \to X$ such that $f(0) = p$ and $f(1) = q$.

Theorem 4.7.3. *If a topological space (X, τ) is path connected, then it is connected.*

Sketch of proof. Indirectly, assume that X is not connected, i.e., $X = A \cup B$, where A and B are disjoint open sets, and none of them is empty. Take two points $p \in A$ and $q \in B$. Since X is path connected, there is a continuous map $f : [0;1] \to X$, $f(0) = p$, $f(1) = q$. By the definition the continuity, $f^{-1}(A)$ and $f^{-1}(B)$ are open subsets of $[0;1]$. Moreover, $f^{-1}(A)$ and $f^{-1}(B)$ are disjoint subsets, $f^{-1}(A) \cup f^{-1}(B) = X$, and none of them is empty. This is a contradiction, since one can show that $[0;1]$ is a connected topological space, see Exercise 4.8.7. □

The converse of Theorem 4.7.3 is not true, indeed, there are crazy topological spaces which are connected, but not path connected. However, for our topological spaces (cell complexes, manifolds) the two properties are equivalent: *a cell complex or a manifold (see later) is connected if and only if it is path connected.*

Theorem 4.7.4. *The continuous image of a connected topological space is connected. The continuous image of a path connected topological space is path connected.*

Proof. Exercise 4.8.8 □

Example 4.7.5 (Non-homeomorphic spaces)**.** How can we see that the topological spaces $\breve{B}^1 = (-1,1)$, $B^1 = [-1;1]$ and S^1 are pairwise not homeomorphic? There are several possible answers to this question, e.g.

- B^1 and S^1 are compact, but $\breve{B}^1$ is not compact, hence, it is not homeomorhic to any of the others.
- Every point of $\breve{B}^1$ and S^1 has a neighborhood homoemorphic to $\breve{B}^1$, but the boundary points -1 and 1 of B^1 do not have such a neighborhood. Hence, B^1 is not homeomorphic to any of the others.[7]

However, there is another nice way to distinguish the three spaces. Consider the complement of a point! The complement $S^1 \setminus \{p\}$ of every point $p \in S^1$ is connected. For every point $p \in \breve{B}^1$, its complement $\breve{B}^1 \setminus \{p\}$ is not connected. Finally, B^1 has points with connected complements ($p = -1$ or $p = 1$), and it has points whose complements are not connected.

4.8 Exercises

Exercise 4.8.1 (Graph of a map)**.** Show that the graph of a continuous map is homeomorphic to the domain of the map! Explained in more detail:

Recall that the *graph* of a map $f : X \to Y$ is defined as

$$\operatorname{graph}(f) = \{(x, f(x)) \in X \times Y\}. \tag{4.8.2}$$

If X and Y are topological spaces, then $\operatorname{graph}(f)$ inherits a subspace topology from the product topological space $X \times Y$. Show that if f is continuos, then $\operatorname{graph}(f)$ is homeomorphic to X!

Exercise 4.8.3 (Balls and spheres)**.** Compose the homoemorphisms given in the proofs of Proposition 4.1.6 and 4.1.9 to

(a) give an explicit homeomorphism between $\breve{B}^n$ and $\mathbb{R}^n$,

(b) give an explicit homeomorphism between $\breve{B}^n$ and $S^n \setminus \{\text{point}\}$,

(c) find a parametrization $g : B^n \to S^n$ which sends the boundary $\partial B^n = S^{n-1}$ to one point of S^n! (This map induces a homeomorphism $\widetilde{g} : B^n/S^{n-1} \to S^n$, see Example 4.3.16.)

Exercise 4.8.4 (Cylinders)**.** (a) Find an explicit homeomorphism between $S^1 \times \mathbb{R}$ and $\mathbb{R}^2 \setminus \{0\}$!

[7] However, this argument is not a proof, since logically it uses the statement we want to prove. Indeed, -1 and 1 in $[-1;1]$ have a neighborhood homoemorphic to $[-1;1]$, hence, when we state that they do not have a neighborhood homeomorphic to $\breve{B}^1$, we already use the fact that $\breve{B}^1$ is not homeomorphic to $[-1;1]$.

(b) What is the quotient of $S^1 \times [0;1]$ with the equivalence relation $(p;0) \sim (q;0)$ and $(p;1) \sim (q;1)$ for every $p, q \in S^1$ (and every other point is equivalent only with itself)? Find an explicit homeomorphism (via a parametrization like in Example 4.3.16).

Exercise 4.8.5 (Torus, Möbius, Klein)**.** (a) Find a parametrization $g : [0;1] \times [0;1] \to T \subset \mathbb{R}^3$ of the the surface of revolution $T \subset \mathbb{R}^3$ described in Example 4.2.6! Verify that the parametrization shows that T is homeomorphic to $S^1 \times S^1$ and also to the quotient of $[0;1] \times [0;1]$ defining the torus $\mathcal{T}^2$!

(b) Find a parametrization $g : [0;1] \times [0;1] \to \mathbb{R}^3$ whose image is $\mathcal{M} \subset \mathbb{R}^3$, and which realizes the quotient $[0;1] \times [0;1] \to \mathcal{M}$!

(c) Find a parametrization $g : [0;1] \times [0;1] \to \mathbb{R}^3$ whose image is the usual picture of the Klein bottle in $\mathbb{R}^3$!

Exercise 4.8.6 (Klein bottle in 3 and 4 dimension)**.** Consider the Klein bottle $\mathcal{K}$ as a quotient of $[0;1] \times [0;1]$ indicated by Figure 10, let $h : [0;1] \times [0;1] \to \mathcal{K}$ be the quotient map. The usual picture of the Klein bottle in $\mathbb{R}^3$ can be considered as the image of a map $f : \mathcal{K} \to \mathbb{R}^3$. A realization of the Klein bottle as a subspace of $\mathbb{R}^4$ is the image of another map $g : \mathcal{K} \to \mathbb{R}^4$, which is a homeomorphism to its image.

(a) Draw in the square $[0;1] \times [0;1]$ the preimage of the self-intersection circle of the 3D picture, i.e., $(f \circ h)^{-1}(S^1)$, where $S^1 \subset \mathbb{R}^3$ is the self-intersection circle of the Klein bottle.

(b) Satirize the square $[0;1] \times [0;1]$ with black and white gradient according to the fourth coordinate of g!

Exercise 4.8.7 (Connectedness and compactness)**.** (a) Using Definition 4.7.1, show that $X = [0;1]$ is connected, i.e., its only decomposition to the disjoint union of open subsets is the trivial one.

(b) Using Definition 4.6.1, show that the interval $X = (0;1)$ is not compact, i.e., find an open cover which does not have a finite subcover.

Exercise 4.8.8 (Connectedness and compactness)**.** Prove Theorem 4.6.6 and Theorem 4.7.4, i.e., show that continuous images of a connected, path connected or compact topological space are also connected, path connected or compact, respectively. More precisely:

(a) Let X be connected topological space. Show that the image $f(X) \subset Y$ of X by a continuous map $f : X \to Y$ is connected. (Use the definition.)

(b) Let X be a path connected topological space. Show that the image $f(X) \subset Y$ of X by a continuous map $f : X \to Y$ is path connected. (Use the definition.)

(c) Let X be compact topological space. Show that the image $f(X) \subset Y$ of X by a continuous map $f : X \to Y$ is compact. (Use the definition.)

Exercise 4.8.9 (Not homeomorphic spaces)**.** Show that the torus $\mathcal{T}^2$, the Möbius band $\mathcal{M}$, the open cylinder $S^1 \times \mathbb{R}$, the closed cylinder $S^1 \times B^1$ and the sphere S^2 are pairwise not homeomorphic! Find topological properties which distinguish them!

Exercise 4.8.10 (Möbius band)**.** Make a Möbius band from a piece of paper, and cut it along its middle cirle, see Figure 14! What happens? Formalize your experience in topological terminology, and explain it using the quotient definition of the Möbius band (instead of the 3D realization) shown by Figure 10.

Exercise 4.8.11 (Cell complexes)**.** Find a cell decomposition for the following spaces, i.e., a cell complex X which is homeomorphic to the following spaces. Use as few cells as possible!

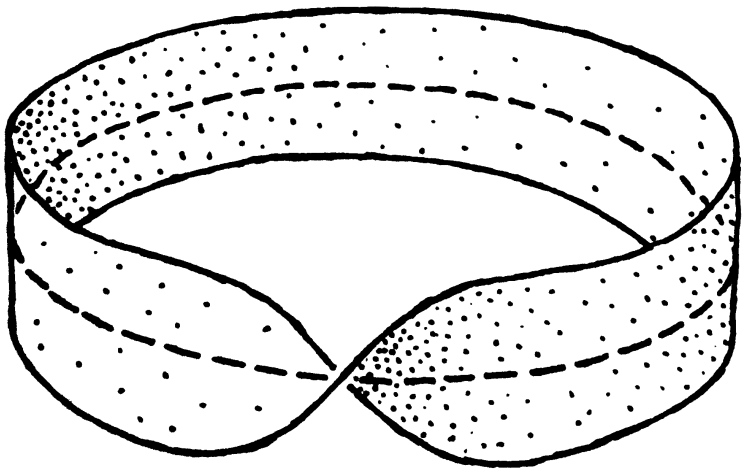


Figure 14: The middle circle of the Möbius band $\mathcal{M} \subset \mathbb{R}^3$. Cut the Möbius band along this circle! What happens? (Graphics by Mátyás Szabari, Új világok teremtése.)

(a) Closed cylinder $S^1 \times B^1$,

(b) Klein bottle $\mathcal{K}$,

(c) Möbius band $\mathcal{M}$,

(d) Solid torus $B^2 \times S^1$.

Exercise 4.8.12 (Quotient spaces)**.** (a) Oliver and Xaver are playing five-a-side, Oliver is the one with a circle, Xaver with the x. The position is shown by Figure 15. Oliver and Xaver both claim to have won. On what surface does Oliver imagine the game and how does Xaver?

(b) Consider the equivalence relations indicated by the pictograms on Figure 16! Determine the quotient spaces!

Exercise 4.8.13 (Quotient spaces)**.** What are the quotient spaces

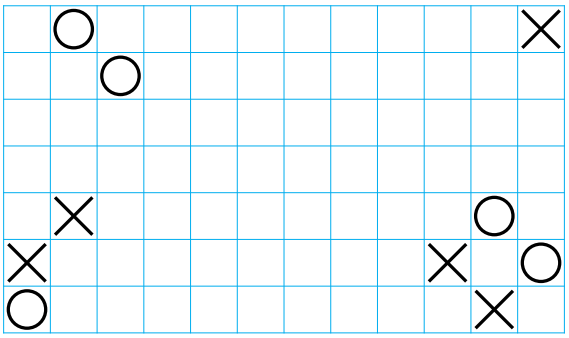


Figure 15: Oliver and Xaver are playing five-a-side, Oliver is the one with a circle, Xaver with the x. Oliver and Xaver both claim to have won. On what surface does Oliver imagine the game and how does Xaver?

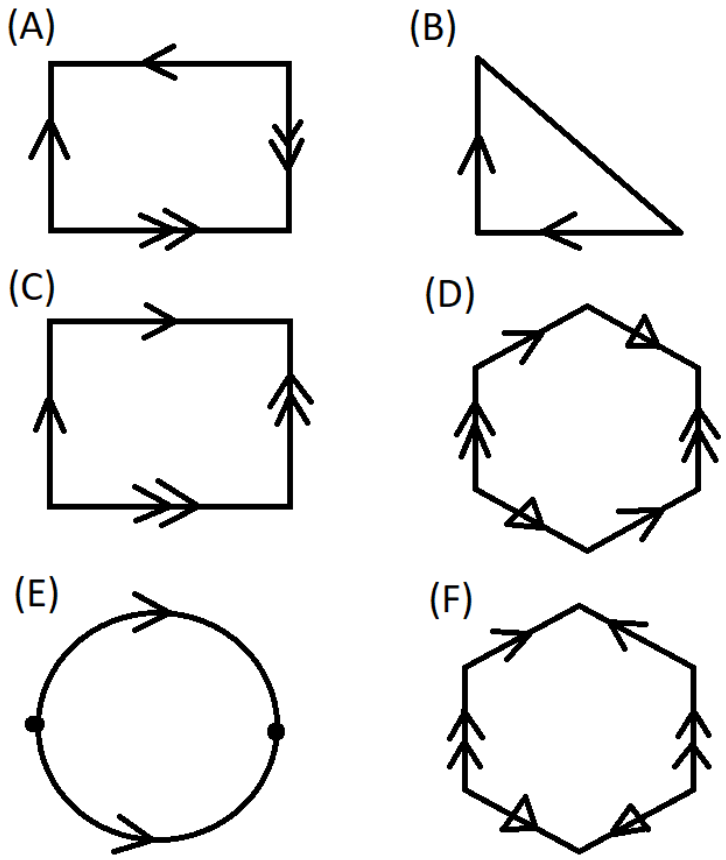


Figure 16: What are the quotient spaces indicated by the pictograms?

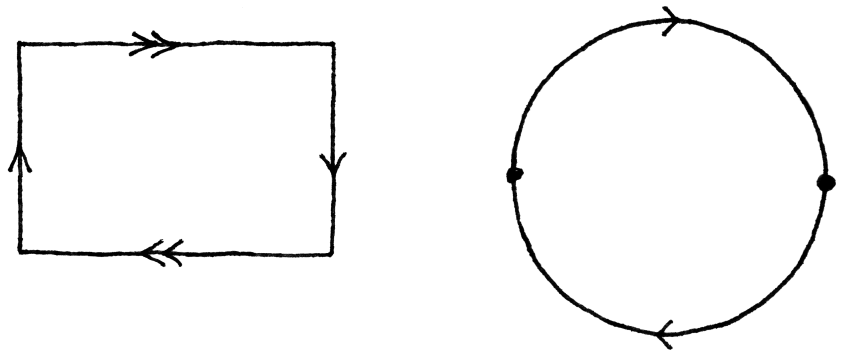


Figure 17: What is this quotient space? Graphics by Mátyás Szabari, Új világok teremtése.

(a)

$$Y = \frac{S^1}{p \sim -p}, \tag{4.8.14}$$

(b)

$$Y = \frac{S^1 \times S^1}{(p,q) \sim (q,p)}? \tag{4.8.15}$$

Exercise 4.8.16 (A crazy example)**.** Let $X = \mathbb{R}_{(1)} \cup \mathbb{R}_{(2)}$ be the union of two disjoint copies of the real line $\mathbb{R}$, and consider the equivalence relation $r_{(1)} \sim r_{(2)}$ if $r_{(i)} < 0_{(i)}$, where $r_{(i)} \in \mathbb{R}_{(i)}$ is the real number on the i-th real line ($i = 1, 2$). (The other real numbers, the two 0's and the positive ones are only equivalent with themselves.)

Consider the sequence $a_n = -\frac{1}{n}$. Where does it converge?

Exercise 4.8.17 (Möbius and Klein in art)**.** Look for works of art that display the Möbius strip or the Klein bottle. How do they use these symbols?

5 Projective spaces

5.1 Real projective plane

Starting from a square, gluing together both pairs of its opposite sides in orientation preserving way (like for the cylinder) results in a torus, while changing the orientation for one pair (like for the Möbius band) results in a Klein bottle, as shown by Figure 10. What happens if we glue both pairs of sides with a reversed orientation, see Figure 17? This picture can be simplified, since the square is homeomorphic to the disc B^2, and the opposite pairs of points on its S^1 boundary are identified. What is this?

We show that this is a projective plane, i.e. the abstract geometric world in which 'the parallel lines meet at infinity'. More precisely, in geometry, the projective plane is an extension of the Euclidean plane $\mathbb{R}^2$ with one extra point for each family of parallel lines, and an extra line joining the extra points. Inspired by the perspective, every family of parallel lines are *declared to meet* at the corresponding extra point, which in this way can be imagined as being at the infinity. The resulted structure of points and lines (the original and the extra points and lines together) has a very symmetric structure: any pair of distinct points determines a unique line (i.e. the line through them) and any pair of distinct lines determines a unique point (i.e. their intersection point). This structure is called *real projective plane* (shortly, *projective plane*), which serves as the scene of the *projective geometry* (in 2D). Although the projective plane is not a topological space by default, it can be endowed with a natural topology making it a topological space homeomorphic to the quotient

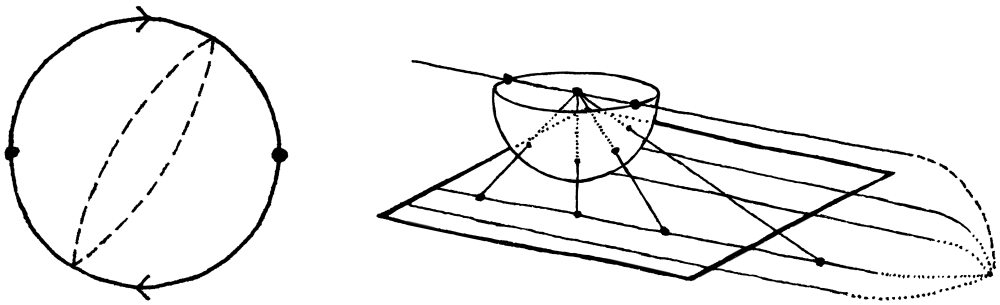


Figure 18: Identification of the projective plane as a quotient topological space. Graphics by Mátyás Szabari, Új világok teremtése.

space on Figure 17. The projective plane, as a topological space, is denoted by $\mathbb{R}P^2$, which also denotes every topological space homeomorphic to it.

To show the homeomorphism, we start with an intuitive argument. Imagine that you are standing on an an infinite plane, and looking around. What do you see? At the height of your eyes, you see a line of the horizon. Its points are virtual points, where the parallels seem to intersect. Since the plane is homeomorphic to the open disc $\breve{B}^2$, the horizon can be considered as its boundary, hence, the plane and the horizon together form a closed disc B^2. At this point of the construction, the parallel lines meet at the boundary points of the closed disc, each family meets at two opposite points. If we want every family of parallels to meet at only one point, we have to identify the opposite points of the S^1 boundary of B^2, like in Figure 18.

This intuitive argument can be made precise by considering the central projection on Figure 7. Previously we used it to show that the open disc $\breve{B}^2$ is homeomorphic to $\mathbb{R}^2$ in two steps: first, $\breve{B}^2$ is homeomorphic to the open southern hemisphere (of S^2), second, the projection from the center of the sphere shows that the open southern hemisphere (of S^2) is homeomorphic to $\mathbb{R}^2$. Try to extend the projection to the S^1 boundary of the hemisphere! The lines joining the center and the boundary points do not intersect the plane, but each of them determines a family of parallel lines on the plane, hence, a point at infinity. Moreover, each line joining the center and a boundary point intersects the boundary at the opposite point as well, see Figure 18. Therefore, the projection extended to the closed ball performs a bijection between the quotient space on Figure 17 and the projective plane, and it defines the topology of the projective plane, as a topological space:

$$\mathbb{R}P^2 = \frac{B^2}{p \sim -p \text{ for } p \in \partial B^2 = S^1}. \tag{5.1.1}$$

$\mathbb{R}P^2$ is (path) connected and compact, since it is a quotient of the (path) connected, compact topological space B^2.

Try to extend the central projection to the whole sphere S^2 from the southern hemisphere! Then, each point of the projective plane (i.e., an original point of the plane or an 'extra point at the infinity') corresponds to an opposite pair of points in S^2, hence we can realize the projective plane as the quotient space

$$\mathbb{R}P^2 = \frac{S^2}{p \sim -p}, \tag{5.1.2}$$

see Figure 19. This realization shows – intuitively – the 'homogeneous' nature of $\mathbb{R}P^2$, i.e., its points are not distinguished, as suggested by the above descriptions. Only the choice of the equator separates them to 'original points of the plane' and 'extra points at the infinity'. This description also indicates that 'the projective geometry is the half of the spherical geometry'. Indeed, on the unit sphere, each pair of 'lines' (i.e., great circle) intersect at two opposite points, which are identified by the quotient map $S^2 \to \mathbb{R}P^2$.

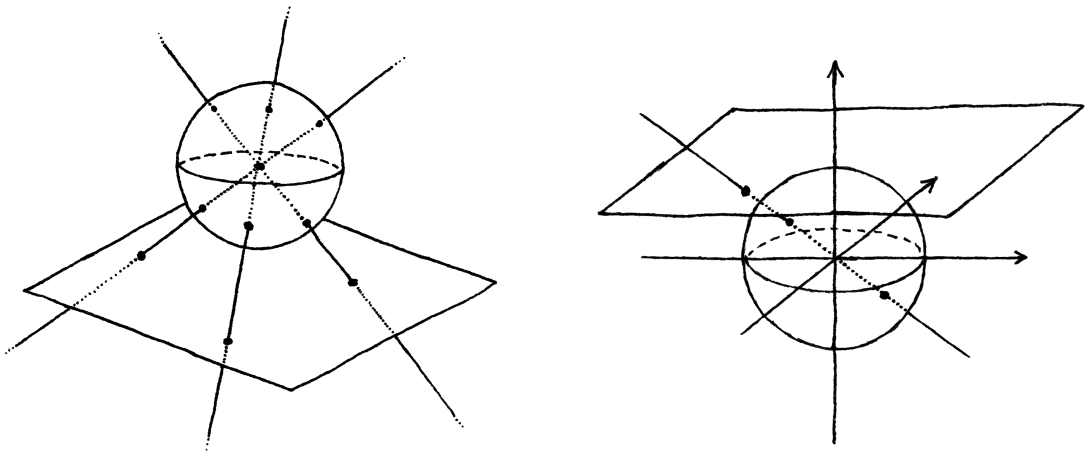


Figure 19: The projective plane, as a quotient of S^2 (left), and as the $z = 1$ plane extended with points at infinity (right). Graphics by Mátyás Szabari, Új világok teremtése.

Now, put the whole configuration into the coordinate system in $\mathbb{R}^3$ such that S^2 is the unit sphere whose center is the origin, this is the center of the projection, and the plane of 'original points' is the $z = 1$ plane, see Figure 19. The points of $\mathbb{R}P^2$ are in one to one correspondence with the directions of the projections, i.e., the lines in $\mathbb{R}^3$ through the origin. Each line is determined by a direction vector $v \in \mathbb{R}^3 \setminus \{0\}$, up to a scalar factor. That is, if $r \in \mathbb{R} \setminus \{0\}$, then v and rv determine the same point of $\mathbb{R}P^2$, hence we have

$$\mathbb{R}P^2 = \{\text{1D linear subspaces in } \mathbb{R}^3\} = \frac{\mathbb{R}^3 \setminus \{0\}}{v \sim rv \text{ for } r \in \mathbb{R} \setminus \{0\}}. \tag{5.1.3}$$

By the last realization, the points of $\mathbb{R}P^2$ can be described with homogeneous coordinates, i.e., the equivalence class of $v = (x, y, z) \neq (0, 0, 0)$ is denoted by

$$p = [v] = [x : y : z] = [rx : ry : rz] \in \mathbb{R}P^2, \tag{5.1.4}$$

where $r \neq 0$.

We mention one more quite surprising description of $\mathbb{R}P^2$: it can be obtained from a Möbius band and a disc glued together along their S^1 boundaries, i.e.,

$$\mathbb{R}P^2 = \mathcal{M} \cup_{S^1} B^2. \tag{5.1.5}$$

It's proof is left to an exercise.

As a summary, the real projective plane is defined by the quotients

$$\begin{aligned}
\mathbb{R}P^2 &= \frac{B^2}{p \sim -p \text{ for } p \in \partial B^2 = S^1} && (5.1.6)\\
&= \frac{S^2}{p \sim -p} && (5.1.7)\\
&= \{\text{1D linear subspaces in } \mathbb{R}^3\} && (5.1.8)\\
&= \frac{\mathbb{R}^3 \setminus \{0\}}{v \sim rv \text{ for } r \in \mathbb{R} \setminus \{0\}} && (5.1.9)\\
&= \mathcal{M} \cup_{S^1} B^2. && (5.1.10)
\end{aligned}$$

Similarly to the Klein bottle, the projective plane cannot be realised as a subspace of $\mathbb{R}^3$, i.e., without self-intersection. Two realizations with a wild self-intersection are shown by Figure 20 (without detailed explanation, just for fun).

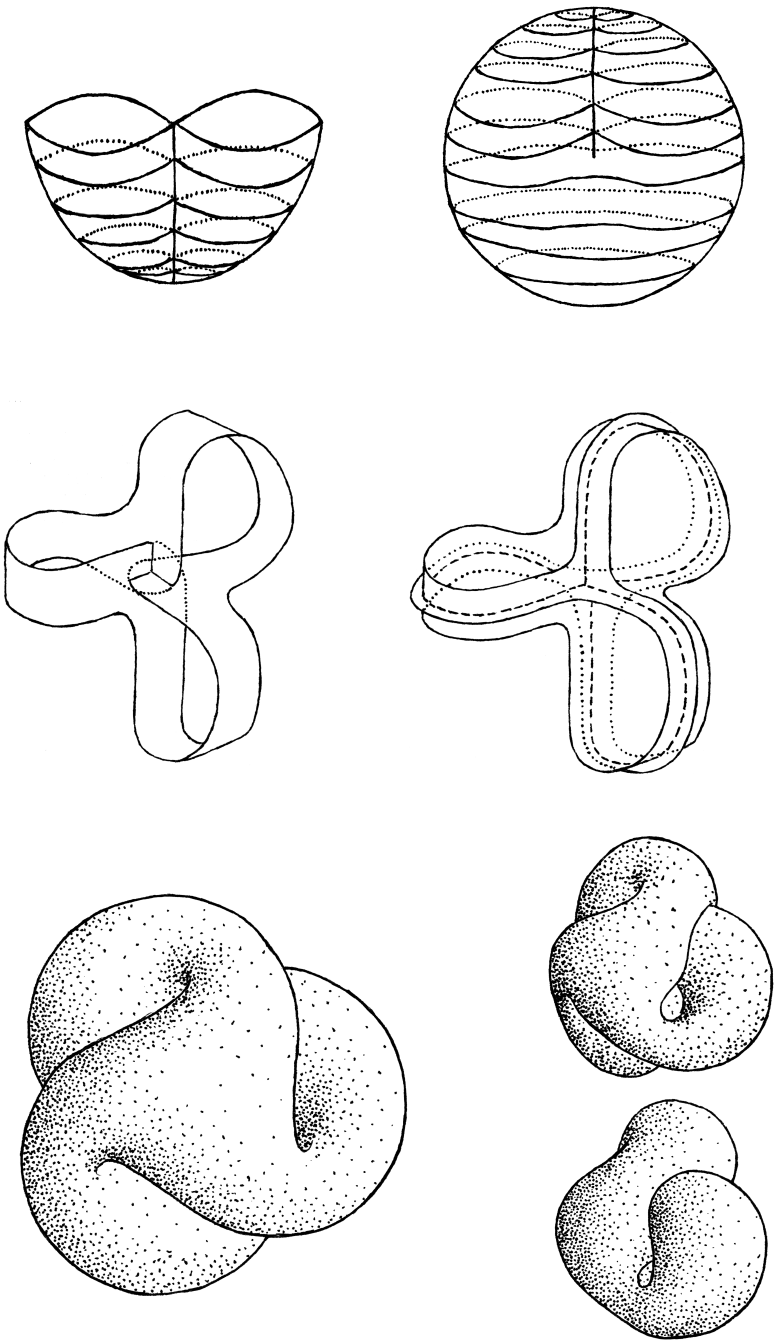


Figure 20: Kind of realizations of $\mathbb{R}P^2$ in $\mathbb{R}^3$ with wild self-intersection. The *cross cap* (above) realizes the gluing of the opposite pairs of points of ∂B^2. The *Boy surface* is the gluing of a Möbius band $\mathcal{M}$ and a disc B^2 along their S^1 boundaries in a tricky way. Graphics by Mátyás Szabari, Új világok teremtése.

5.2 Real projective spaces

Real projective spaces in higher dimension are defined in the same way as the projective plane. From the projective geometric approach, $\mathbb{R}P^n$ is obtained from $\mathbb{R}^n$ extended with extra points at the infinity, each extra point corresponds to a family of parallel lines intersecting at that point. The extra points form a hyperplane at the infinity, actually, an $\mathbb{R}P^{n-1}$. Exactly the same projections as in case of the projective plane show the following realizations of $\mathbb{R}P^n$ as quotient topological space:

$$\begin{aligned}
\mathbb{R}P^n &= \frac{B^n}{p \sim -p \text{ for } p \in \partial B^n = S^{n-1}} && (5.2.1)\\
&= \frac{S^n}{p \sim -p} && (5.2.2)\\
&= \{\text{1D linear subspaces in } \mathbb{R}^{n+1}\} && (5.2.3)\\
&= \frac{\mathbb{R}^{n+1} \setminus \{0\}}{v \sim rv \text{ for } r \in \mathbb{R} \setminus \{0\}}. && (5.2.4)
\end{aligned}$$

By the last realization, the points of $\mathbb{R}P^n$ can be described with homogeneous coordinates, i.e., the equivalence class of $v = (x_1, \ldots, x_{n+1}) \neq (0, \ldots, 0)$ is denoted by

$$p = [v] = [x_1 : x_2 : \cdots : x_{n+1}] = [rx_1 : rx_2 : \cdots : rx_{n+1}] \in \mathbb{R}P^n, \tag{5.2.5}$$

where $r \neq 0$.

$\mathbb{R}P^0$ is only one point, and $\mathbb{R}P^1 = S^1$. Check all the definitions in this case!

Based on the quotient realizations, the real projective spaces can be given as cell complexes, with exactly one cell in every dimension and with the property that the m-skeleton is the projective space of dimension m, i.e., $\mathrm{sk}_m(\mathbb{R}P^n) = \mathbb{R}P^m$. For $n = 0, 1$ it is clear: a one-cell E^1 glued to the 0-cell $E^0 = \mathbb{R}P^0$ forms a circle $\mathbb{R}P^1 = S^1$. How should we glue the two-cell $E^2 = B^2$, i.e., how should the gluing map

$$\phi^2 : \partial E^2 = S^1 \to \mathbb{R}P^1 = S^1 \tag{5.2.6}$$

be defined? From the frist quotient realization of $\mathbb{R}P^2$ it is clear that ϕ^2 is the quotient map, identifying the opposite pairs of points p and $-p$ of S^1. It can be expressed in an elegant way, considering the unit circle $S^1 \subset \mathbb{C}$, as $\phi^2(z) = z^2$. Hence,

$$\mathbb{R}P^2 = \mathbb{R}P^1 \cup_{\phi^2} B^2. \tag{5.2.7}$$

To build $\mathbb{R}P^3$ as a cell complex, we start from its 2-skeleton $\mathbb{R}P^2$, and glue one 3-cell $E^3 = B^3$ along the quotient map

$$\phi^3 : \partial E^3 = S^2 \to \mathbb{R}P^2, \tag{5.2.8}$$

defined by $\phi^3(p) = \{p, -p\} \in \mathbb{R}P^2$. Once $\mathbb{R}P^{n-1}$ is already constructed as a cell complex, $\mathbb{R}P^n$ can be obtained by gluing one n-cell to it along the quotient map

$$\phi^n : \partial B^n = S^{n-1} \to \mathbb{R}P^{n-1}, \tag{5.2.9}$$

results in

$$\mathbb{R}P^n = \mathbb{R}P^{n-1} \cup_{\phi^n} B^n. \tag{5.2.10}$$

Intuitively, this decomposition fits to the construction of $\mathbb{R}P^n$ as a space $\mathbb{R}^n$ (here, the interior of B^n) extended with a hyperplane at the infinity (here, $\mathbb{R}P^{n-1}$). The construction can be also summarized as

$$\mathbb{R}P^n = B^0 \cup_{\phi^1} B^1 \cup_{\phi^2} B^2 \cup_{\phi^3} \cdots \cup_{\phi^n} B^n. \tag{5.2.11}$$

5.3 Complex projective spaces

Analog constructions work for the *complex projective spaces* $\mathbb{C}P^n$. Intuitively, $\mathbb{C}P^n$ is the extension of $\mathbb{C}^n$ with extra points corresponding to the families of parallel complex lines. Considering $\mathbb{C}^n$ as the $z_{n+1} = 1$ hyperplane of $\mathbb{C}^{n+1}$, its extension $\mathbb{C}P^n$ can be identified with the complex lines in $\mathbb{C}P^{n+1}$ through the origin, i.e., the 1 dimensional complex linear subspaces. Each of them is spanned by its complex direction vector $v \in \mathbb{C}^{n+1} \setminus \{0\}$, which is determined up to a nonzero scalar factor: $v \sim wz$ for $w \in \mathbb{C} \setminus \{0\}$. The equivalence classes can be labeled by homogeneous coordinates $[v] = [z_1 : z_2 : \cdots : z_{n+1}]$. Each equivalence class can be represented by a unit vector $v \in S^{2n+1}$, that is, $\|v\| = 1$. Indeed, the unit sphere in $\mathbb{C}^{n+1} = \mathbb{R}^{2n+2}$ is $2n+1$ dimensional. The unit vector v is determined up a scalar factor of unit length, i.e., $v \sim uv$, where $u \in S^1 \subset \mathbb{C}$, that is, $|u| = 1$. Hence, we have

$$\begin{aligned} \mathbb{C}P^n &= \{\text{1D complex linear subspaces in } \mathbb{C}^{n+1}\} && (5.3.1)\\ &= \frac{\mathbb{C}^{n+1} \setminus \{0\}}{v \sim wv \text{ for } w \in \mathbb{C} \setminus \{0\}} && (5.3.2)\\ &= \frac{S^{2n+1}}{v \sim uv \text{ for } u \in S^1 \subset \mathbb{C}}. && (5.3.3) \end{aligned}$$

The complex projective line $\mathbb{C}P^1$ is homeomorphic to S^2, as we proved it in Section 2.2. Intuitively, it can be considered as 'a complex line $\mathbb{C}$ extended with one point at infinity', i.e., by the intuitive argument, $\mathbb{C} \cup \{\infty\} = S^2$. $\mathbb{C}P^1$ is also called Riemann sphere, and Bloch sphere in physics. Its defining quotient map $S^3 \to \mathbb{C}P^1$ is exactly the Hopf map. Hence, the quotient map $S^{2n+1} \to \mathbb{C}P^n$ is a generalization of the Hopf map (Hopf fibration), and the quotient map $S^1 \to \mathbb{R}P^1 = S^1$, $p \mapsto \{p, -p\}$ is the real analog of the Hopf map.

How can $\mathbb{C}P^n$ be realizes as a cell complex? In case of $\mathbb{R}P^n$ we used the quotient map $B^n \to \mathbb{R}P^n$. What is the analog construction in the complex case? Let us generalize the method in Section 2.2 used to identify $\mathbb{C}P^1$ and S^2!

Consider a unit vector $v = (z_1, \ldots, z_{n+1}) \in S^{2n+1} \subset \mathbb{C}^{n+1}$. By a multiplication with $u \in S^1 \subset \mathbb{C}$, i.e., by a simultaneous rotation of the complex numbers z_j, it can be reached that z_{n+1} is rotated to the positive real direction, i.e., $uz_{n+1} \in \mathbb{R}^+$ (or 0). Assuming that $z_{n+1} \in \mathbb{R}^+$, the other complex coordinates $z_1, \ldots, z_n$ determine z_{n+1}, and they can be chosen as a point of B^{2n}, since $\sum_{j=1}^n |z_j|^2 \leq 1$. Hence, $(z_1, \ldots, z_n) \in B^{2n}$ uniquely determines the equivalence class $[v] \in \mathbb{C}P^n$ – except if $z_{n+1} = 0$, which happens exactly if $\sum_{j=1}^n |z_j|^2 = 1$. Indeed, in this latter case, a simultaneous rotation uv means the rotation of $z_1, \ldots, z_n$, since $uz_{n+1} = 0$. In this case $(z_1, \ldots, z_n) \in \partial B^{2n} = S^{2n-1}$, hence, we have

$$\mathbb{C}P^n = \frac{B^{2n}}{v \sim uv \text{ for } v \in \partial B^{2n} = S^{2n-1},\ u \in S^1 \subset \mathbb{C}}. \tag{5.3.4}$$

This quite complicated quotient allows us to realize $\mathbb{C}P^n$ as a cell complex of dimension $2n$, with exactly one cell in *every even dimension*, and with $\mathrm{sk}_{2k}(\mathbb{C}P^n) = \mathrm{sk}_{2k+1}(\mathbb{C}P^n) = \mathbb{C}P^k$. So the cells are

$$E^0 = B^0,\ E^2 = B^2,\ E^4 = B^4, \ldots \tag{5.3.5}$$

The gluing

$$\phi^2 : \partial B^2 = S^1 \to B^0 = \mathbb{C}P^0 \tag{5.3.6}$$

is the constant map, resulted in

$$\mathbb{C}P^1 = \mathbb{C}P^0 \cup_{\phi^2} B^2 = S^2. \tag{5.3.7}$$

The 4D cell is glued to this along the quotient map

$$\phi^4 : \partial B^4 = S^3 \to \mathbb{C}P^1, \tag{5.3.8}$$

which is actually the Hopf map, resulted in

$$\mathbb{C}P^2 = \mathbb{C}P^1 \cup_{\phi^4} B^4. \tag{5.3.9}$$

In general, if the $2n-1$ skeleton $\mathbb{C}P^{n-1}$ is already built up, then the $2n$-cell is glued to it along the quotient map

$$\phi^{2n} : \partial B^{2n} = S^{2n-1} \to \mathbb{C}P^{n-1}, \tag{5.3.10}$$

results in

$$\mathbb{C}P^n = \mathbb{C}P^{n-1} \cup_{\phi^{2n}} B^{2n}. \tag{5.3.11}$$

The construction can be also summarized as

$$\mathbb{C}P^n = B^0 \cup_{\phi^2} B^2 \cup_{\phi^4} B^4 \cup_{\phi^6} \cdots \cup_{\phi^{2n}} B^{2n}. \tag{5.3.12}$$

Importantly, there is one cell in every even dimension. This fact has far-reaching consequences in the physical applications.

5.4 Projectors

Let $\Psi \in \mathbb{C}^n$ be a unit vector, i.e., $\|\Psi\| = 1$, in other words, $\Psi \in S^{2n-1} \subset \mathbb{C}^n$. Considering Ψ as a column vector, the (orthogonal) projector onto the 1D subspace spanned by Ψ is

$$P_\Psi = \Psi^\dagger \cdot \Psi = |\Psi\rangle\langle\Psi|, \tag{5.4.1}$$

the right side is the bra-cat notation. Obviously,

- P_Ψ is hermitien, i.e., $P_\Psi^\dagger = P_\Psi$.
- One of the eigenvalues of P_Ψ is 1, corresponding to the eigenspace spanned by Ψ.
- The other eigenvalues are 0, the corresponding eigenspace is the orthogonal complement of Ψ with respect to the hermitian inner product on $\mathbb{C}^n$, i.e., $\Psi^\perp = \{v \in \mathbb{C}^n \mid \Psi^\dagger \cdot v = 0\}$.
- $\mathrm{tr}(P_\Psi) = 1$.
- If $\Psi' = u \cdot \Psi$ ($u \in \mathbb{C}$, $|u| = 1$), then $P_{\Psi'} = P_\Psi$. In other words, we have a bijection

  $$[\Psi] \mapsto P_\Psi, \tag{5.4.2}$$

  which expresses that the 1D projectors are in one-to-one correspondence with the 1D subspaces of $\mathbb{C}^n$.

Let us denote by $\mathbb{C}^{n\times m}$ the space of $n \times m$ complex matrices, it is a complex vectorspace of dimension $n \cdot m$. The space $\mathrm{Herm}(n) \subset \mathbb{C}^{n\times n}$ of $n \times n$ hermitian matrices is a real vector space of dimension n^2 (why, and why is it not a complex vectorspace?). The space $\mathcal{P}_n \subset \mathrm{Herm}(n)$ of 1D projectors inherits a subspace topology from $\mathrm{Herm}(n) \cong \mathbb{R}^{n^2}$. The correspondence $[\Psi] \mapsto P_\Psi$ defines a homeomorphism between the complex projective space $\mathbb{C}P^{n-1}$ and the space $\mathcal{P}_n$ of 1D projectors. In this way $\mathbb{C}P^{n-1}$ is realised as a subspace of $\mathbb{R}^{n^2}$. Moreover, it is a subspace of $\mathbb{R}^{n^2} - 1$, since the trace is 1.

In quantum mechanics, the *pure states* (or projective states) are the 1D projectors, hence, their space is homeomorphic to the complex projective space $\mathbb{C}P^{n-1}$. The *mixed states* (or *density matrices*) are the convex combinations of the pure states in $\mathrm{Herm}(n)$. For $n = 2$, the space $\mathcal{P}_2$ of pure states is homeomorphic to $\mathbb{C}P^1 \cong S^2$, it is the Bloch sphere. We show that the space of mixed states is a ball B^3, called *Bloch ball*. It is not trivial, since, on one hand, there are many other topological spaces with boundary S^2, and, on the other hand, the mixed states are defined via convex combinations of pure states, which is a definitely non-topological construction and it deeply depends on the ambient vector space.

To show that the mixed states form a ball B^3, consider the space $\mathcal{P}_2 \cong S^2$ of pure states as a subspace of $\mathrm{Herm}(2) = \mathbb{R}^4$. We know that $\mathrm{tr}(P_\Psi) = 1$, and for a hermitian matrix $H = a\sigma_0 + x\sigma_x + y\sigma_y + z\sigma_z$, the trace is $\mathrm{tr}(H) = 2a$. Hence, for pure states, $a = 1/2$. Since the eigenvalues $\lambda_{1,2} = a \pm \sqrt{x^2+y^2+z^2}$ are 0 and 1, $\sqrt{x^2+y^2+z^2} = 1/2$ for pure states. Hence, the set of pure states is a metric sphere S^2 of radius $1/2$ in $\mathbb{R}^3 \subset \mathrm{Herm}(2)$ consists of the matrices with $a = 1/2$. We conclude that the set of convex combinations (i.e., the *convex hull*) of a metric sphere in $\mathbb{R}^3$ is a ball B^3. More details are in the next section.

5.5 Special unitary and special orthogonal groups

This topic was eventually dropped from the course. However, I already wrote it down, I recommend to read it.

Recall that the set of *orthogonal matrices* is

$$O(n) = \{O \in \mathbb{R}^{n\times n} \mid O^T \cdot O = I_n\}, \tag{5.5.1}$$

where I_n is the $n \times n$ identity matrix. It follows from the definition that the determinant of an orthogonal matrix O is either $+1$ or -1. The set of *special orthogonal matrices* is

$$SO(n) = \{O \in O(n) \mid \det(O) = 1\}. \tag{5.5.2}$$

The defining equation $O^T \cdot O = I_n$ expresses that the columns of O form an orthonormal basis in $\mathbb{R}^n$ (with respect to the standard inner product), hence, the orthogonal matrices are exactly the congruences of $\mathbb{R}^n$ leaving the origin fixed. The special orthogonal matrices are the rotations of $\mathbb{R}^n$, leaving the origin fixed.

Similarly, the set of *unitary matrices* is

$$U(n) = \{U \in \mathbb{C}^{n\times n} \mid U^\dagger \cdot U = I_n\}. \tag{5.5.3}$$

It follows from the definition that $|\det(U)| = 1$ holds for unitary matrices. The set of *special unitary matrices* is

$$SU(n) = \{U \in U(n) \mid \det(U) = 1\}. \tag{5.5.4}$$

$O(n)$, $SO(n)$, $U(n)$ and $SU(n)$ are topological spaces, indeed, they inherit the subspace topology from $\mathbb{R}^{n\times n} = \mathbb{R}^{n^2}$, and from $\mathbb{C}^{n\times n} = \mathbb{C}^{n^2}$, respectively. They are also groups, because all the four sets are closed for matrix multiplication and inverse, and they contain the identity matrix. Since the group operations are continuous maps, these spaces are *topological groups*.[8]

Let us describe the topological type of $O(n)$, $SO(n)$, $U(n)$ and $SU(n)$ for the smallest values on n! Obviously,

$$O(1) = S^0 = \mathbb{Z}_2 = \{-1, +1\} \subset \mathbb{R}, \tag{5.5.5}$$

$$SO(1) = SU(1) = \{1\}, \tag{5.5.6}$$

$$U(1) = \{u \in \mathbb{C} \mid |u| = 1\} = S^1 \subset \mathbb{C}. \tag{5.5.7}$$

Also,

$$SO(2) = S^1, \tag{5.5.8}$$

they can be identified via the homeomorphism

$$\mathbb{R}^2 \supset S^1 \ni \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x & -y \\ y & x \end{pmatrix} \in SO(2). \tag{5.5.9}$$

Similarly, we have

$$SU(2) = S^3 \tag{5.5.10}$$

via the homeomorphism

$$\mathbb{C}^2 \supset S^3 \ni \begin{pmatrix} z \\ w \end{pmatrix} \mapsto \begin{pmatrix} z & -\overline{w} \\ w & \overline{z} \end{pmatrix} \in SU(2). \tag{5.5.11}$$

In both cases, the first column can be an arbitrary vector of unit length, and it uniquely determines the second column extending it to a spacial orthogonal (respectively, special unitary) matrix.

What is the 'shape' of $SO(3)$? One might guess it is S^2 – but it is false. S^2 is just the set of possible positions of the first column of a matrix $O \in SO(3)$. Intuitively, since the rotations around the 3 axes of $\mathbb{R}^3$ are 3 independent generators, $SO(3)$ is expected to be 3 dimensional.

[8] Moreover, $O(n)$, $SO(n)$, $U(n)$ and $SU(n)$ are also *Lie groups*, meaning that they are smooth manifolds and groups, and the group operations are smooth maps – definitions are coming soon.

Proposition 5.5.12. *$SO(3)$ is homeomorphic to the projective space $\mathbb{R}P^3$.*

Proof. Shortly, every element of $SO(3)$ is a rotation around an axis, see exercises. Hence, an element $O \in SO(3)$ is determined by its axis and angle $\alpha \in [-\pi; \pi]$. Associate to O the vector v on the axis with length $\|v\| = \alpha$. The corresponding vectors are in the closed ball B^3 of radius π centered at the origin. However, the rotations with $\alpha = \pi$ and $\alpha = -\pi$ around the same axis coincide, hence, the opposite pairs of points on the boundary $\partial B^3 = S^2$ are glued together, producing an identification of $SO(3)$ with $\mathbb{R}P^3$.

A more precise argument can be done from the other direction: consider the vectors $v \in B^3$, where $B^3 \subset \mathbb{R}^3$ is the closed ball of radius π centered at the origin. We define a map

$$\begin{aligned} f : B^n &\to SO(3) && (5.5.13)\\ f(v) &= R_v, && (5.5.14) \end{aligned}$$

where R_v is the rotation around the (oriented) axis of v with (oriented) angle $\alpha = \|v\|$. The map f is a continuous bijection, except if $\|v\| = \pi$, then $R_v = R_{-v}$. Hence, f induces a bijection

$$\widetilde{f} : \mathbb{R}P^3 \to SO(3), \tag{5.5.15}$$

which is in turn a homeomorphism. □

Remark 5.5.16. We note that by the Euler–Rodrigues formulas, the rotation R_v is expressed explicitly as

$$R_v = \exp \begin{pmatrix} 0 & -z & -y \\ z & 0 & -x \\ -y & x & 0 \end{pmatrix}, \tag{5.5.17}$$

where $v = (x, y, z)$.

Since $SU(2) = S^3$ and $SO(3) \cong \mathbb{R}P^3$, we have a homeomorphism

$$SO(3) \cong \frac{SU(2)}{U \sim -U}. \tag{5.5.18}$$

An explicit description of the quotient map $SU(2) \to SO(3)$ is given as an exercise.

The Lie groups $SO(3)$ and $SU(2)$ has a central role in the description of the particles.

5.6 Exercises

Exercise 5.6.1 (Real projective plane)**.** (a) Show that

$$\mathcal{M} \cup_{S^1} B^2 \cong \mathbb{R}P^2. \tag{5.6.2}$$

(b) Show that $\mathbb{R}P^2$ can be realised as a subspace of $\mathbb{R}^5$. (Hint: projectors)

(c) Show that $\mathbb{R}P^2$ can be realised as a subspace of $\mathbb{R}^4$. (Hint: point (a))

Exercise 5.6.3 (Rotations of $\mathbb{R}^3$)**.** Show that every element of $SO(3)$ is a rotation around an axis!

Exercise 5.6.4 ($SU(2)$ and $SO(3)$)**.** Give an explicit description of the quotiont map $SU(2) \to SO(3)$ using the hermitian matrices! Consider the 2×2 hermitian matrices of trace 0 in the Pauli basis, i.e., $H = x\sigma_x + y\sigma_y + z\sigma_z$. Their space $\mathrm{Herm}_0(2)$ is identified with $\mathbb{R}^3$. For $U \in SU(2)$ define the conjugation map

$$\begin{aligned} C_U : \mathrm{Herm}_0(2) &\to \mathrm{Herm}_0(2) && (5.6.5)\\ C_U(H) &= U \cdot H \cdot U^{-1}. && (5.6.6) \end{aligned}$$

(a) Show that C_U is a linear map, and write its matrix $\widetilde{C}_U$ in the Pauli basis.

(b) Show that $\widetilde{C}_U \in SO(3)$.

(c) Show that the map

$$\begin{aligned} f : SU(2) &\to SO(3) && (5.6.7)\\ f(U) &= \widetilde{C}_U && (5.6.8)\\ & && (5.6.9) \end{aligned}$$

is a continuous and surjective map, and $f^{-1}(f(U)) = \{U, -U\}$ for every $U \in SU(2)$.

(d) Which property of f should be checked to conclude that the bijection

$$\frac{SU(2)}{U \sim -U} \to SO(3) \tag{5.6.10}$$

induced by f is a homeomorphism?

Exercise 5.6.11 ($O(n)$ and $U(n)$)**.** (a) Show that $O(n)$ is not connected.

(b) Show that $O(n)$ is homeomorphic to $SO(n) \times O(1)$.

(c) Show that $U(n)$ is homeomorphic to $SU(n) \times U(1)$.

(d) Show that the above homeomorphisms are not group ismomorphisms, i.e., they do not preserve the multiplication. (In fact, as a group, $U(n)$ is not isomorphic to $SU(n) \times U(1)$, and $O(n)$ is not isomorphic to $SO(n) \times O(1)$.)

6 Physical examples

6.1 Vector fields as topological spaces

Many quantities in physics are described by vector fields. The dimension and the range of possible values of these fields is defined by the underlying physical phenomena, which also influences the topological properties of these fields. Consider the following examples:

6.1.1 Magnetism

In classical magnetostatics, magnetic materials are described by the magnetization field $\boldsymbol{M}(\boldsymbol{r}) : \mathbb{R}^3 \to \mathbb{R}^3$. The magnetization describes the density of magnetic moment at point $\boldsymbol{r}$. In solids, primarily the electrons are responsible for the observable magnetic properties. The magnetic moment may have classical origin, traditionally corresponding to microscopic current loops in the material, which is known as the orbital magnetic in the modern description. Quantum-mechanical effects are essential for describing magnetism. First, the origin of the electron's own angular momentum or spin has its roots in relativistic quantum mechanics, and the associated spin magnetic moment in solids is typically considerably larger than the orbital magnetic moment. Second, the Bohr-van Leeuwen theorem excludes the formation of a net magnetic moment in thermal equilibrium in classical statistical mechanics. Magnetic ordering observed in ferromagnets, antiferromagnets and ferrimagnets can only be explained by quantum-mechanical effects, particularly the exchange interaction.

In ferromagnets, the magnitude of magnetic moments is typically constant, and the external field only determines their orientation. It is useful to introduce the direction of magnetic moments $\boldsymbol{m}$ as $\boldsymbol{M}(\boldsymbol{r}) = M_\mathrm{s}\boldsymbol{m}(\boldsymbol{r})$, where M_s is the saturation magnetization. Since $\boldsymbol{m}$ is a unit vector, it is a map to the two-sphere: $\boldsymbol{m} : \mathbb{R}^3 \to S^2$.

An example for the free energy of ferromagnets is the following:

$$F = \int \mathcal{A}(\boldsymbol{\nabla}\boldsymbol{m})^2 - \mathcal{K}(m^z)^2 - M_\mathrm{s}\boldsymbol{m}\boldsymbol{B}\mathrm{d}^3\boldsymbol{r} + \iint \frac{\mu_0}{8\pi} M_\mathrm{s}^2 \boldsymbol{m}^T(\boldsymbol{r}) |\boldsymbol{r}-\boldsymbol{r}'|^{-3} \left(I - 3\frac{(\boldsymbol{r}-\boldsymbol{r}')(\boldsymbol{r}-\boldsymbol{r}')^T}{|\boldsymbol{r}-\boldsymbol{r}'|^2} \right) \boldsymbol{m}^T(\boldsymbol{r}')\,\mathrm{d}^3\boldsymbol{r}\mathrm{d}^3\boldsymbol{r}'. \tag{6.1.1}$$

The first term is the exchange interaction, which promotes a uniform direction of the magnetic moments in a ferromagnet. It is of quantum-mechanical origin, but it does not require spin–orbit coupling, making it typically the strongest interaction term. The second term is the magnetocrystalline anisotropy, which aligns a magnetic moment along a specific spatial direction (here, the z direction). It is a consequence of spin–orbit coupling in relativistic quantum mechanics. The third term describes the coupling of the magnetization to the external field. The fourth term is the dipolar interaction between the magnetic moments. The last two terms are also present in classical magnetostatics.

6.1.2 Liquid crystals

Liquid crystals consist of large molecules, which typically have rod-like, disk-like or cone-like shapes. The shapes of the molecules determine their interactions and ordering. Ordering may be restricted to the orientation of the molecules (nematic ordering), or it may extend to their positions (smectic ordering). Rod-like molecules are described by their director $\boldsymbol{n}$, which is typically treated as a unit vector $\boldsymbol{n} : \mathbb{R}^3 \to S^2$. In many molecules, rotating them by 180° results in the same atomic configuration. This corresponds to identifying opposite directions on the sphere, making the director an element of the projective plane $\mathbb{R}P^2$.

An example for the free energy describing liquid crystals is the following (Phys. Rev. E **90**, 042502):

$$F = \int \frac{K_1}{2} \left(\mathrm{div}\boldsymbol{n}\right)^2 + \frac{K_2}{2} \left(\boldsymbol{n} \cdot \mathrm{rot}\boldsymbol{n} - q_0\right)^2 + \frac{K_3}{2} \left(\boldsymbol{n} \cdot \mathrm{rot}\boldsymbol{n}\right)^2 - \frac{\varepsilon_a}{2} \left(\boldsymbol{n} \cdot \boldsymbol{E}\right)^2 - \frac{\chi_a}{2} \left(\boldsymbol{n} \cdot \boldsymbol{H}\right)^2 \mathrm{d}^3\boldsymbol{r}. \tag{6.1.2}$$

6.1.3 Electromagnetic fields

Classical electrodynamics describes the electric and magnetic fields in space, $\boldsymbol{E}\left(\boldsymbol{r}, t\right) : \mathbb{R}^3 \times \mathbb{R} \to \mathbb{R}^3$ and $\boldsymbol{B}\left(\boldsymbol{r}, t\right) : \mathbb{R}^3 \times \mathbb{R} \to \mathbb{R}^3$. Typically the magnitude of these fields can be arbitrary, and the components oscillate with different phases in time. Under special circumstances, the possible values of the fields may be restricted. For example, electromagnetic standing waves confined to a plane, which show evanescent decay along the third spatial direction, admit a solution of the Maxwell equations where all components are in the same phase (Science **361**, 993); see the corresponding exercise. In this case, it is sufficient to treat the direction of the fields in S^2. It is also possible to reduce the space of electric or magnetic fields to S^3 (Nat. Commun. **12**, 6785).

6.1.4 Superconductors

Superconductors are described by a complex order parameter $\psi\left(\boldsymbol{r}\right) : \mathbb{R}^3 \to \mathbb{C}$. Typically both the magnitude and the phase of the order parameter may vary. This is described by the Ginzburg–Landau free-energy functional:

$$F = \int \frac{1}{4m} \left| \left(\frac{\hbar}{\mathrm{i}} \boldsymbol{\nabla} - 2e\boldsymbol{A} \right) \psi \right|^2 + \alpha \left|\psi\right|^2 + \beta \left|\psi\right|^4 + \frac{\boldsymbol{B}^2}{2\mu_0} \mathrm{d}^3\boldsymbol{r}. \tag{6.1.3}$$

In homogeneous superconductors, the magnitude of the order parameter is fixed by the temperature-dependent parameters $\alpha < 0$ and $\beta > 0$. In this case, it is sufficient to consider the phase of the order parameter, which takes values on the circle S^1. A phase gradient corresponds to a supercurrent flowing through the system.

6.2 Topology of the state of spaces

6.2.1 Classical mechanics

The fundamental space in classical mechanics is the phase space P. In Hamiltonian mechanics, an element of the phase space is $v = \{q_i, p_i\}$, where q_i and p_i are generalized coordinates and momenta.

Typically this means that $P = \mathbb{R}^{2N}$, where N is the number of particles. However, the generalized coordinate may be an angle variable restricted to the circle, while the corresponding momentum is the angular momentum. The coordinates may also be restricted to a sphere S^2, e.g., for the magnetic moments or macromolecules in liquid crystals discussed above. There are also quantum-mechanically inspired classical models where the phase space is discrete. One example is the Ising model, where the variable σ_i may only take two values, e.g., $\{0, 1\}$, $\{-1, 1\}$ or $\{\text{up}, \text{down}\}$. This is used to describe magnetic ordering in strongly anisotropic materials. One of its generalizations is the p-state Potts model, where $\sigma_i \in \{0, \ldots, p-1\}$.

A physical observable is a real-valued function on the phase space $A : \mathbb{P} \to \mathbb{R}$. The possible values of the measurement are in the range of A.

The system is characterized by its state ϱ. This state must determine the results of the measurements for any physical observable A. Mathematically, ϱ is a probability measure on P:

- $\int \mathbb{I}(X) \, \mathrm{d}\varrho \geq 0, \forall X \subset P$,
- $\int \mathbb{I}(\emptyset) \, \mathrm{d}\varrho \geq 0$,
- $\int \mathbb{I}(\cup_{k=1}^{\infty} X_k) \, \mathrm{d}\varrho = \sum_{k=1}^{\infty} \int \mathbb{I}(X_k) \, \mathrm{d}\varrho$ for disjoint sets,
- $\int \mathrm{d}\varrho = 1$.

ϱ defines the probabilities of the possible outcomes of the measurement. The expectation value of an observable is $\langle A \rangle = \int A \mathrm{d}\varrho$.

Pure states ϱ_{v_0} are defined as $\int \mathbb{I}(X) \, \mathrm{d}\varrho_{v_0} = 1$ if $v_0 \in X$ and $\int \mathbb{I}(X) \, \mathrm{d}\varrho_{v_0} = 0$ if $v_0 \notin X$, i.e., they are the point measures. In physics, the preferred notation for probability measures uses the probability density function $\varrho : P \to \mathbb{R}^+$, with $\int A \mathrm{d}\varrho = \int A(v) \, \varrho(v) \, \mathrm{d}v$. For pure states, formally one can write $\varrho_{v_0} = \delta(v - v_0)$, although this is not an actual function. Since there is a one-to-one correspondence between the pure states ϱ_{v_0} and the elements of the phase space v_0, the space of pure states D_{pure} is homeomorphic to P.

Mixed states are any probability measure on P. A well-known example is the Boltzmann distribution with probability density function $\varrho = \mathrm{e}^{-\beta H(v)}$, where $\beta > 0$ is a parameter describing the inverse temperature and H is an observable, corresponding to the energy of the system. From a topology point of view, the space of all states D is typically a complicated function space. For discrete phase spaces, it may have a geometric interpretation; see the corresponding exercise.

6.2.2 Quantum mechanics

In quantum mechanics, the fundamental space is a Hilbert space $\mathcal{H}$. In finite dimensions considered here, a Hilbert space is any vector space equipped with a scalar product $\langle ., . \rangle : \mathcal{H} \times \mathcal{H} \to \mathbb{C}$

- $\langle \psi, \varphi \rangle = \langle \varphi, \psi \rangle^* \, \forall \psi, \varphi \in \mathcal{H}$,
- $\langle \psi, c_1 \varphi_1 + c_2 \varphi_2 \rangle = c_1 \langle \psi, \varphi_1 \rangle + c_2 \langle \psi, \varphi_2 \rangle \, \forall \psi, \varphi_1, \varphi_2 \in \mathcal{H}, c_1, c_2 \in \mathbb{C}$,
- $\langle \psi, \psi \rangle \geq 0$,
- $\langle \psi, \psi \rangle = 0 \iff \psi = 0$.

A physical observable is represented by a self-adjoint operator $A : \mathcal{H} \to \mathcal{H}$ on this Hilbert space, i.e., $\langle \psi, A\varphi \rangle = \langle A\psi, \varphi \rangle \, \forall \psi, \varphi \in \mathcal{H}$. The possible values of a measurement are in the spectrum of A, which contains real numbers due to the operator being self-adjoint.

A quantum-mechanical state is again a measure defining the probabilities of the possible outcomes of a measurement, namely a positive-operator valued measure. It may be conveniently described by the density operator $\varrho : \mathcal{H} \to \mathcal{H}$, which is a positive semidefinite operator with $\mathrm{Tr}\varrho = 1$. The expectation value of an observable is given by $\langle A \rangle = \mathrm{Tr}(A\varrho)$.

A pure state satisfies the further constraint $\varrho^2 = \varrho$. It can be shown that all pure states can be identified with an orthogonal projection onto a one-dimensional subspace, $\varrho_\psi = |\psi\rangle \langle\psi|$ for $\psi \in \mathcal{H}, \|\psi\| = 1$. If $\mathcal{H} = \mathbb{C}^n$, then the space of pure states is the projective space $D_{\text{pure}} = \mathbb{C}P^{n-1}$.

A mixed state satisfies $\mathrm{Tr}\varrho^2 < 1$, while $\mathrm{Tr}\varrho^2 = 1$ for pure states. A mixed state may be written as a convex combination of pure states, $\varrho = \sum_i c_i \left|\psi_i\right\rangle \left\langle\psi_i\right|, c_i \in \mathbb{R}^+, \sum_i c_i = 1$. However, the decomposition into pure states is not unique, see the corresponding exercise. The space of all states D is generally complicated.

As an example, consider $\mathcal{H} = \mathbb{C}^2$. The space of pure states is $\mathbb{C}P^1 = S^2$ the Bloch sphere. To describe the mixed states, note that any self-adjoint operator on $\mathcal{H}$ may be written as $A = a_0\sigma_0 + \boldsymbol{a}\boldsymbol{\sigma}$, where $a_0 \in \mathbb{R}, \boldsymbol{a} \in \mathbb{R}^3$ and $\sigma_0, \boldsymbol{\sigma}$ are the Pauli matrices. The eigenvalues of A are $a_0 \pm \|\boldsymbol{a}\|$, while $\mathrm{Tr}A = 2a_0$. In order to obtain a density matrix, it is easy to see that $a_0 = \frac{1}{2}$, $\|\boldsymbol{a}\| \leq \frac{1}{2}$ must hold. Consequently, any state may be written as $\varrho = \frac{1}{2}\left(\sigma_0 + \boldsymbol{d}\boldsymbol{\sigma}\right)$, where $\boldsymbol{d} \in B^3$ is in the Bloch ball.

7 Homotopy

7.1 Continuos deformation

We cheated a bit by saying that homeomorphism means continuous deformation of spaces. For example, $X = S^1$ and $Y = \mathbb{R}^2 \setminus \{0\}$ are not homeomorphic, but they can be deformed to each other continuously via the 'continuous family' of continuous maps

$$f_t(x) = t\frac{x}{\|x\|} + (1-t)x \tag{7.1.1}$$

indeed, $f_0(x) = x \in \mathbb{R}^2 \setminus \{0\}$ and $f_1(x) = x/\|x\| \in S^1$. We actually feel some similarity between the shape of S^1 and $\mathbb{R}^0 \setminus \{0\} = S^1 \times \mathbb{R}$, essentially they have the same topology. Similarly, B^n can be deformed continuously to a single point (the origin) by the family of maps

$$f_t(x) = (1-t)x, \tag{7.1.2}$$

according to the intuition that the topology of B^n is 'trivial'.

How can we formulate these notions precisely? As it turns out, continuous deformation is expressed by the concept of homotopy. While homeomorphism is rather a 'continuous identification of topological spaces'.

7.2 Homotopy between continuous maps

Definition 7.2.1. Let X and Y be topological spaces, and let $f, g : X \to Y$ be continuous maps. f and g are *homotopic* if there is a continuous map $H : X \times [0;1] \to Y$ such that $H(x,0) = f(x)$ and $H(x,1) = g(x)$. The map H is called a *homotopy* between f and g.

It is convenient to introduce the notation $H(x,t) = H_t(x)$, hence, $H_0 = f$ and $H_1 = g$, and H_t can be considered as a continuous family between f and g.[9] f is continuously deformed to g via the continuous maps $H_t : X \to Y$.

As it turns out, homotopy is an equivalence relation between the continuous maps $X \to Y$. For example, the reversed homotopy $\widetilde{H}_t = H_{1-t}$ joins g and f, showing the symmetry of the homotopy: f is homotopic to g if and only if g is homotopic to f. (How can you prove that the homotopy is transitive, i.e., if f is homotopic to g and g is homotopic to h, than f is homotopic to h?) We denote by $f \sim g$ the fact that f and g are homotopic maps.

We deonte by $[X, Y]$ the set of homotopy classes of smooth maps, i.e.,

$$[X,Y] = \frac{\{f : X \to Y \text{ continuous maps}\}}{\text{homotopy}} \tag{7.2.2}$$

The homotopy class of a continuous map $f : X \to Y$ is denoted by $[f] \in [X, Y]$.

'Topological classification' in (condensed matter) physics usually means characterisation of the homotopy classes of certain continuous maps.

[9] One can try to define the homotopy as a path in the space of continuous maps joining f and g, i.e., a continuous map $[0;1] \to \{f : X \to Y \text{ continuous}\}$, $t \mapsto H_t$. However, to make it precise, we should define the 'topological space of continuous maps between X and Y'. The braver can try it!

Example 7.2.3 (Nullhomotopy)**.** Any continuous map $f : X \to \mathbb{R}^n$ is homotopic to a constant map $g(x) \equiv 0$, indeed,

$$H_t(x) = (1-t)f(x) \tag{7.2.4}$$

is a homotopy between f and g.

Similarly, any continuous map $f : \mathbb{R}^n \to X$ is homotopic to a constant map, indeed,

$$H_t(x) = f((1-t)x) \tag{7.2.5}$$

satisfies $H_0(x) = f(x)$ and $H_1(x) \equiv f(0)$ is a constant map.

A map homotopic to a constant map (i.e., a map whose image is one point) is called *nullhomotopic*. We showed that every map from or to $\mathbb{R}^n$ is nullhomotopic. In the same way we can see that any map from or to B^n is nullhomotopic. In other words, each set $[X, B^n]$, $[B^n, Y]$, $[X, \mathbb{R}^n]$, $[\mathbb{R}^n, Y]$ has only one element, namely, the homotopy class of the constant map '0'. Shortly we write

$$[X, B^n] = [B^n, Y] = [X, \mathbb{R}^n] = [\mathbb{R}^n, Y] = 0. \tag{7.2.6}$$

Example 7.2.7 (Punctured plane)**.** Consider the continuous maps $f : S^1 \to \mathbb{R}^2 \setminus \{0\}$ from a circle to the punctured plane. It is intuitively clear that the inclusion map $f(x) = x$ is not nullhomotopic, indeed, a nullhomotopy would 'get stuck' in the origin, which does not belong to the target space. This argument will be made precise later, but we can already make some observations.

First, observe that any continuous map $f : X \to \mathbb{R}^2 \setminus \{0\}$ from an arbitrary topological space X is homotopic to a map to $S^1 \subset \mathbb{R}^2 \setminus \{0\}$ via the homotopy

$$H_t(x) = (1-t)f(x) + t\frac{f(x)}{\|f(x)\|}, \tag{7.2.8}$$

for which it is important that $H_t(x)$ is never 0, i.e., $H_t : X \to \mathbb{R}^2 \setminus \{0\}$ for every $t \in [0; 1]$. Does this observation imply that we have a bijection

$$[X, \mathbb{R}^2 \setminus \{0\}] \leftrightarrow [X, S^1] \ ? \tag{7.2.9}$$

Yes, because a homotopy H between two maps $f, g : X \to \mathbb{R}^2 \setminus \{0\}$ is also homotopic to a homotopy H' joining the corresponding maps $f', g' : X \to S^1$ (where $f' \sim f$ and $g' \sim g$ in $\mathbb{R}^2 \setminus \{0\}$).

In particular, we proved that

$$[S^1, \mathbb{R}^2 \setminus \{0\}] = [S^1, S^1 \times \mathbb{R}] \leftrightarrow [S^1, S^1]. \tag{7.2.10}$$

One expects that these sets are in one to one correspondence with the integers $\mathbb{Z}$, i.e., each homotopy class is completely characterised by the *winding number* of a map counting the number of turns around the origin. In other words, two maps are homotopic if and only if they have the same winding number. This implies our original expectation, that the inclusion map $f : S^1 \to \mathbb{R}^2 \setminus \{0\}$, $f(x) = x$ is not nullhomotopic. Indeed, the winding number of the inclusion f is 1, while the winding number of the constant map is 0. The identification $[S^1, S^1] \leftrightarrow \mathbb{Z}$ via the winding number will be shown more precisely later, using a precise definition for the winding number as a particular case of a the *degree* of maps.

Example 7.2.11 (Punctured space)**.** The same argument shows that

$$[X, \mathbb{R}^{n+1} \setminus \{0\}] = [X, S^n \times \mathbb{R}] \leftrightarrow [X, S^n]. \tag{7.2.12}$$

In particular,

$$[S^n, \mathbb{R}^{n+1} \setminus \{0\}] = [S^n, S^n \times \mathbb{R}] \leftrightarrow [S^n, S^n], \tag{7.2.13}$$

and this set will be identified with $\mathbb{Z}$ via the degree, which is a higher dimensional generalization of the winding number.

Example 7.2.14 (Maps to spheres)**.** If a continuous map $f : X \to S^n$ is not surjective, then it is nullhomotopic. Indeed, f avoids a point $p \in S^n$ (i.e., $p \notin f(X)$), and $S^n \setminus \{p\} \cong \mathbb{R}^n$, hence, f is homotopic to a constant map in $S^n \setminus \{p\}$.

If X is a cell complex with $\dim(X) < n$, then every continuous map $f : X \to S^n$ is nullhomotopic, i.e., $[X, S^n] = 0$. To show it, one may expect that 'a continuous map cannot increase the dimension', hence, f cannot be surjective onto S^n, since $\dim(S^n) > \dim(X)$. Then the previous argument implies that f is nullhomotopic.

However, the intuitive requirement that the image of a continuous map cannot have higher dimension than the domain is not correct! The famous *Peano curves* and other *space-filling curves* are counterexamples: they are continuous and surjective maps from an interval $[0; 1]$ onto the square $[0; 1]^2$, actually, they are certain type of *fractals*. Check them on internet!

We just notice that it is possible that a continuous map $f : X \to S^n$ (where $\dim(X) < n$) is surjective. But it is still true that every such map is nullhomotopic! It can be shown for example in these two steps:

1. Each continuous map f is homotopic to a smooth map (see later), and

2. a smooth maps cannot increase the dimension, i.e., its image $f(X)$ cannot be equal to the whole target space S^n.

Example 7.2.15 (Cell complexes)**.** A similar argument shows that for two cell complexes X and Y with $\dim(X) = k < n = \dim(Y)$, every continuous map $f : X \to Y$ is homotopic to a map $g : X \to \mathrm{sk}_k(Y)$. In other words, every map can be continuously deformed into the k-skeleton of Y. In other words, f be ousted from every cell of dimension higher than k by a homotopy.

To see it, let $E^m = B^m$ be a cell of dimension $m > k$ of Y, it is glued to the $m - 1$ skeleton along a map $\phi^m : \partial E^m \to \mathrm{sk}_{m-1}(Y)$. On expects that f cannot be surjective onto the interior of E^m, because $m > k$, but as we noticed, it is not true in general. However, it can be fixed by a homotopy: f is homotopic to a map f' which is not surjective onto the interior of E^m, hence, it avoids a point p. Then, f' can be 'blown out' from p to the boundary of E^m, i.e., f' is homotopic to a map avoiding the interior of E^m. The same can done for every cell of dimension higher then k, resulting in a map $g : X \to \mathrm{sk}_k(Y)$.

We notice that this argument *does not imply* that $[X, Y] = [X, \mathrm{sk}_k(Y)]$. Why? Although any map $f : X \to Y$ is homotopic to a map $g : X \to \mathrm{sk}_k(Y)$, if two maps f_1 and f_2 are homotopic in Y, then the corresponding maps g_1 and g_2 are not necessarily homotopic in $\mathrm{sk}_k(Y)$. Indeed, the homotopy is map $H : X \times [0; 1] \to Y$, and $\dim(X \times [0; 1]) = k + 1$. Hence, H is not necessarily homotopic to a map $X \to \mathrm{sk}_k(Y)$. By the above argument, H is homotopic to a map $K : X \to \mathrm{sk}_{k+1}(Y)$, but it is not enough to obtain a homotopy between g_1 and g_2 in $\mathrm{sk}_k(Y)$. Instead, we can conclude that

$$[X, Y] \leftrightarrow [X, \mathrm{sk}_{k+1}(Y)]. \tag{7.2.16}$$

Example 7.2.17 (Maps of the circle)**.** In particular,

$$[S^1, Y] \leftrightarrow [S^1, \mathrm{sk}_2(Y)] \tag{7.2.18}$$

holds for cell complexes Y. Indeed, every continuous map $f : S^1 \to Y$ is homotopic to a map $g : S^1 \to \mathrm{sk}_1(Y)$, and a homotpy between two maps $f_1, f_2 : S^1 \to Y$ can be realised in the 2-skeleton $\mathrm{sk}_2(Y)$. That is, if f_1 and f_2 are homotopic in Y, then the corresponding maps $g_1, g_2 : S^1 \to \mathrm{sk}_1(Y)$ are homotopic in $\mathrm{sk}_2(Y)$.

The set of homotopy classes $[S^1, X]$ measures the 'holes' of X. Indeed, by the following proposition, a map from the circle to X is nullhomotopic if and only if it can be extended to the disc, i.e., if the map 'does not bypass a hole'. (This idea will be continued later, in the construction of the fundamental group.)

Proposition 7.2.19. *$f : S^1 \to X$ is nullhomotopic if and only if it has a continuous extension $\widetilde{f} : B^2 \to X$, $\widetilde{f}|_{\partial B^2 = S^1} = f$.*

Proof. A nullhomotopy is a continuous map $H : S^1 \times [0;1] \to X$ from the cylinder to X such that $H_0 = f$ and H_1 is a constant map. Since H_1 maps the upper boundary circle of the cylinder to one point, it induces a continuous map $\widetilde{H} : S^1 \times [0;1]/\sim \to X$, where $S^1 \times [0;1]/\sim$ is the quotient space obtained from the cylinder by contracting its upper boundary $S^1 \times \{1\}$. But this quotient space is homeomorphic to a disc B^2. On Diagram 7.2.20, the quotient map is denoted by g. We conlude for a map H there is a map $\widetilde{H}$ if and only if H is constant on the upper boundary. (This is a universal property of the quotients.) Taking $\widetilde{H} = \widetilde{f}$ proves the proposition.

$$\begin{array}{ccccc} & & B^2 & & \\ & \overset{g}{\nearrow} & & \overset{\widetilde{H}}{\searrow} & \\ S^1 \times [0;1] & & \xrightarrow{\qquad H \qquad} & & X \end{array} \tag{7.2.20}$$

□

7.3 Homotopy equivalence between topological spaces

Above we discussed continuous deformation of continuous maps (i.e., homotopy), now we study continuous deformation of topological spaces, called homotopy equivalence.

Definition 7.3.1. Two topological spaces X and Y are *homotopy equivalent* (or, of the *same homotopy type*) if there are continuous maps $f : X \to Y$ and $g : Y \to X$ such that $f \circ g : Y \to Y$ is homotopic to the identity map of Y and $g \circ f : X \to X$ is homotopic to the identity map of X. The maps f and g are called a *homotopy equivalences* between X and Y, they are homotopy inverses of each other. Notation: $X \sim_{h.e.} Y$ means that X is homotopy equivalent to Y.

Intuitively, two spaces X and Y are homotopy equivalent if they can be transformed into one another by bending, shrinking and expanding operations.

If two topological spaces are homeomorphic, then they are homotopy equivalent (why?). Therefore, homotopy equivalence is a weaker equivalence of topological spaces than the homeomorphism.

Example 7.3.2 (Contractible topological spaces)**.** $\mathbb{R}^n$ (and also B^n) is homotopy equivalent to a point $\{p\}$. What are the homotopy equivalences between them? $f : \mathbb{R}^n \to \{p\}$ is the constant map $f(x) \equiv p$, this is the only map to a one-point space. $g : \{p\} \to \mathbb{R}^n$ can be chosen for example as $g(p) = 0$. Then, $(f \circ g)(p) = p$ is already the identity map of $\{p\}$, and $(g \circ f)(x) \equiv 0$ is a constant map. We saw that any map to $\mathbb{R}^n$ is homotopic to the constant map, in particular,

$$H_t(x) = tx \tag{7.3.3}$$

is a homotopy between $g \circ f$ and the identity map of $\mathbb{R}^n$.

A space X is *contractible* if it is homotopy equivalent to a point. This notion formalizes the intuitive argument that X is topologically trivial. We showed that $\mathbb{R}^n$ and B^n are contractible, in particular, they are homotopy equivalent to each other. Contractible spaces have several different characterizations, see exercises.

Example 7.3.4 (Punctured plane)**.** The punctured plane $\mathbb{R}^2 \setminus \{0\}$ is homotopy equivalent to S^1. Let $f : S^1 \to \mathbb{R}^2 \setminus \{0\}$ be the inclusion map $f(x) = x$, and let $g : \mathbb{R}^2 \setminus \{0\} \to S^1$ be the normalization map $g(x) = x/\|x\|$. Then, $(g \circ f)(x) = x$ is already the identity map of S^1, and $(g \circ f)(x) = x/\|x\|$ is homotopic to the identity map of $\mathbb{R}^2 \setminus \{0\}$ via

$$H_t(X) = tx + (1-t)\frac{x}{\|x\|}. \tag{7.3.5}$$

The same argument shows that $\mathbb{R}^n \setminus \{0\}$ is homotopy equivalent to S^{n-1}. And $S \times [0;1]$ is also homotopy equivalent to S^1.

Investigating the set of homotopy classes $[X, Y]$, each space can be replaced by a homotopy equivalent one, if it is more convenient to study. More precisely:

Proposition 7.3.6. *if X is homotopy equivalent to X' and Y is homotopy equivalent to Y', then*

$$[X, Y] \leftrightarrow [X', Y']. \tag{7.3.7}$$

We present a detailed proof for this – very intuitive – proposition.

Lemma 7.3.8. *Compositions of homotopic maps are homotopic. That is, if the continuous maps $f_1, f_2 : X \to Y$ are homotopic, and $g_1, g_2 : Y \to Z$ are also homotopic, then $g_1 \circ f_1$ and $g_2 \circ f_2$ are homotopic, as maps from X to Z.*

Proof of Lemma 7.3.8. If H is a homotopy between f_1 and f_2 and K is a homotopy between g_1 and g_2, then $K_t \circ H_t$ defines a homotopy between $g_1 \circ f_1$ and $g_2 \circ f_2$. □

Proof of Proposition 7.3.6. Let $f : X \to X'$ be a homotopy equivalence, let $f' : X' \to X$ be a homotopy inverse of f. Let $g : Y \to Y'$ be a homotopy equivalence, let $g' : Y' \to Y$ be a homotopy inverse of g. For a map $h : X \to Y$ we define $A(h) : X' \to Y'$ as

$$A(h) := g \circ h \circ f', \tag{7.3.9}$$

and for a map $k : X' \to Y'$ we define $B(k) : X \to Y$ as

$$B(k) := g' \circ k \circ f. \tag{7.3.10}$$

Observe that if h_1 and h_2 are homotopic maps ($X \to Y$), then $A(h_1)$ and $A(h_2)$ are also homotopic ($X' \to Y'$), and also $B(k_1)$ is homotopic to $B(k_2)$ if k_1 is homotopic to k_2. Moreover, $B(A(h))$ is homotopic to h, and $A(B(k))$ is homotopic to k. Indeed,

$$B(A(h)) = g' \circ g \circ h \circ f' \circ f, \tag{7.3.11}$$

where $g' \circ g$ is homotopic to the identity of Y, and $f' \circ f$ is homotopic to the identity of X, hence, by Lemma 7.3.8, $B(A(h))$ is homotopic to h. (The homopotopy of $A(B(k))$ and k is similar.)

We define the maps (between sets)

$$\begin{aligned} [X, Y] &\to [X', Y'] && (7.3.12)\\ \alpha([h]) &:= [A(h)], && (7.3.13) \end{aligned}$$

and

$$\begin{aligned} [X', Y'] &\to [X, Y] && (7.3.14)\\ \beta([k]) &:= [B(k)]. && (7.3.15) \end{aligned}$$

By the above observations, α and β are well defined, and they are inverses, $\beta = \alpha^{-1}$. In particular, they are bijections between $[X, Y]$ and $[X', Y']$, proving the proposition.

□

7.4 Fundamental group

The fundamental group is a tool to distinguish topological spaces. It measures the 'holes' of a topological space. (What is the hole? Is it a property of the material or of the outside world that permeates it? Since the topological spaces are considered without assuming an outside word, the fundamental group answers this philosophical question: the holes are inner properties, they can be recognised without referring to the outside word. See this video.)

The first candidate to measure the holes of a topological space X would be the set of homotopy classes $[S^1, X]$. Indeed, by Proposition 7.2.19, a map from the circle to X is nullhomotopic if and only if it can be extended to the disc, i.e., if the map 'does not bypass a hole'. Although $[S^1, X]$ really measures the holes of X in this sense, it is just a set, but not a group in general. For the fundamental group we have to consider also a base point $b_0 \in X$.

Definition 7.4.1. • A *loop based at* $b_0 \in X$ is a continuous map $f : [0;1] \to X$ such that $f(0) = f(1) = b_0$. Equivalently, by taking the quotient $[0;1] \to S^1$, a loop f can be considered as a continuous map $f : S^1 \to X$ with $f(p_0) = b_0$, where $p_0 \in S^1$ is a fixed point.

- Two loops f and g are *homotopic* as loops based at b_0 if there is a homotopy $H : S^1 \times [0;1] \to X$, $H_0 = f$, $H_1 = g$ such that H_s is a loop based at b_0 for every $s \in [0;1]$. That is, $H_s(p_0) = b_0$. Equivalently: $H : [0;1] \times [0;1] \to X$, $H_0 = f$, $H_1 = g$, $H_s(0) = H_s(1) = b_0$.

- The *concatenation* of two loops $f, g : [0;1] \to X$ based at b_0 is the loop $f * g$ defined as

$$(f * g)(t) = \begin{cases} f(2t) & \text{if} \quad 0 \le t \le \frac{1}{2} \\ g(2t-1) & \text{if} \quad \frac{1}{2} \le t \le 1. \end{cases} \tag{7.4.2}$$

  Thus the loop $f * g$ first follows the loop f with 'twice the speed' and then follows g with 'twice the speed'.

For a given X, the set of homotopy classes of loops based at b_0 is denoted by $\pi_1(X, b_0)$. The concatenation of loops induces an operation (product) on $\pi_1(X, b_0)$:

$$[f] \cdot [g] = [f * g], \tag{7.4.3}$$

Indeed, for homotopic loops $f_1 \sim f_2$ and $g_1 \sim g_2$ the concatenations are also homotopic, i.e., $f_1 * g_1 \sim f_2 * g_2$, hence the operation is well defined. One can show that $\pi_1(X, b_0)$ endowed with this product is a group, see exercises.

Definition 7.4.4. The *fundamental group* of X with respect to the base point $b_0 \in X$ is the set of homotopy classes

$$\pi_1(X, b_0) = \frac{\text{loops in } X \text{ based at } b_0}{\text{homotopy}} \tag{7.4.5}$$

endowed with the product $[f] \cdot [g] = [f * g]$.

What is the identity element in $\pi_1(X, b_0)$? This is the homotopy class $[i]$ of the constant loop $i(t) \equiv b_0$. What is the inverse of $[f]$? It is the homotopy class $[\bar{f}]$ of the reversed loop $\bar{f}(t) := f(1-t)$, that is, the same loop, but traversed in the opposite direction. See exercises.

Although the base point b_0 cannot be avoided from the definition, finally its choice is not relevant in many cases:

- If X is path connected, then the fundamental groups with respect to different base points are isomorphic, i.e., $\pi_1(X, b_0) \simeq \pi_1(X, b_0')$ for every $b_0, b_0' \in X$.[10] Hence, for path connected spaces the base point can be omitted from the notation, we simply write $\pi_1(X)$.

- If furthermore $\pi_1(X)$ is an Abelian group, then there is a bijection $\pi_1(X) \leftrightarrow [S^1, X]$, because in this case two loops are homotopic as loops if and only if they are homotopic as maps $S^1 \to X$. This is not the case if $\pi_1(X)$ is non-Abelian, see exercises.

The fundamental group π_1 associates a group $\pi_1(X)$ to every topological space X. Moreover, it associates a group homomorphism[11] to every continuous map $f : X \to Y$, defined by

$$\pi_1(f) : \pi_1(X) \to \pi_1(Y) \tag{7.4.6}$$

$$\pi_1(f)([g]) = [f \circ g] \tag{7.4.7}$$

for every element $[g] \in \pi_1(X)$. One can check that $\pi_1(f)$ is well defined and it is really a homomorphism, that is, it preserves the identity element, the inverse and the multiplication.

The next propositions are the modified versions of Proposition 7.3.6 and Lemma 7.3.8, and the same proof works by considering also the base points.

[10] $G \simeq H$ denotes the isomorphism of two groups G and H.

[11] A *homomorphism* between two groups G and H is a map $f : G \to H$ which preserves the product and the inverse: $f(g \cdot h) = f(g) \cdot f(h)$ and $f(g^{-1}) = (f(g))^{-1}$ for every $g, h \in G$. In particular, $f(1_G) = 1_H$. A group homomorphism f is an *isomorphism* if it is a bijection. Two groups are isomorphic is there is an isomorphism between them. Isomorphic groups are 'the same' from the point of view of algebra.

Proposition 7.4.8.
- *If two continuous maps $f, f' : X \to Y$ are homotopic, then they induce the same homomorphism $\pi_1(f) = \pi_1(f') : \pi_1(X) \to \pi_1(Y)$.*
- *If X and Y are homotopy equivalent spaces, then $\pi_1(X) \simeq \pi_1(Y)$.*

The following examples have to be compared with the examples in Section 7.2 and 7.3, in many cases the only difference is the presence of the base point.

Example 7.4.9 (Contractible spaces)**.** Consider $X = \mathbb{R}^n$ or $X = B^n$ or, more generally, any contractible space. Every map $f : Y \to X$ is nullhomotopic, and it holds also for the loops. Hence, $\pi_1(\mathbb{R}^n) = \pi_1(B^n) = 0$, where 0 is a simplified notation for the one element group $\{i\}$ (where i is the identity element usually denoted by 0 or 1). More generally, $\pi_1(X) = 0$ for every contractible space.

A path connected space X is *simply connected* if $\pi_1(X) = 0$, that is, every loop is nullhomotopic. Equivalently, X is simply connected is every map $f : S^1 \to X$ is nullhomotopic, i.e., $[S^1, X] = 0$. As we saw, every contractible space is simply connected, but its reverse is not true, as the next example shows.

Example 7.4.10 (Spheres)**.** Consider the speheres S^n, $n \geq 2$. We saw that every map $f : S^1 \to S^n$ is nullhomotopic, and this holds for loops as well (i.e., considering the base point). Hence, $\pi_1(S^n) = 0$ if $n \geq 2$, that is, the spheres of dimension at least 2 are simply connected. However, they are not contractible, for example, there are not nullhomotopic maps $f : S^n \to S^n$, as we will see later.

Example 7.4.11 (Circle and punctured space)**.** We mentioned (however, we have not yet proved) that

$$[S^1, \mathbb{R}^2 \setminus \{0\}] \leftrightarrow [S^1, S^1] \leftrightarrow \mathbb{Z}. \tag{7.4.12}$$

This holds also with considering the base point, moreover, one can check that the concatenation of loops results in the addition of the corresponding integers, called winding numbers, expressing the number of times a loop bypasses around the hole at the origin. Hence, we have

$$\pi_1(\mathbb{R}^2 \setminus \{0\}) \simeq \pi_1(S^1) \cong \mathbb{Z}. \tag{7.4.13}$$

To illustrate the power of the induced homomorphism, we show precisely the intuitively clear fact that the disc cannot be continuously retracted to its boundary circle. More precisely:

Proposition 7.4.14. *There is no continuous map $f : B^2 \to \partial B^2 = S^1$ such that $f(p) = p$ holds for every $p \in \partial B^2 = S^1$.*

Proof. Assume indirectly that we have such map f. Let $\iota : S^1 \to B^2$ denote the inclusion map of the boundary, that is, $\iota(p) = p$. The condition $f(p) = p$ for every $p \in \partial B^2 = S^1$ can be reformulated as $f \circ \iota : S^1 \to S^1$ is the identity map, i.e., $f \circ \iota = \mathrm{id}_{S^1}$. The induced homomorphisms are

$$\pi_1(\iota) : \pi_1(S^1) = \mathbb{Z} \quad \to \quad 0 = \pi_1(B^2), \tag{7.4.15}$$
$$\pi_1(f) : \pi_1(B^2) = 0 \quad \to \quad \mathbb{Z} = \pi_1(S^1), \tag{7.4.16}$$

Consider the composition

$$\pi_1(f \circ \iota) = \pi_1(f) \circ \pi_1(\iota) : \pi_1(S^1) = \mathbb{Z} \to \mathbb{Z} = \pi_1(S^1). \tag{7.4.17}$$

On the one hand, this homomorphism is the identity map of $\mathbb{Z}$, since $f \circ \iota$ is the identity map of the circle. On the other hand, this homomorphism is the constant zero, since already $\pi_1(\iota)$ is the constant zero homomorphism. This a contradiction proves that there is no such f. □

One can easily show that

Proposition 7.4.18. *The fundamental group of a product is the product of the fundamental groups, i.e.,*

$$\pi_1(X \times Y) = \pi_1(X) \times \pi_1(Y). \tag{7.4.19}$$

Example 7.4.20 (Torus)**.** Therefore,

$$\pi_1(S^1 \times S^1) = \mathbb{Z} \oplus \mathbb{Z} = \mathbb{Z}^2, \tag{7.4.21}$$

where the *direct sum* $\mathbb{Z} \oplus \mathbb{Z} = \mathbb{Z} \times \mathbb{Z}$ is just another notation for the 'product' in case of Abelian groups. Which loops do we have? $(0;0)$ is the identity element, the homotopy class of the constant loop. $(0;1)$ is represented by one of the circles – say, the meridian –, while $(1;0)$ is the other one, the longitude. $(1;1)$ is the concatenation of the meridian and the longitude, but it is homotopic to the circle corresponding to the diagonal of $[0;1] \times [0;1]$ (a Hopf circle). Other pairs of integers $(n;m) \in \mathbb{Z} \oplus \mathbb{Z}$ also have nice representatives called *torus knots*, e.g., the loop representing $(2;3)$ is the trefoil knot, see Figure 21.

Example 7.4.22 (Plane with two holes)**.** Consider $X = \mathbb{R}^2 \setminus \{p, q\}$, where $p \neq q \in \mathbb{R}^2$. What is the fundamental group of X? One may expect it is $\mathbb{Z} \oplus \mathbb{Z}$, like for the torus. $(1;0)$ and $(0;1)$ would be the loops around p and q, respectively, in the positive direction. Then $(1;1)$ would be the concatenation of the two loops. However, their concatenation depends on the order, the two possibilities are not homotopic! See Figure 21.

For the right answer, let us denote by a the loop around p, and by b the loop around q, in the positive direction. We saw that their concatenations are not homotopic, that is, $ab \neq ba$. Hence, $\pi_1(X)$ is not Abelian. Every loop is homotopic to a sequence of concatenations of a, b, a^{-1} and b^{-1}, therefore, they can be labeled by a word built from these four letters. The only simplification rule is that neighboring aa^{-1}, bb^{-1}, $a^{-1}a$ or $b^{-1}b$ pairs of letters can be canceled (or inserted). An example for the multiplication is

$$ab \cdot b^{-1}a^{-1}b = abb^{-1}a^{-1}b = aa^{-1}b = b. \tag{7.4.23}$$

The group of words built by two letters and their inverses is called *free group generated by two elements*, it is denoted by $F\langle a, b\rangle$. Therefore,

$$\pi_1(\mathbb{R}^2 \setminus \{p, q\}) \simeq F\langle a, b\rangle. \tag{7.4.24}$$

7.5 Fundamental group of cell complexes

How can the fundamental group of a cell complex be determined?

Example 7.5.1 (Torus)**.** Both the torus $\mathcal{T}^2 = S^1 \times S^2$ and the space $\mathbb{R}^2 \setminus \{p, q\}$ have two holes, intuitively. Why are their fundamental groups different? What is the relationship between the corresponding groups $F\langle a, b\rangle$ and $\mathbb{Z} \oplus \mathbb{Z}$?

We can try to characterize $\pi_1(\mathcal{T}^2)$ in the same way as for $\mathbb{R}^2 \setminus \{p, q\}$: let us write a for a loop around the meridian, and b around the longitude. Then, every loop in $\mathcal{T}^2$ can be labeled by a word built from the letters a, b, a^{-1} and b^{-1}. However, in this case, $ab = ba$, in fact, the two loops can be deformed to each other along the 2-cell of the torus, see Figure 21. Hence, every word can be rearranged as $a^n b^m$ with $(n, m) \in \mathbb{Z} \oplus \mathbb{Z}$, and the multiplication of the words results in the addition of the exponents.

This shows that if we introduce a new simplification rule $ab = ba$ for the words in the group $F\langle a, b\rangle$, then the resulted group becomes isomorphic to $\mathbb{Z} \oplus \mathbb{Z}$. The new rule, called a relation, is coming from the cell decomposition. The 1-skeleton $\mathrm{sk}_1(\mathcal{T}^2)$ of the torus consists of two circles glued together at one point (a 'figure-8' form), it is actually homotopy equivalent with $\mathbb{R}^2 \setminus \{p, q\}$. Its fundamental group is the free group $F\langle a, b\rangle$. Then, the 2-cell is glued along a map $\partial B^2 = S^1 \to \mathrm{sk}_1(\mathcal{T}^2)$. Up to homotopy, this gluing map is given by the word $aba^{-1}b^{-1}$, that is, the *commutator* of a and b. The gluing of the 2-cell kills the corresponding loop, that is, $aba^{-1}b^{-1}$ becomes nullhomotopic by the gluing of the 2-cell. We write

$$\pi_1(\mathcal{T}^2) \simeq F\langle a, b \mid aba^{-1}b^{-1} = 1\rangle \simeq \mathbb{Z} \oplus \mathbb{Z}. \tag{7.5.2}$$

For the general case, recall that a *free group generated by the elements of a (finite) set* G is the set of words built from the letters $g_i \in G$ and their formal inverses g_i^{-1}. It is denoted by $F\langle G\rangle$.

The only simplification rule is canceling (or inserting) neighboring pairs $g_i g_i^{-1}$ or $g_i^{-1} g_i$. The free group $F\langle G\rangle$ is not Abelian, if G has more than one element (i.e., more than one generators). Let $R \subset F\langle G\rangle$ be an arbitrary subset of words. The group *presented by the set of generators G and relations R* is the group obtained from $F\langle G\rangle$ by introducing the new simplification rules $r_i = 1$ for every words $r_i \in R$. This group is denoted by $F\langle G \mid R\rangle$.

For a precise definition of $F\langle G \mid R\rangle$, observe that if r_i and r_j are declared to be 1, then also their product $r_i \cdot r_j$ has to be 1, the inverses r_i^{-1} and r_j^{-1} has to be 1, and also every conjugation $a r_j a^{-1}$ with arbitrary $a \in F\langle G\rangle$ has to be 1. The subset obtained from R by extending it with every possible products, inverses and conjugates is the *normal subgroup* generated by R, denoted by $N(R)$. Then, $F\langle G \mid R\rangle$ is defined as the quotient group

$$F\langle G \mid R\rangle := \frac{F\langle G\rangle}{N(R)}. \tag{7.5.3}$$

Now we are able to describe the fundamental group of cell complexes. Assume that X is a path connected cell complex.

- If $\dim(X) = 1$, then X is a graph, and it is homotopy equivalent to a *wedge* of circles, that is, some circles glued together at one point: $X = \vee_{i=1}^{k} S^1$. Its fundamental group is the free group generated by k elements, that is, one generator g_i corresponds to every circle:

$$\pi_1(X) = F\langle g_1, \dots, g_k\rangle \tag{7.5.4}$$

- If $\dim(X) = 2$, then the fundamental group of the 1-skeleton is a free group. The gluing map of the 2-cell $E_j^2 = B^2$ is a continuous map

$$\phi_j^2 : \partial B^2 = S^1 \to \mathrm{sk}_1(X). \tag{7.5.5}$$

  Since it is a loop, it represents an element (a word)

$$r_j := [\phi_j^2] \in \pi_1(\mathrm{sk}_1(X)) = F\langle g_1, \dots, g_k\rangle \tag{7.5.6}$$

  Since this loop becomes nullhomotopic along $E_j^2 = B^2$, the word r_j is a relation. It is possible to show that every relation is in this form. Hence, the fundamental group of a 2D cell complex X is presented by the generators corresponding to the circles in the 1-skeleton and the relations corresponding to the gluing maps of the 2-cells, that is,

$$\pi_1(X) = F\langle g_1, \dots, g_k \mid r_1, \dots, r_l\rangle, \tag{7.5.7}$$

  where l is the number of 2-cells.

- If $\dim(X) > 2$, then the gluing of the higher dimensional cells B^n $(n > 2)$ do not modify the fundamental group, that is,

$$\pi_1(X) = \pi_1(\mathrm{sk}_2(X)). \tag{7.5.8}$$

This is, indeed, the same argument as used in Example 7.2.17, also considering the base point: every loop in X is homotopic to a loop in $\mathrm{sk}_1(X)$, and if two loops are homotopic in X, then the homotopy can be realized in $\mathrm{sk}_2(X)$.

Example 7.5.9 (Circle)**.** For S^1, there is one generator a and no relations, hence,

$$\pi_1(S^1) \simeq F\langle a\rangle \simeq \mathbb{Z}. \tag{7.5.10}$$

Indeed, the word $a^n \in F\langle a\rangle$ can be identified with the exponent $n \in \mathbb{Z}$, respecting the multiplication: $a^n a^m = a^{n+m}$.

Example 7.5.11 (Balls and spheres)**.** The 2-disc B^2 is obtained from S^1 by gluing a 2-cell (B^2 itself) along the identity map $\partial B^2 \to S^1$. hence, it gives the relation $a = 1$, killing the generator, results in

$$\pi_1(B^2) \simeq F\langle a \mid a = 1\rangle \simeq 0, \tag{7.5.12}$$

where the 0 in the right side is actually the identity element 1[12].

Every sphere S^n and ball B^n ($n \geq 2$) can be obtained from B^2 by gluing new cells of dimension ≥ 2, therefore, their fundamental group is trivial. One can also build S^n from one 0-cell and one n-cell, which also shows that $\pi_1(S^n) = 0$, since there are no 1-cells (no generators).

Example 7.5.13 (Projective spaces)**.** What is the fundamental group of the real projective plane $\mathbb{R}P^2$? From its usual cell decomposition in Figure 17, we have one 1-cell a, which forms a circle, hence, this is one generator, and there is one 2-cell glued along a map going twice around a. Hence,

$$\pi_1(\mathbb{R}P^2) \simeq F\langle a \mid a^2 = 1\rangle \simeq \mathbb{Z}_2, \tag{7.5.14}$$

indeed, this group has only two distinct elements 1 and a with $a^2 = 1$.

$\mathbb{R}P^n$ can be obtained from $\mathbb{R}P^2$ by gluing cells of dimension higher than 2, hence, $\pi_1(\mathbb{R}P^n) = \mathbb{Z}_2$ for every $n \geq 2$.

$\pi_1(\mathbb{C}P^n) = 0$, since it has a cell decomposition without 1-cells.

Example 7.5.15 (Klein bottle)**.** The usual cell decomposition of $\mathcal{K}$ in Figure 10 has two 1-cells, and both form a circle. Hence, these are the two generators a (corresponding to the simple arrow, it is the 'longitude' of $\mathcal{K}$) and b (corresponding to the double arrow, it is the 'meridian' of $\mathcal{K}$). There is one 2-cell, the corresponding relation is induced by its gluing map: $baba^{-1} = 1$, hence,

$$\pi_1(\mathcal{K}) \simeq F\langle a, b \mid baba^{-1} = 1\rangle \tag{7.5.16}$$

This group is not Abelian, since $ab \neq ba$. However, a^2 and b commute, that is, $a^2 b = ba^2$. Indeed, from the relation, $bab = a$, hence,

$$a^2 b = (bab)ab = ba(bab) = ba^2. \tag{7.5.17}$$

This fact can be understood by 'reduplicating' the Klein bottle, obtaining a torus with longitude a^2, see Figure 21.

7.6 Covering spaces, covering maps

This topic was eventually dropped from the course. However, it helps to create a more complete picture.

Definition 7.6.1. Let X and Y be topological spaces.

- A continuous map $f : X \to Y$ is a *covering map*[13] if every $p \in Y$ has a neighborhood $U \subset Y$ whose preimage $f^{-1}(U)$ is a the disjoint union on subsets $V_i \subset X$ such that $f|_{V_i} : V_i \to U_i$ is a homeomorphism for every j. X is a *covering space* of Y.
- A covering $f : X \to Y$ is a *universal covering* of Y, if X is simply connected, i.e., X is path connected and $\pi_1(X) = 0$.

Some properties, without proof:

Proposition 7.6.2. • *Any covering map $f : X \to Y$ is surjective.*

[12] In Abelian groups the 'multiplication' is usually denoted by $+$, the identity element is denoted by 0, and the inverse of an element a is $-a$. It can be sometimes confusing, when we switch between the multiplicative and additive notation.

[13] Not to be confused with open coverings $X = \bigcup_i U_i$ of topological spaces.

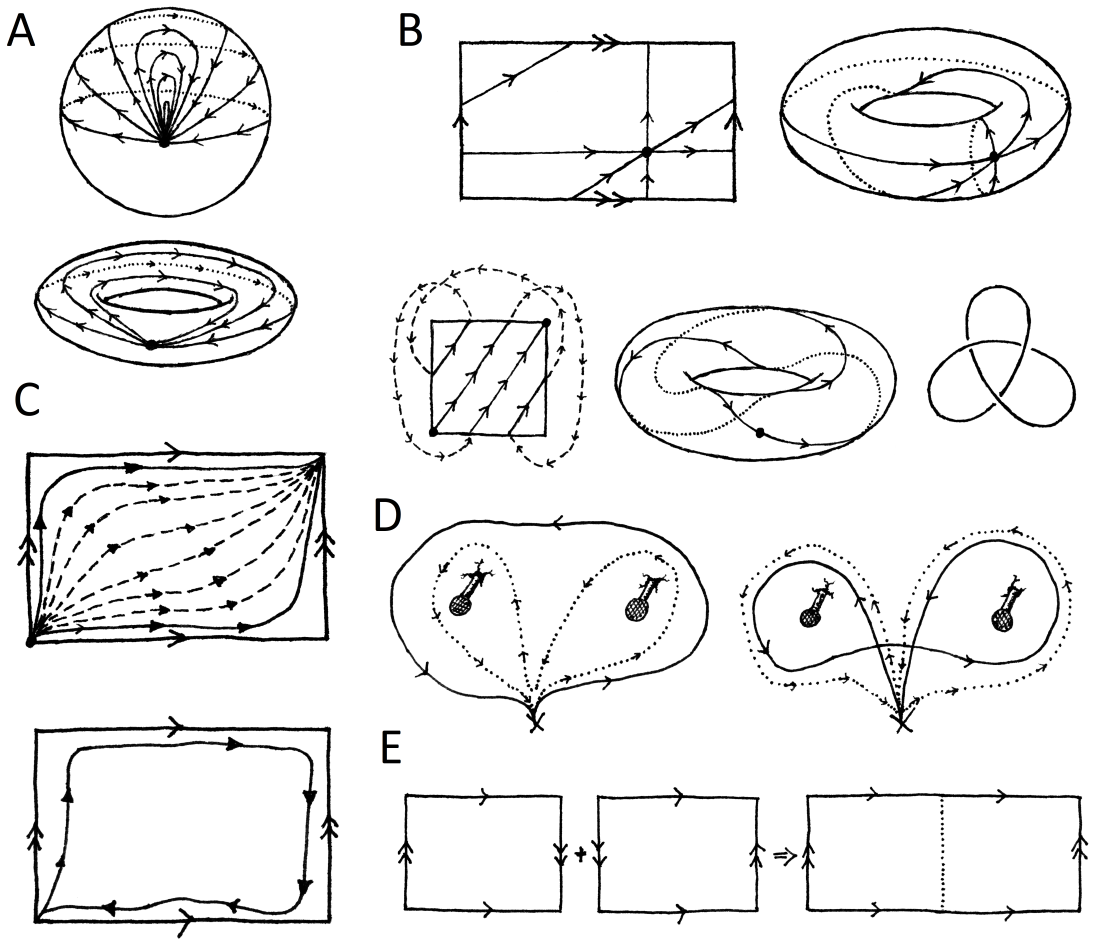


Figure 21: Loops in spaces. (A) In S^2, every loop is nullhomotopic, but in $\mathcal{T}^2$ there are nontrivial loops. (B) The loops in $\mathcal{T}^2$ representing $(0;1)$, $(1;0)$ and $(1;1)$ in $\pi_1(\mathcal{T}^2)$. The trefoil knot represents $(2;3)$. (C) $\pi_1(\mathcal{T}^2)$ is Abelian, because $ab = ba$, in other words, $aba^{-1}b^{-1} = 1$. (D) In $\pi_1(\mathbb{R}^2 \setminus \{p, q\})$, $ab \neq ba$. (E) The double covering $\mathcal{T}^2 \to \mathcal{K}$. Graphics by Mátyás Szabari, Új világok teremtése.

- *If X is path connected, then $f^{-1}(p)$ has the same cardinality for every $p \in X$. This cardinality is called the degree of covering f*[14].
- *Not every space X admits a universal covering $u : \widetilde{X} \to X$, but our spaces does (connected finite cell complexes, manifolds). The universal covering space $\widetilde{X}$ is unique up to homeomorphism.*

Example 7.6.3 (Trivial covering)**.** Every space Y has a covering $f : X \to Y$ with $X = \bigcup_i Y_i$ a disjoint union of copies of Y. This is the trivial covering. This possibility can be avoided by restricting our study to path connected covering spaces X.

Example 7.6.4 (Non-trivial coverings)**.**
- The quotient map $f : \mathbb{R} \to \mathbb{R}/\mathbb{Z} \cong S^1$ is a covering map. Since $\pi_1(\mathbb{R}) = 0$, this is the universal covering of S^1.
- The quotient map $f : \mathbb{R}^2 \to \mathbb{R}^2/\mathbb{Z}^2 \cong \mathcal{T}^2$ is a covering map. Since $\pi_1(\mathbb{R}^2) = 0$, this is the universal covering of the torus $\mathcal{T}^2$. See Figure 11. Similarly, $f : \mathbb{R}^n \to \mathbb{R}^n/\mathbb{Z}^n \cong \mathcal{T}^n$ is the universal covering of the n-torus $\mathcal{T}^n$.
- The quotient map $f : S^1 \mapsto \mathbb{R}P^1 \cong S^1$ is a covering of S^1 with itself, but it is not the universal covering, since $\pi_1(S^1) \neq 0$. In this case, $f(z) = z^2$ for $z \in S^1 \in \mathbb{C}$, it is a double cover (the degree is 2). We can also give an n-fold cover $f : S^1 \to S^1$, $f(z) = z^n$.
- The quotient map $f : S^n \to \mathbb{R}P^n$ is a covering map. Since $\pi_1(S^n) = 0$ for $n \geq 2$, then it is the universal covering of the real projective space $\mathbb{R}P^n$.
- In particular, the quotient map $f : SU(2) \to SO(3)$ is the universal covering of $SO(3)$. This fact has many consequences in particle physics, but also in engineering. For example, it explains the *Gimbal lock* occurring if the 3D rotations are parametrized by the 3 angles: any map $\mathcal{T}^3 \to SO(3)$ is not a covering map. See Wikipedia.

Observe that in every examples of universal covering $u : \widetilde{X} \to X$, the fundamental group $\pi_1(X)$ acts on $\widetilde{X}$, and X is homeomorphic to the space of orbits of the action, i.e.,

$$X \cong \frac{\widetilde{X}}{\pi_1(X)}. \tag{7.6.5}$$

For example, $\mathcal{T}^2 \cong \mathbb{R}^2/Z^2$, and $\pi_1(\mathcal{T}^2) \simeq \mathbb{Z}^2$; or $\mathbb{R}P^2 \cong S^2/\mathbb{Z}_2$, and $\pi_1(\mathbb{R}P^2) \simeq \mathbb{Z}_2$. This is the case in general, as the following theorem shows (without proof).

Theorem 7.6.6. *Let $u : \widetilde{X} \to X$ be the universal covering of X. Let $b_0 \in X$ be an arbitrary base point. Then,*

- *Given an arbitrary point $\widetilde{b}_0 \in \widetilde{X}$ with $u(\widetilde{b}_0) = b_0$, any path $f : [0;1] \to X$, $f(0) = b_0$ has a unique lift $\widetilde{f} : [0;1] \to \widetilde{X}$, with $\widetilde{f}(0) = \widetilde{b}_0$ and $f = u \circ \widetilde{f}$.*
- *If $f[0;1] \to X$ is a loop based at b_0, then its unique lift $\widetilde{f}$ is a loop in $\widetilde{X}$ (based at $\widetilde{b}_0$) if and only if f is nullhomotopic, that is, $[f] = 1 \in \pi_1(X)$ is the identity element. (Otherwise the end point $\widetilde{f}(1)$ is an element of $\widetilde{X}$ with $u(\widetilde{f}(1)) = b_0$.)*
- *If f is a loop based at b_0, then the endpoint $\widetilde{f}(1)$ of its unique lift $\widetilde{f}$ depends on the class $[f] \in \pi_1(X)$. More precisely, given two loops f and g based at b_0, then for their lifts $\widetilde{f}(1) = \widetilde{g}(1)$ if and only if $[f] = [g] \in \pi_1(X)$.*
- *Therefore, there is a bijection*
$$\pi_1(X) \leftrightarrow u^{-1}(b_0) \tag{7.6.7}$$
defined by $[f] \mapsto \widetilde{f}(1)$.

[14]Not to be confused with the degree, which will be defined for arbitrary smooth maps between manifolds of the same dimension.

- *Let the action of $\pi_1(X)$ on $\widetilde{X}$ be defined by the deck transformations: for a point $p \in \widetilde{X}$ and an element $[f] \in X$, consider f as a loop based at $u(p)$, and let $[f] \cdot p$ be defined as the end point $\widetilde{f}(1)$ of the unique lifting $\widetilde{f}$. Then we have the homeomorphism*

$$X \cong \frac{\widetilde{X}}{\pi_1(X)}. \tag{7.6.8}$$

Moreover, we have the following lifting property.

Theorem 7.6.9. *Let $u : \widetilde{Y} \to Y$ be the universal covering of Y, and let $f : X \to Y$ be a continuous map. If X is simply connected, then f can be lifted, that is, there is a map $\widetilde{f} : X \to \widetilde{Y}$ such that $f = u \circ \widetilde{f}$.*

Example 7.6.10 (Maps to the torus)**.** Let $Y = \mathcal{T}^n$ be the n-torus (in particular, for $n = 1$ it is S^1), and let X be any simply connected space, for example, a sphere S^m ($m \geq 2$). Then, $[X, Y] = 0$. In fact, every map $f : X \to Y$ can be lifted to a map $\widetilde{f} : X \to \mathbb{R}^n$, which is nullhomotopic.

Example 7.6.11 (Maps to the real projective spaces)**.** If X is simply connected, then

$$[X, \mathbb{R}P^n] \leftrightarrow [X, S^n]. \tag{7.6.12}$$

In fact, every map $f : X \to \mathbb{R}P^n$ can be lifted to a map $\widetilde{f} : X \to S^n$, and also a homotopy between two maps can be lifted.

7.7 Higher homotopy groups

This topic was eventually dropped from the course. However, it helps to create a more complete picture.

The 1st homotopy group is the fundamental group $\pi_1(X)$. The higher homopopy groups $\pi_n(X)$ of a topological space X can be defined similarly. Instead of loops we consider *spheroids* based at $b_0 \in X$. A spheroid can be considered as either a continuous map

$$f : S^n \to X \tag{7.7.1}$$

such that $f(p_0) = b_0$ holds for the base points, or a continuous map

$$f : B^n \to X \tag{7.7.2}$$

such that $f(\partial B^n) = b_0$. The two approaches are equivalent via the quotient $B^n/\partial B^n \cong S^n$. Two spheroids are homotopic if there is a homotopy H between them such that H_t is a spheroid based at b_0 for every $t \in [0; 1]$.

The elements of $\pi_n(X)$ are the homotopy classes of the spheroids based at b_0. The concatenation of two spheroids can be defined similarly as for loops (how?). Then $\pi_n(X)$ is a group, it is the n-th homotopy group of X. Moreover, $\pi_n(X)$ is always an Abelian group, if $n \geq 2$. Similarly to the $n = 1$ case, $\pi_n(X)$ does not depend on the base point if X is path connected.

$\pi_0(X)$ is not a group (in general). It is the set of path connected components of X. Indeed, $\pi_0(X)$ is the set of homotopy classes of maps (0-spheroids) $f : S^0 = \{-1; 1\} \to X$ with $f(-1) = b_0$. Two such maps f and g are homotopic if and only if $f(1)$ can be joined with $g(1)$ by a path in X.

Example 7.7.3 (Spheres)**.** $\pi_n(S^n) \simeq \mathbb{Z}$, that is, the same as the set of homotopy classes $[S^n, S^n]$.

$\pi_m(S^n) = 0$ for $m < n$.

One might expect that $\pi_m(S^n) = 0$ holds also for $m > n$, but it is not true. For example, $\pi_3(S^2) \simeq \mathbb{Z}$, it is generated by the homotopy class $[h]$ of the Hopf map $h : S^3 \to S^2$.

Example 7.7.4 (Projective spaces)**.** By the lifting property, $\pi_m(\mathbb{R}P^n) \simeq \pi_m(S^n)$ for $m \geq 2$.

Similarly for the fundamental group, any continuous map $f : X \to Y$ induces a homomorphism $\pi_n(f) : \pi_n(X) \to \pi_n(Y)$ via the composition of the spheroids with f. Homotopic maps induce the same homomorphism. If X and Y are homotopy equivalent, then $\pi_n(X) \simeq \pi_n(Y)$.

One may expect that the homotopy groups $\pi_n(X)$ together characterise X up to homotopy equivalence. However, it holds only in a restricted version.

Theorem 7.7.5 (Whitehead)**.** *Let X and Y be cell complexes. If for a continuous map $f : X \to Y$ the induced homomorphisms $\pi_n(f) : \pi_n(X) \to \pi_n(Y)$ are isomorphisms for every n, then f is a homotopy equivalence, in particular, X and Y are homotopy equivalent spaces.*

In the special case if $\pi_n(X) = 0$ for all n, then X is contractible. Indeed, in this case the constant map $X \to \{\text{point}\}$ induces isomorphisms for every n, hence it is a homotopy equivalence by Whitehead's theorem.

7.8 Exercises

Exercise 7.8.1 (Homotopy equivalence and homeomorphism)**.** (a) Classify the topological spaces in Figure **??** up to homotopy equivalence and up to homeomorphism!

(b) Extend the classes with the following topological spaces: closed disc B^2, closed interval B^1, closed cylinder $S^1 \times B^1$, torus $\mathcal{T}^2 = S^1 \times S^1$, solid torus $S^1 \times B^2$, point B^0, circle S^1, Möbius band $\mathcal{M}$.

(c) Determine the fundamental group of each class!

FOLYTATNI!

8 Physical examples

8.1 Eigenstates of periodic crystals

The properties of quasiparticle excitations (electrons, phonons, magnons or superconducting excitations) are most commonly investigated in periodic crystals in solid-state physics. The electronic excitations correspond to eigenstates of the single-particle Hamiltonian $H = \frac{\boldsymbol{p}^2}{2m} + V$, where the potential is periodic: $V(\boldsymbol{r}) = V(\boldsymbol{r} + \boldsymbol{R}) \, \forall \boldsymbol{R} = \sum_{i=1}^{d} n_i \boldsymbol{a}_i$, where the $\boldsymbol{a}_i$ are basis vectors of the lattice in d-dimensional space and $n_i \in \mathbb{Z}$.

Finding the eigenstates corresponds to diagonalizing the Hamiltonian H. While the potential is already diagonal in the coordinate representation given above, the momentum is not; it corresponds to $\boldsymbol{p} = -\mathrm{i}\hbar\boldsymbol{\nabla}$. More information can be gained by introducing the Fourier basis $\Psi_{\boldsymbol{k}}(\boldsymbol{r}) = \mathrm{e}^{\mathrm{i}\boldsymbol{k}\boldsymbol{r}}$ for $\boldsymbol{k} \in \mathbb{R}^d$. These functions are eigenstates of the momentum operator, $\boldsymbol{p}\Psi_{\boldsymbol{k}}(\boldsymbol{r}) = \hbar\boldsymbol{k}\Psi_{\boldsymbol{k}}(\boldsymbol{r})$. We express the matrix elements of the Hamiltonian in this basis,

$$\begin{aligned}\langle \Psi_{\boldsymbol{k}} | H | \Psi_{\boldsymbol{k}'} \rangle &= \int \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\boldsymbol{r}} \left[\frac{\boldsymbol{p}^2}{2m} + V(\boldsymbol{r}) \right] \mathrm{e}^{\mathrm{i}\boldsymbol{k}'\boldsymbol{r}} \mathrm{d}^3\boldsymbol{r} = \frac{\hbar^2\boldsymbol{k}^2}{2m} \int \mathrm{e}^{-\mathrm{i}(\boldsymbol{k}-\boldsymbol{k}')\boldsymbol{r}} \mathrm{d}^3\boldsymbol{r} + \int \mathrm{e}^{-\mathrm{i}(\boldsymbol{k}-\boldsymbol{k}')\boldsymbol{r}} V(\boldsymbol{r}) \, \mathrm{d}^3\boldsymbol{r} \\ &= \frac{\hbar^2\boldsymbol{k}^2}{2m} \mathcal{V} \delta_{\boldsymbol{k},\boldsymbol{k}'} + \sum_n \int_{\mathrm{UC}} \mathrm{e}^{-\mathrm{i}(\boldsymbol{k}-\boldsymbol{k}')(\boldsymbol{r}+\boldsymbol{R}_n)} V(\boldsymbol{r} + \boldsymbol{R}_n) \, \mathrm{d}^3\boldsymbol{r}, \end{aligned} \tag{8.1.1}$$

where $\mathcal{V}$ is the volume of the solid used as a normalization constant, where we used the orthogonality of the momentum eigenstates. In the second part, we partitioned the integral over the whole volume into an integral over a unit cell, and a summation over the unit cells. Using the periodicity $V(\boldsymbol{r} + \boldsymbol{R}_n) = V(\boldsymbol{r})$, the second term may be further transformed as

$$\frac{\hbar^2\boldsymbol{k}^2}{2m} \mathcal{V} \delta_{\boldsymbol{k},\boldsymbol{k}'} + \sum_n \mathrm{e}^{-\mathrm{i}(\boldsymbol{k}-\boldsymbol{k}')\boldsymbol{R}_n} \int_{\mathrm{UC}} \mathrm{e}^{-\mathrm{i}(\boldsymbol{k}-\boldsymbol{k}')\boldsymbol{r}} V(\boldsymbol{r}) \, \mathrm{d}^3\boldsymbol{r} = \frac{\hbar^2\boldsymbol{k}^2}{2m} \mathcal{V} \delta_{\boldsymbol{k},\boldsymbol{k}'} + N \delta_{\boldsymbol{k}-\boldsymbol{k}',\boldsymbol{G}_m} V_{\boldsymbol{G}_m}, \tag{8.1.2}$$

where N is the number of unit cells and $\boldsymbol{G}_m = \sum_{i=1}^{d} m_i \boldsymbol{b}_i$ is a so-called reciprocal-lattice vector, where the $\boldsymbol{b}_i$ are the basis vectors and $m_i \in \mathbb{Z}$. The reciprocal-lattice vectors satisfy $\boldsymbol{G}_m \boldsymbol{R}_n = 2\pi \sum_{i=1}^{d} m_i n_i$, and as such $\mathrm{e}^{-\mathrm{i}\boldsymbol{G}_m \boldsymbol{R}_n} = 1$.

In summary, this means that in the Fourier basis the Hamiltonian assumes a block-diagonal form: the kinetic energy is diagonal in $\boldsymbol{k}$, while the potential energy only has matrix elements between wave vectors differing by any reciprocal-lattice vector $\boldsymbol{G}_m$, and these matrix elements are the Fourier series coefficients $V_{\boldsymbol{G}_m}$ of the periodic potential. Due to the block-diagonal shape, the eigenvectors of the Hamiltonian may be obtained via linear combination from a single block,

$$\psi_{n\boldsymbol{k}}(\boldsymbol{r}) = \sum_m \left[u_{n\boldsymbol{k},\boldsymbol{G}_m} \mathrm{e}^{\mathrm{i}\boldsymbol{G}_m \boldsymbol{r}} \right] \mathrm{e}^{\mathrm{i}\boldsymbol{k}\boldsymbol{r}}, \tag{8.1.3}$$

where the $u_{n\boldsymbol{k},\boldsymbol{G}_m}$ are complex coefficients. Performing the summation over m results in

$$\psi_{n\boldsymbol{k}}(\boldsymbol{r}) = u_{n\boldsymbol{k}}(\boldsymbol{r}) \mathrm{e}^{\mathrm{i}\boldsymbol{k}\boldsymbol{r}}, \tag{8.1.4}$$

where $u_{n\boldsymbol{k}}(\boldsymbol{r}) = u_{n\boldsymbol{k}}(\boldsymbol{r} + \boldsymbol{R}_n)$. This is known as Bloch's theorem: any eigenstate of a lattice-periodic Hamiltonian may be written as a product of a lattice-periodic function with a momentum eigenstate.

From a topological point of view, we are interested in the possible values of $\boldsymbol{k}$ in this expression. Originally, $\boldsymbol{k} \in \mathbb{R}^d$ can be an arbitrary vector. Since we separated the reciprocal-lattice vector, now $\boldsymbol{k}$ only takes values inside a single unit cell in reciprocal space, and we identified $\boldsymbol{k}'$ with $\boldsymbol{k}$ if $\boldsymbol{k} - \boldsymbol{k}' = \boldsymbol{G}_m$. Taking the quotient topology according to this relation results in T^d, the d-dimensional torus. This is most simply visualized by drawing the parallelepiped which has the $\boldsymbol{b}_i$ as its edges, and gluing the opposite edges and faces together. In solid-state physics, because of symmetry reasons it is more common to define the unit cell via Voronoi tessellation of the reciprocal lattice, resulting in the so-called Brillouin zone. Although the Brillouin zone is typically geometrically not a parallelepiped, it is topologically still equivalent to a torus.

8.2 Topological insulator in one dimension

Polyacetylene is a hydrocarbon consisting of a single chain of carbon atoms. The bonds between neighbouring carbon atoms are alternatingly strong and weak. Su, Schrieffer and Heeger proposed a model Hamiltonian for describing the eigenstates of this molecule,

$$H = \sum_{j=1}^{N} v^* |j, B\rangle \langle j, A| + v |j, A\rangle \langle j, B| + w |j, B\rangle \langle j+1, A| + w^* |j+1, A\rangle \langle j, B|, \tag{8.2.1}$$

where A and B denote two different atoms in the unit cell, v is the hopping parameter between atoms inside the same unit cell, and w is the hopping parameter between atoms in neighbouring unit cells. The difference between v and w makes it possible to distinguish between strong and weak bonds. Compared to the previous section, instead of coordinate eigenstates we used a localized basis $|j, A/B\rangle$ describing wave functions localized at single atoms which are orthogonal to each other. Expressing the Hamiltonian in a countable set of such eigenstates constitutes the tight-binding approximation. We use periodic boundary conditions, meaning that $j = N + 1$ denotes $j = 1$ at the other end of the chain. This makes the Hamiltonian invariant under translation by a unit cell.

Similarly to the previous section, it is useful to introduce the Fourier basis,

$$|k, A/B\rangle = \frac{1}{\sqrt{N}} \sum_j \mathrm{e}^{\mathrm{i}kj} |j, A/B\rangle. \tag{8.2.2}$$

Note that it is sufficient to consider $k \in [0, 2\pi[= S^1$ here, since shifting k by a reciprocal-lattice vector $m2\pi$ results in the same wave function due to the considered atomic basis. The matrix elements of the Hamiltonian in this basis read

$$\langle k, A | H | k', B\rangle = \frac{1}{N} \sum_{j,j_1,j_2} \left(v \mathrm{e}^{-\mathrm{i}kj_1} \mathrm{e}^{\mathrm{i}k'j_2} \delta_{j,j_1} \delta_{j,j_2} + w^* \mathrm{e}^{-\mathrm{i}kj_1} \mathrm{e}^{\mathrm{i}k'j_2} \delta_{j+1,j_1} \delta_{j,j_2} \right) = \frac{1}{N} \sum_j \mathrm{e}^{-\mathrm{i}(k-k')j} \left(v + w^* \mathrm{e}^{-\mathrm{i}k} \right) = v + w^* \mathrm{e}^{-\mathrm{i}k}. \tag{8.2.3}$$

By calculating the other matrix elements, the Hamiltonian in Fourier space may be expressed as

$$H_{\boldsymbol{k}} = \begin{bmatrix} 0 & v + w^* \mathrm{e}^{-\mathrm{i}k} \\ v^* + w \mathrm{e}^{\mathrm{i}k} & 0 \end{bmatrix} = d_x(k)\,\sigma^x + d_y(k)\,\sigma^y, \tag{8.2.4}$$

with

$$d_x(k) = |v| \cos\varphi_v + |w| \cos(k + \varphi_w), \tag{8.2.5}$$
$$d_y(k) = |v| \sin\varphi_v + |w| \sin(k + \varphi_w). \tag{8.2.6}$$

As discussed earlier, the eigenvalues of this Hamiltonian are $\varepsilon_\pm(k) = \pm\|\boldsymbol{d}(k)\|$. We will consider Hamiltonians with $\|\boldsymbol{d}(k)\| \neq 0\ \forall k$. This is satisfied for $|w| \neq |v|$. We will call these insulators. Assume that the number of electrons corresponds to the number of unit cells N. Each electron occupies a single-particle state, starting from the lowest-energy state, and each state may contain a maximum of one electron. This way, all states with negative eigenvalues will be occupied, so exciting the system will require moving an electron from a negative-energy state to a positive-energy state, which always costs a finite energy regardless of the chain length, like in an insulator. A metal is found in this description for $\|\boldsymbol{d}(k)\| = 0$, where the highest occupied and the lowest unoccupied states have the same energy.

This means that the parameter space for $\boldsymbol{d}$ is $\mathbb{R}^2 \setminus \{\boldsymbol{0}\}$, which is homotopy equivalent to S^1. The graph of $\boldsymbol{d}(k)$ on this space is a circle. We can distinguish between two types of homotopically inequivalent graphs: one that does not wind around the origin (nullhomotopic) for $|w| < |v|$, and one that winds around the origin exactly once for $|w| > |v|$. The difference between these two classes can be observed when looking at the spectrum of the Hamiltonian for free boundary conditions where Bloch's theorem does not apply. For $w = 0$, there are N states with eigenvalue $|v|$ and N states with eigenvalue $-|v|$, just like for the periodic boundary conditions discussed above. For $v = 0$, we find $N - 1$ states with eigenvalue $|w|$ and $N - 1$ states with eigenvalue $-|w|$, and 2 eigenstates with 0 eigenvalue. Changing the parameters but preserving the winding number of $\boldsymbol{d}(k)$ does not change the number of eigenstates with 0 energy. For this reason, the case with $|w| > |v|$ is called a topological insulator, which contains 2 topological states for free boundary conditions inside the gap of the bulk insulator. Looking at the wave functions, it can be seen that these states are exponentially localized at the two ends of the chain, since they cannot exist inside the bulk. The connection between the presence of the end states and the topology of the band structure is called bulk-boundary correspondence.

Note that the Fourier-space Hamiltonian H_k only contains terms proportional to σ^x and σ^y, not to σ^0 and σ^z. This is an example of the so-called chiral symmetry, where $\exists U$ unitary operator such that $UHU^{-1} = -H$. Here, this unitary operator is represented by σ^z after Fourier transformation. Due to this symmetry, the eigenvalues come in $\pm\varepsilon_k$ pairs.

The Hamiltonian may be extended by further terms such as

$$\Delta H_1 = \sum_{j=1}^{N} \varepsilon\left(|j,A\rangle\langle j,A| + |j,B\rangle\langle j,B|\right) + t\left(|j,A\rangle\langle j+1,A| + |j,B\rangle\langle j+1,B| + \text{h.c.}\right). \tag{8.2.7}$$

This results in a term proportional to σ^0 in H_k. This term breaks the chiral symmetry mentioned above, but does not change the topological classification. However, for free boundary conditions the end states move away from zero energy. Introducing a term proportional to σ^z, for example by considering different on-site energies, would change the space of $\boldsymbol{d}$ vectors to $\mathbb{R}^3 \setminus \{\boldsymbol{0}\}$, where the fundamental group consists of only a single element.

Adding hoppings between further neighbours to the Hamiltonian, such as

$$\Delta H_2 = \sum_{j=1}^{N} u\,|j,B\rangle\langle j+2,A| + u^*\,|j+2,A\rangle\langle j,B|, \tag{8.2.8}$$

does not change the topology of the parameter space. The resulting graph of $\boldsymbol{d}(k)$ may wind multiple times around the origin. For free boundary conditions, this will lead to an increase in the number of zero-energy modes proportional to the winding number.

8.3 Exchange statistics of indistinguishable particles

Consider a quantum-mechanical system describing a single particle on a Hilbert space $\mathcal{H}$. For the mathematical description of the particle, we use the elements of the Hilbert space $\psi \in \mathcal{H}$. In order to perform calculations, we use a representation of these elements. In coordinate representation, ψ is described by a function $\psi(\boldsymbol{r}) : \mathbb{R}^d \to \mathbb{C}$, where d is the spatial dimension of the system. In coordinate representation, the momentum operator is described by the differential operator $-\mathrm{i}\hbar\boldsymbol{\nabla}$ acting on the wave function $\psi(\boldsymbol{r})$. In order to be able to apply the momentum operator to the elements of the Hilbert space, we only consider wave functions in the domain of this operator, i.e., everywhere differentiable wave functions, which means that they are also continuous. These wave functions may also be defined to be single-valued actual functions.

Consider two particles, which are described by the wave function $\psi(\boldsymbol{r}_1, \boldsymbol{r}_2) : \mathbb{R}^{2d} \to \mathbb{C}$. However, there is a concept in quantum mechanics which states that elementary particles are indistinguishable; i.e., one particle being at position $\boldsymbol{r}_1$ and the other particle at position $\boldsymbol{r}_2$ should be physically equivalent to one particle being at $\boldsymbol{r}_2$ and the other at $\boldsymbol{r}_1$. This means that the points $(\boldsymbol{r}_1, \boldsymbol{r}_2)$ and $(\boldsymbol{r}_2, \boldsymbol{r}_1)$ must be identified in the space $\mathbb{R}^{2d}$ in order to obtain the actual configuration space of two indistinguishable particles. Furthermore, we will leave out those points in set $D = \mathbb{R}^d$ from the configuration space where the two positions coincide. This quotient results in

$$\frac{\mathbb{R}^{2d} \setminus D}{(\boldsymbol{r}_1, \boldsymbol{r}_2) = (\boldsymbol{r}_2, \boldsymbol{r}_1)} = \mathbb{R}^d \times \mathbb{R}^+ \times \mathbb{R}P^{d-1}. \tag{8.3.1}$$

This can be understood the following way. Instead of the coordinates of the two particles, it is advisable to use the centre of mass $(\boldsymbol{r}_1 + \boldsymbol{r}_2)/2$ and the relative position $\boldsymbol{r}_1 - \boldsymbol{r}_2$. The centre of mass gives $\mathbb{R}^d$ since it is unaffected by the equivalence relation. The relative coordinate is described by its length $|\boldsymbol{r}_1 - \boldsymbol{r}_2| \in \mathbb{R}^+$, and the normalized relative coordinate with the equivalence relation above gives the real projective space $\mathbb{R}P^{d-1}$.

On this configuration space, we want to preserve the continuity of the wave functions. It turns out that this is generally incompatible with defining single-valued functions on the configuration space. This is due to the non-trivial topology of the configuration space, which is homotopy equivalent to $\mathbb{R}P^{d-1}$. Along a closed curve, the wave function may obtain a phase factor $\mathrm{e}^{\mathrm{i}\alpha}$; note that closed curves on the configuration space include going from $(\boldsymbol{r}_1, \boldsymbol{r}_2)$ to $(\boldsymbol{r}_2, \boldsymbol{r}_1)$ in $\mathbb{R}^{2d}$. The best we can do while preserving continuity is to make α depend only on the homotopy class of the curve taken in the configuration space, so continuously deforming the curve does not change the phase. Note that this phase factor does not make $(\boldsymbol{r}_1, \boldsymbol{r}_2)$ and $(\boldsymbol{r}_2, \boldsymbol{r}_1)$ physically inequivalent, since the physical properties are encoded in the pure state $|\psi\rangle\langle\psi|$ constructed from the wave function which does not depend on the phase.

The fundamental groups of the configuration spaces are

$$\pi_1\left(\mathbb{R}P^1\right) = \mathbb{Z}, \tag{8.3.2}$$

$$\pi_1\left(\mathbb{R}P^n\right) = \mathbb{Z}_2 \text{ for } n \geq 2. \tag{8.3.3}$$

Since we assigned the phase factors to the homotopy classes, the mapping from the fundamental group to the phases $\pi_1 \to U(1)$ must be a group homomorphism for consistency. Such homomorphisms are called (unitary) representations of the group. Denote moving from $(\boldsymbol{r}_1, \boldsymbol{r}_2)$ to $(\boldsymbol{r}_2, \boldsymbol{r}_1)$ in $\mathbb{R}^{2d}$ by $\sigma \in \pi_1$. Due to the relation $\sigma^2 = e$, the group $\mathbb{Z}_2$ has only two unequivalent unitary representations, $\alpha(\sigma) = 0$ and $\alpha(\sigma) = \pi$. In contrast, for the group $\mathbb{Z}$ $\alpha(\sigma)$ may take any value in $[0, 2\pi[$. This may be imagined by considering the positions of the two particles in $\mathbb{R}^d$ and identifying σ with their exchange in a counterclockwise manner. In two dimensions, this may be repeated arbitrary many times, and the wave function always picks up a phase factor of $\mathrm{e}^{\mathrm{i}\alpha}$ during this exchange. However, in three or more dimensions the curves may be rotated out from the plane, meaning that exchanging the positions of the atoms twice is always homotopy equivalent to not changing the positions at all.

This procedure may be generalized to N indistinguishable particles. The original space $\mathbb{R}^{Nd}$ must be factorized by any permutation of the positions of the particles. The obtained configuration

space S typically does not have a simple geometric interpretation, but it may be constructed as a cell complex. For $d > 2$, we obtain $\pi_1(S) = S_N$, the permutation group. This is easy to understand, since to any permutation of the positions in $\mathbb{R}^d$ we assign a homotopy class in the fundamental group. For $d = 2$, it again matters whether we exchange two particles in a clockwise or counterclockwise manner, and the fundamental group is $\pi_1(S) = B_N$, the braid group. These groups can be understood by introducing the generators $\sigma_1, \ldots, \sigma_{N-1}$, where σ_k exchanges the positions of particles k and $k+1$ in a clockwise direction. These generators satisfy

$$\sigma_k \sigma_{k+1} \sigma_k = \sigma_{k+1} \sigma_k \sigma_{k+1}, \tag{8.3.4}$$

$$\sigma_k \sigma_l = \sigma_l \sigma_k \text{ for } |k-l| \geq 2, \tag{8.3.5}$$

$$\sigma_k^{-1} = \sigma_k \tag{8.3.6}$$

for the permutation group, which results in $N!$ elements. Only the first two conditions are satisfied for the braid group, which results in infinite elements. The braid group may be imagined by considering the positions of the particles in $\mathbb{R}^2$, and plotting their path along the third spatial direction as the particles are exchanged. These paths are braided with each other.

The unitary representations of the permutation group and the braid group are the same as those for $N = 2$ listed above, regardless of the number of particles. From $\sigma_k \sigma_{k+1} \sigma_k = \sigma_{k+1} \sigma_k \sigma_{k+1}$ follows $\alpha(\sigma_k) = \alpha(\sigma)$, which can also be deduced from the particles being indistinguishable. It turns out that there is a close relationship between the physical properties of the particles, particularly their dynamics, and the representation chosen for their description. Particles described by $\alpha(\sigma) = 0$ are bosons, while those described by $\alpha(\sigma) = \pi$ are fermions. In two spatial dimensions, further exotic particles can be found which are known as anyons. While anyons are not elementary particles, certain excitations in solids behave like anyons. Indications for this exotic behaviour has been observed in two-dimensional electron gases, and certain phenomena in the fractional quantum Hall effect may be explained based on anyons.

9 Manifolds

'n dimensional beings live in n dimensional manifolds' – as my topology professor András Szűcs said.

Manifolds are topological spaces locally look like $\mathbb{R}^n$. While on topological spaces, in general, only the continuity of maps is meaningful, for manifolds the differential calculus can be adapted from $\mathbb{R}^n$.

9.1 Basic definitions and examples

Definition 9.1.1. A topological space M is a *topological manifold* if every point $p \in M$ has an open neighborhood $U \subset M$ which is homoemorphic to an open subset $V \subset \mathbb{R}^n$.[15]

A homoemorphism $\phi : U \to V$ is called a *local chart*.

Remark 9.1.2.

- We obtain an equivalent definition of topological manifolds if we additionally assume that U (hence, V) is homoemorphic to an open ball $\check{B}^n \cong \mathbb{R}^n$. Indeed, the open set $V \subset \mathbb{R}^n$ has a subset homeomorphic to an open ball containing $\phi(p)$.

- ϕ can be considered as a local coordinate system. Indeed, U inherits the coordinates from $V \subset \mathbb{R}^n$ via ϕ. More precisely, a coordinate function $x_i : \mathbb{R}^n \to \mathbb{R}$ induces a coordinate function $x_i \circ \phi : U \to \mathbb{R}$ on $U \subset M$.

- It can be shown that if a topological manifold M is connected, then the dimension n of $\mathbb{R}^n$ in the definition is the same for every point $p \in M$. It is the *dimension* of the manifold M.

[15] This definition holds for many topological spaces, which we do not want to call manifolds, see e.g. Exercise 4.8.16. To avoid them, we extend the definition of topological manifolds with two additional conditions: (1) M is a *Hausdorff* topological space, that is, every pair of points $p, q \in M$ admits disjoint open neighborhoods (U_p of p and U_q of q whith $U_p \cap U_q = \emptyset$); (2) the topology of M has a *countable basis*, i.e., countably many open sets $U_1, U_2, \cdots \subset M$ such that every open set of M can be written as a union of some of them.

Hence, in contrast of the general topological spaces, connected topological manifolds have a well defined dimension. If we want to emphasize that $\dim(M) = n$, we shortly write M^n.

Definition 9.1.3. Let M be a topological manifold, and let $\phi_1 : U_1 \to V_1$, $\phi_2 : U_2 \to V_2$ be two local charts. The *transition map* $\psi_{1,2}$ is defined as

$$\begin{aligned} \psi_{1,2} : V_1 \supset \phi_1(U_1 \cap U_2) &\to \phi_2(U_1 \cap U_2) \subset V_2 && (9.1.4) \\ \psi_{1,2} &= \phi_2 \circ \phi_1^{-1} && (9.1.5) \end{aligned}$$

Importantly, the transition maps are maps between open subsets of $\mathbb{R}^n$. Therefore, the differentiability of the transition maps is meaningful. Recall that a map $f : \mathbb{R}^n \to \mathbb{R}^k$ is *smooth* if it is arbitrary many times differentiable. See more details in Section 9.3.

Definition 9.1.6. A *smooth atlas* on a topological manifold M is a collection of local charts (U_i, ϕ_i) such that

- $\bigcup_i U_i = M$ holds for the open subsets $U_i \subset M$, and
- for the local charts $\phi_i : U_i \to V_i \subset \mathbb{R}^n$ (homeomorphisms!), every transition map $\psi_{i,j}$ is smooth at every point of its domain.

A *smooth manifold* is a topological manifold M endowed with a smooth atlas.

Example 9.1.7 (Manifolds)**.** Most of our examples for topological spaces are also topological, moreover, smooth manifolds. On $\mathbb{R}^n \cong \breve{B}^n$, the identity map is local chart, which forms a smooth atlas with only one chart.

On S^n, the stereographic projection $\phi : U = S^n \setminus \{p\} \to \mathbb{R}^n$ from any point $p \in S^n$ is a local chart. Choosing two opposite points, the corresponding two charts form a smooth atlas, since the transition maps are smooth, see exercises.

$\mathbb{R}P^n$ can be covered by an atlas consists of $n+1$ local charts coming from the homogeneous coordinates. Recall that the points can be written as $p = [x_1 : x_2 : \cdots : x_{n+1}] \in \mathbb{R}P^n$, where $(x_1, \ldots, x_{n+1}) \neq (0, \ldots, 0)$. Define the open subsets

$$U_i = \{p = [x_1 : x_2 : \cdots : x_{n+1}] \in \mathbb{R}P^n \mid x_i \neq 0\} \subset \mathbb{R}P^n \tag{9.1.8}$$

for $i = 1, \ldots, n+1$, and the homoemorphisms

$$\begin{aligned} \phi_i : U_i &\to \mathbb{R}^n && (9.1.9) \\ \phi(p) &= \left(\frac{x_1}{x_i}, \ldots \frac{x_{n+1}}{x_i}\right), && (9.1.10) \end{aligned}$$

where the i-th coordinate $x_i/x_i = 1$ is omitted. The local charts (U_i, ϕ_i) form a smooth atlas on $\mathbb{R}P^n$, since the transition maps are smooth, see exercises. In the same way we obtain a smooth atlas on $\mathbb{C}P^n$ with local charts $\phi_i : U_i \to C^n \cong \mathbb{R}^{2n}$, hence, $\mathbb{C}P^n$ is a smooth manifold of (real) dimension $2n$.

It is not hard to define local charts and smooth atlas(es) on the torus $\mathcal{T}^2$ and on the Klein bottle $\mathcal{K}$, see exercises.

According to our definition, the closed balls B^n, the closed cylinder $S^1 \times B^1$, the Möbius band $\mathcal{M}$ and the solid torus $S^1 \times B^2$ are not manifolds, because their boundary points do not have any open neighborhood homoemorphic to $\mathbb{R}^n$. However, we extend the definition below for manifolds with boundary.

We also met topological spaces which are definitely not manifolds. For example, the space of rational numbers $\mathbb{Q} \subset \mathbb{R}$ is not a manifold, non of its points admit an open neighborhood in $\mathbb{Q}$ homeomorphic to $\mathbb{R}^n$. Also, $X := \{xy = 0\} \subset \mathbb{R}^2$ is not a manifold, the origin does not admit an open neighborhood homeomorphic to $\mathbb{R}^n$. Another (counter)example is the 1-skeleton of the torus, which is two circles glued together at one point.

Definition 9.1.11. A topological space M is a (topological) *manifold with boundary* if every point $p \in M$ has an open neighborhood $U \subset M$ which is homoemorphic to an open subset $V \subset \mathbb{R}^n$ or to an open subset $V \subset \mathbb{R}^n_+ = \{(x, y) \in \mathbb{R}^{n-1} \times \mathbb{R} \mid y \geq 0\}$.

The *boundary* ∂M of M consists of the points of M which do not have a neighborhood homeomerphic to an open subset of $\mathbb{R}^n$.

Remark 9.1.12.

- The boundary ∂M of a manifold with boundary is always a manifold (without boundary) of one less dimension, that is, $\dim(\partial M) = \dim M - 1$.
- The boundary operation satisfies the Leibniz rule:

$$\partial(M \times N) \cong \partial M \cup_{\partial M \times \partial N} \partial N, \tag{9.1.13}$$

  where M and N are two manifolds with boundary. Verify this for the balls $M = B^m$, $N = B^n$ with the smallest dimensions m and n, see exercises.
- The notion of boundary *of a manifold with boundary* is very confusing, compared with the boundary *of a subset of a topological space*. On the one hand, the two notions are completely different, for example, the boundary of a manifold is an absolute concept, it does not depend on the ambient space. The two notions definitely do not agree in general, for example, if we consider $M \subset M$, then its boundary, as a subset of itself, is empty. On the other hand, the two notions agree in cases when M is considered as a subspace of $\mathbb{R}^n$ of the same dimension, for example, $B^n \subset \mathbb{R}^n$ or $S^1 \times B^2 \subset \mathbb{R}^3$. In Hungarian we distinguish the two things: 'az M sokaság *pereme*' and 'az $Y \subset X$ részhalmaz *határa*'.
- Another confusion: according to our definitions, every manifold is a manifold with boundary, but a manifold with nonempty boundary is not a manifold. We say *manifold without boundary* if we want to emphasize that $\partial M = \emptyset$, that is, M is a manifold.
- A compact manifold without boundary is called *closed manifold*. Yes, it is one more confusing terminology, sorry. Closed manifold are the 'finite, but borderless spaces' intuitively, see my popular science video Space creation (with English subtitles).

Example 9.1.14.

- Closed manifolds: S^n, $\mathcal{T}^n$, $\mathcal{K}$, $\mathbb{R}P^n$, $\mathbb{C}P^n$ etc.
- Compact manifolds with nonempty boundary: B^n, $\mathcal{M}$, $S^1 \times B^1$, $S^1 \times B^2$ etc. What are the boundaries of these spaces? Verify the Leibniz rule!
- Non-compact manifolds without boundary: $\mathbb{R}^n \cong \breve{B}^n$, open cylinder $S^1 \times \mathbb{R}$ etc.
- Non-compact manifolds with non-empty boundary: $\mathbb{R}^n_+$, $S^1 \times [0; 1[$ etc. What are the boundaries of these spaces? Verify the Leibniz rule!

Definition 9.1.15. A smooth atlas on a manifold M is *oriented*, if the Jacobian determinant of the transition maps is positive at each point, that is, $\det(d\psi_{i,j})_p > 0$ for every i, j and every point p of the domain of $\psi_{i,j}$.

A smooth manifold M is *oriented* if its smooth atlas is oriented. A topological manifold is *orientable* is it admits an oriented smooth atlas.

Example 9.1.16 (Orientability)**.** One can verify that S^n, $\mathbb{C}P^n$ and $\mathcal{T}^2$ are orientable manifolds, see exercises. $\mathcal{M}$, $\mathcal{K}$ and $\mathbb{R}P^2$ are non-orientable. The answer for the orientability question of the real projective spaces $\mathbb{R}P^n$ is quite surprising, see exercises.

A two dimensional manifold M^2 (surface) is orientable if it allows a consistent definition of 'clockwise' and 'anticlockwise' at each point of the surface. If M^2 is in $\mathbb{R}^3$, then it is equivalent with the continuous choice of a nowhere zero normal vector field. Along the a Möbius band 'clockwise' is changed into 'counterclockwise' after running through the midle circle, and coming back to the starting point. Equivalently, 'outward' normal direction is changed to 'inward' along the midle circle. A 2-manifold M is orientable if and only if it does not contain a Möbius band. See FIGURA!!!.

Remark 9.1.17. Technically, a *smooth structure* on a topological manifold M is a maximal smooth atlas, that is, a smooth atlas extended with all local charts which are compatible with the atlas, i.e., the new transition maps are still smooth. Similarly, an *orientation* on a smooth manifold is a maximal oriented smooth atlas. Every orientable smooth manifold has exactly two different orientations.

Definition 9.1.18. Let M^m and N^n be smooth manifolds. A map $f : M \to N$ is *smooth* at $p \in M$ if there are local charts $\phi : U \to V$ on an open neighborhood $U \subset M$ of p and $\rho : W \to Z$ on an open neighborhood $W \subset N$ of $f(p)$ such that the composition

$$\rho \circ f \circ \phi^{-1} : \mathbb{R}^m \supset V \to Z \subset \mathbb{R}^n \tag{9.1.19}$$

is smooth.

The map f is smooth if it is smooth at every $p \in M$.

Although this definition suggest that the smoothness of a map depends on the local chart, course it does not. Indeed, by the chain rule and the smoothness of the transition maps, the composition (9.1.19) is smooth for every pair of charts around p and $f(p)$, if it is smooth for one of them. See exercises.

Definition 9.1.20. A map $f : M \to N$ between smooth manifolds is a *diffeomorphism* is it is a smooth bijection whose inverse $f^{-1} : N \to M$ is also smooth.

Two manifolds M and N are *diffeomorphic* if there is a diffeomorphism $f : M \to N$.

Like homeomorphism for topological spaces, diffeomorphism is the natural equivalence of smooth manifolds. If M and N are diffeomorphic, then they have the same dimension $\dim M = \dim N$.

Example 9.1.21 (Diffeomorphism)**.** The aim of this very technilcal example is to clarify a very common confusion.

The map $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^3$ is a homeomorphism, but not a diffeomorphism, since its inverse $f^{-1}(y) = \sqrt[3]{y}$ is not smooth at $y = 0$.

Consider a topological manifold M and endow it with two different smooth atlases. We obtain two smooth manifolds M_1 and M_2 with the same underlying topological manifold M. What does it mean if the two atlases are compatible, that is, if their union is also a smooth atlas? In other words, if the two atlases determine the same smooth structure (that is, maximal smooth atlas) on M. One can easily deduce that this holds if and only if the identity map $f : M \to M$, $f(p) = p$ is a diffeomorphism between M_1 and M_2.

So what if we define two smooth atlases on the topological manifold $\mathbb{R}$? The first one consists of only one chart, the identity map $\phi_1 : \mathbb{R} \to \mathbb{R}$, $\phi_1(x) = x$, results in the smooth manifold $\mathbb{R}_1$. The second one also consists of only one chart, the map $\phi_2 : \mathbb{R} \to \mathbb{R}$, $\phi_2(x) = x^3$, results in the smooth manifold $\mathbb{R}_2$. Recall that ϕ_2 is a local chart, since it is a homeomorphism. The two local charts are not compatible, since one of the the transition maps is not smooth. Hence, the union of the two atlases is not a smooth atlas. The two atlases define different smooth structures on $\mathbb{R}$, that is, the identity map of $\mathbb{R}$ is not a diffeomorphism between the smooth manifolds $\mathbb{R}_1$ and $\mathbb{R}_2$.

Nevertheless, $\mathbb{R}_1$ and $\mathbb{R}_2$ are diffeomorphic. Indeed, the map $f : \mathbb{R}_2 \to \mathbb{R}_1$, $f(x) = x^3$ is a diffeomorphism. Let's check it! f is a bijection, hence, we only have to check that both f and f^{-1} are smooth. By definition this means that the maps

$$\phi_1 \circ f \circ \phi_2^{-1} \text{ and } \phi_2 \circ f^{-1} \circ \phi_1^{-1} \tag{9.1.22}$$

are smooth as maps $\mathbb{R} \to \mathbb{R}$ in the common sense. This is true, since both maps are the identity of $\mathbb{R}$. Hence, as a map $\mathbb{R}_2 \to \mathbb{R}_1$, $x \mapsto x^3$ is a diffeomorphism, because of the choice of the atlases.

In summary, ϕ_1 and ϕ_2 define two different smooth structures on $\mathbb{R}$, but the resulted smooth manifolds $\mathbb{R}_1$ and $\mathbb{R}_2$ are diffeomorphic.

9.2 Classification of manifolds

The problem of classification of manifolds includes the classification of topological manifolds up to homeomorphism and smooth manifolds up to diffeomorphism. Of course it is enough to focus on the connected manifolds, the others are disjoint unions of the connected ones.

In dimension one the only closed manifold is the circle S^1. More precisely, every closed connected topological 1-manifold is homeomorphic to S^1, and every every closed connected smooth 1-manifold is diffomorphic to S^1. All the non-closed connected 1-manifolds are $\mathbb{R}$ and $[0;1[$. Every 1-manifold (that is, these three ones) is orientable.

The classification of topological and smooth manifolds coincide in dimension two as well. All the orientable, closed, connected 2-manifolds are homeomorphic (diffeomorphic) to the genus g surface A_g for some $g = 0, 1, 2, \dots$. These surfaces are constructed as follows. $A_0 = S^2$, $A_1 = \mathcal{T}^2$. To construct A_g, we start with a sphere with g holes, that is, $X := S^2 \setminus \bigsqcup_{i=1,\dots,g} \breve{B}^2{}_i$, and we take g pieces of holey tori $Y_i = \mathcal{T}^2 \setminus \breve{B}^2$. The surface A_g is the quotient space obtained by gluing the pieces Y_i to X along the S^1 boundaries. FIGURA!!!

The non-orientable, closed, connected 2-manifolds also can be listed in an infinite family of surfaces A'_g, $g = 1.2, \dots$. These surfaces are constructed similarly: starting with a sphere with g holes $X := S^2 \setminus \bigsqcup_{i=1,\dots,g} \breve{B}^2{}_i$, we take g pieces of Möbius bands $Y_i = \mathcal{M}$. The surface A'_g is the quotient space obtained by gluing the pieces Y_i to X along the S^1 boundaries. Hence, $A'_1 = \mathbb{R}P^2$, $A'_2 = \mathcal{K}$. The non-orientable surfaces A'_g cannot be embedded to $\mathbb{R}^3$, that is, there is no subspace of $\mathbb{R}^3$ homeomorphic to A'_g.

Theorem 9.2.1 (Classification of 2-manifolds).

- *Every closed, connected topological 2-manifold is homeomorphic to exactly one of the A_g or A'_g surfaces.*
- *Every closed, connected smooth 2-manifold is diffeomorphic to exactly one of the A_g or A'_g surfaces.*

But what about the mixed constructions? What if I cut 5 holes on S^2, and I glue 3 handles ($\mathcal{T}^2 \setminus \breve{B}^2$) and 2 Möbius bands? Which surface is this from the families A_g or A'_g? See exercises.

In higher dimensions the complete classification of manifolds is hopeless. Moreover, the topological and smooth classifications do not coincide anymore. There are smooth manifolds which are homeomorphic, but not diffeomorphic to each other. E.g., there are 28 'exotic 7-spheres', that is, 28 smooth manifolds $\Sigma_0, \dots, \Sigma_{27}$ which are all homoemorphic to the sphere S^7, but they are not diffeomorphic to each other (one of them, say Σ_0, is diffeomorphic to the standard sphere S^7).

Moreover, there are infinitely many, indeed, continuum many pairwise non-diffeomorphic smooth manifolds homeomorphic to $\mathbb{R}^4$. These exotic 4-spaces also can be interesting as potential space-time models in general relativity theory. There are also topological manifolds which do not admit any smooth atlas, that is, there is no smooth manifold homeomorphic to such a topological manifold.

9.3 Differential calculus in Euclidean spaces

We have already defined whether is a map $f : M \to N$ between smooth manifolds smooth. But what is the derivative (differential) of a smooth map? To answer this question, first we summarize the differential calculus in $\mathbb{R}^n$.

Consider a map

$$f = \begin{pmatrix} f_1 \\ \vdots \\ f_n \end{pmatrix} : \mathbb{R}^m \to \mathbb{R}^n \tag{9.3.1}$$

defined in an open neighborhood of a point $p \in \mathbb{R}^m$. The Jacobian matrix (total derivative) of f at p is an $n \times m$ matrix $J_f(p)$ such that the linear map $\mathbb{R}^m \to \mathbb{R}^n$, $v \mapsto J_f(p) \cdot v$ is the best linear approximation of f around p, that is,

$$\lim_{v \to 0} \frac{f(p+v) - f(p) - J_f(p) \cdot v}{\|v\|} = 0. \tag{9.3.2}$$

The map f is differentiable at p if such matrix $J_f(p)$ exists. In this casse tha partial derivatives $\partial_j f(p)$ also exist, and they are the columns of the Jacobian $J_f(p)$, that is,

$$J_f(p) = (\partial_1 f(p), \dots, \partial_m f(p)) = (\partial_j f_i(p))_{i,j}. \tag{9.3.3}$$

Recall that the partial derivative $\partial_j f_i(p) \in \mathbb{R}$ of f_i at p is the derivative of the one variable function $t \mapsto f_i(p + te_j)$ at $t = 0$, where e_j is the j-th standard unit basis vector of $\mathbb{R}^m$. The partial derivative $\partial_j f(p) \in \mathbb{R}^n$ of f at p is the column vector of components $\partial_j f_i(p)$.

In the particular case $m = 1$, $f : \mathbb{R} \to \mathbb{R}^n$ is a curve, and $J_f(p)$ is a column vector, it is the tangent vector (velocity) $f'(p)$ of f at p. In the particular case $n = 1$, $f : \mathbb{R}^m \to \mathbb{R}$ is a function (scalar field), and $J_f(p)$ is a row vector, it is the transpose of the gradient vector $\operatorname{grad}_p(f)$. If $m = n$, then $J_f(p)$ is a square matrix.

$\partial_j f(p) = J_f(p) \cdot e_i$. More generally, for every $v \in \mathbb{R}^m$ the corresponding directional derivative $\partial_v f(p) \in \mathbb{R}^n$ is the derivative of $t \mapsto f(p + tv)$ at $t = 0$, and

$$\partial_v f(p) = J_f(p) \cdot v \tag{9.3.4}$$

holds. The operator ∂_v is linear, that is, $\partial_v(af + bg)(p) = a\partial_v f(p) + b\partial_v g(p)$ for every maps $f, g : \mathbb{R}^m \to \mathbb{R}^n$ and $a, b \in \mathbb{R}$. For functions $f, g : \mathbb{R}^m \to \mathbb{R}$, the operator ∂_v satisfies the Leibniz rule:

$$\partial_v(f \cdot g)(p) = \partial_v f(p) \cdot g(p) + f(p) \cdot \partial_v g(p). \tag{9.3.5}$$

More generally, we can define the derivative of a map along a curve. Let $\gamma : \mathbb{R} \to \mathbb{R}^m$ be a curve with $\gamma(0) = p$, $\gamma'(0) \neq 0$, and let $f : \mathbb{R}^m \to \mathbb{R}^n$ be a differential map, as above. The derivative of f along γ at p is defined as $\partial_\gamma f(p) := (f \circ \gamma)'(0) \in \mathbb{R}^n$. In fact, by the chain rule,

$$\partial_\gamma f(p) = \partial_{\gamma'(0)} f(p) = J_f(p) \cdot \gamma'(0). \tag{9.3.6}$$

Let $f : \mathbb{R}^m \to \mathbb{R}^n$ be defined on an open subset $U \subset \mathbb{R}^m$. f is differentiable, if it is differentiable at every point $p \in U$. f is said to be of differentiability class C^1 if f is differentiable and its derivative map $p \mapsto J_f(p)$ is continuous. In general, f is of differentiability class C^k if it is k-times differentiable and its k-th derivative is continuous. f is called smooth (C^∞) if it is of class C^k for every $k \in \mathbb{N}$, that is, f is differentiable arbitrary many times.

According to these classes we can define manifolds of class C^k: these are topological manifolds endowed with an atlas whose transition maps are of class C^k. Maps of class C^0 are the continuous maps, manifolds of class C^0 are the topological manifolds. Smooth manifolds are manifolds of class C^∞. For simplicity, we do not interested in the classes in between, $0 < k < \infty$.

Theorem 9.3.7 (Inverse function theorem)**.** *Let $f : \mathbb{R}^m \to \mathbb{R}^m$ be a smooth map defined in an open neighborhood of $p \in \mathbb{R}^m$. Then, $\det(J_f(p)) \neq 0$ if and only if f is a local diffeomorphism, that is, there exists a neighborhood U of p such that $f|_U : U \to f(U)$ is a diffeomorphism.*

Note that in this case (i.e., for open subsets of $\mathbb{R}^m$) diffeomorphism only means that $f|_U$ is a smooth bijection whose inverse is also smooth. If $\det(J_f(p)) > 0$, then f is orientation preserving at p (hence, on U) and f reverses the orientation if $\det(J_f(p)) < 0$.

9.4 Tangent spaces and tangent maps

We define the tangent spaces T_pM of a smooth manifold M at its points $p \in M$. T_pM is a (real) vector space of dimension $\dim(T_pM) = \dim M$. Its elements are intuitively the tangent vectors $v \in T_pM$ of M at p. There are several equivalent definitions of T_pM, but a good picture about the concept of tangent spaces can be obtained from their relations. In fact, although the first definition is very intuitive, it is too special, but the next two ones are too abstract. In the fourth definition we define a basis for T_pM, with respect to a local chart of M.

Given a smooth map $f : M^m \to N^n$, its 'total derivative' at $p \in M$ is defined as a linear map $T_pf : T_pM \to T_{f(p)}N$, called tangent map. In a pair of basis of T_pM and $T_{f(p)}N$ induced by local charts, the matrix of T_pf is the Jacobian of the corresponding $\mathbb{R}^m \to \mathbb{R}^n$ map.

Example 9.4.1 (Preliminary examples for tangent spaces)**.** Let $M = \mathbb{R}^n$. Then its tangent space $T_p\mathbb{R}^n$ at $p \in \mathbb{R}^n$ can be identified with $T_0\mathbb{R}^n$, hence, with $\mathbb{R}^n$, via $v \mapsto p + v$.

Let $M = S^n \subset \mathbb{R}^{n+1}$ be the unit sphere. Its tangent space T_pS^n at $p \in S^n$ is

$$T_pS^n = \{(p, v) \in \{p\} \times \mathbb{R}^{n+1} \mid v \perp p\} \tag{9.4.2}$$

It is a subspace of $\mathbb{R}^{n+1}$, the first coordinate p is meant to make these spaces disjoint formally.

First definition: manifolds in $\mathbb{R}^N$: Assume that $M^m \subset \mathbb{R}^n$ and an open neighborhood U of $p \in M$ is parametrized by a smooth map $\mathbf{r} : \mathbb{R}^m \supset V \to U \subset \mathbb{R}^n$ such that the rank of the Jacobian $J_\mathbf{r}$ is m at every point of V. (The parametrization $\mathbf{r}$ is the inverse of a local chart $\phi : U \to V$.) Assume that $p = \mathbf{r}(x)$. Then T_pM is the image of the Jacobian of $\mathbf{r}$, more precisely, the image of the linear map $\mathbb{R}^m \to \mathbb{R}^n$, $v \mapsto J_\mathbf{r}(x) \cdot v$. In other words, T_pM is the linear subspace of $\mathbb{R}^n$ spanned by the partial derivatives $\partial_i\mathbf{r}$, $i = 1, \dots, m$.

It is not clear from this approach that the tangent spaces T_pM really belong to the manifold M and not to its embedding $M \subset \mathbb{R}^n$ or to the parametrization $\mathbf{r}$.

Second definition: curves: Intuitively, the tangent vectors $v \in T_pM$ are the tangent vectors $\gamma'(0)$ of smooth curves at p, that is, smooth maps $\gamma : (-\epsilon, \epsilon) \to M$ with $\gamma(0) = p$. However, $\gamma'(0)$ is not yet defined, since we haven't defined the derivative of smooth maps between smooth manifolds. We want to define it below, using the tangent vectors.

Instead of $\gamma'(0)$ we define equivalence classes $[\gamma]$ of smooth curves $\gamma : (-\epsilon, \epsilon) \to M$ with $\gamma(0) = p$. For the equivalence relation we fix a local chart $\phi : M \supset U \to V \subset \mathbb{R}^m$ on an open neighborhood U of p. Then γ_1 and γ_2 are defined to be equivalent if $(\phi \circ \gamma_1)'(0) = (\phi \circ \gamma_2)'(0)$ holds for the curves $\phi \circ \gamma_i : \mathbb{R} \to \mathbb{R}^m$. It can be shown that this property holds for two curves γ_1 and γ_2 with respect to a local chart ϕ if and only if this holds for them with respect to every local chart around p, see exercises. Hence, the equivalence of two smooth curves at p does not depend on the choice of the local chart.

Intuitively, the equivalence of two smooth curves γ_1 and γ_2 means that they are expected to have the same tangent vector $\gamma_1'(0) = \gamma_2'(0)$ at p. According to this intuition we define the tangent vectors as equivalence classes $[\gamma]$ of smooth curves, the tangent space T_pM is the set of these equivalence classes. From this approach it is not clear that T_pM is a vector space.

Third definition: directional derivations: In $\mathbb{R}^m$, the (tangent)vectors $v \in \mathbb{R}^m \cong T_pM$ can be identified with the corresponding directional derivations ∂_v of smooth functions $f : \mathbb{R}^m \to \mathbb{R}$ at $p \in \mathbb{R}^m$. This motivates the following concept. A directional derivation D on M at $p \in M$ associates a number $D(f) \in \mathbb{R}$ for each smooth function $f : M \to \mathbb{R}$ such that

- D is linear: $D(af + bg) = aDf) + bD(g)$ holds for every smooth map $f, g : M \to \mathbb{R}$ and real numbers $a, b \in \mathbb{R}$,
- D satisfies the Leibniz rule: $D(f \cdot g) = D(f) \cdot g(p) + f(p) \cdot D(g)$.

The derivations on M at p form a vector space. Indeed, the set $C^\infty(M)$ of smooth functions $f : M \to \mathbb{R}$ is a vector space, and the derivations at p form a subspace of the vector space of linear functionals $C^\infty(M) \to \mathbb{R}$.

The tangent space T_pM is defined as the vector space of derivations on M at p. Let us compare it with the second definition! Every smooth curve $\gamma : (-\epsilon, \epsilon) \to M$, $\gamma(0) = p$ induces a derivation $D = \partial_\gamma$ defined as $\partial_\gamma(f) := (f \circ \gamma)'(0)$. One can show that

1. ∂_γ is a derivation at p, that is, it is linear and it satisfies the Leibniz rule,
2. ∂_γ depends only on the equivalence class $[\gamma]$ of γ at p,
3. the correspondence $\gamma \mapsto \partial_\gamma$ induces a bijection between the equivalence classes of curves at p and the derivations at p,

see exercises. Therefore, the second and third definitions of tangent vectors are equivalent.

Fourth definition: basis of the tangent space: Given a local chart $\phi : M \supset U \to V \subset \mathbb{R}^m$, there is an associated basis $\partial_1^\phi(p), \dots, \partial_m^\phi(p)$ of T_pM, hence, the tangent vectors are the linear combinations

$$v = \sum_{i=1}^{m} v_i \partial_i^\phi(p). \tag{9.4.3}$$

The basis vectors $\partial_i^\phi(p)$ are defined as follows:

- Using the second definition, $\partial_i^\phi(p)$ is defined as the equivalence class $[\gamma]$ of the curve

$$\gamma(t) = \phi^{-1}(\phi(p) + te_i) \tag{9.4.4}$$

  at p, where e_i is the i-th unit basis vector of $\mathbb{R}^m$.

- Using the third definition, $D = \partial_i^\phi(p)$, as a derivation at p, acts on the smooth function $f : M \to \mathbb{R}$ as

$$D(f) = \partial_i(f \circ \phi^{-1})(\phi(p)). \tag{9.4.5}$$

It can be shown that the two constructions are equivalent via the correspondence $\gamma \mapsto \partial_\gamma$, and the vectors $\partial_1^\phi(p), \dots, \partial_m^\phi(p)$ form a basis of T_pM. This basis depends on the choice of the local chart ϕ. Given two local charts ϕ_1, ϕ_2 around p, the base change matrix between the two basis $(\partial_i^{\phi_1}(p))$ and $(\partial_i^{\phi_2}(p))$ is the Jacobian $J_{\phi_2 \circ \phi_1^{-1}}(\phi_1(p))$ of the transition map $\phi_2 \circ \phi_1^{-1}$ at $\phi_1(p)$.

Given a smooth map $f : M^m \to N^n$, its 'total derivative' at $p \in M$ is defined as a linear map $T_pf : T_pM \to T_{f(p)}N$, called tangent map, as follows:

- Using the second definition, for a curve $\gamma : (-\epsilon, \epsilon) \to M$, $\gamma(0) = p$, the image $T_pf([\gamma])$ of its equivalence class $[\gamma]$ is defined as the equivalence class $[f \circ \gamma]$ of the curve $f \circ \gamma : (-\epsilon, \epsilon) \to N$ at $f(p)$. It can be shown that $[f \circ \gamma]$ depends only on $[\gamma]$.

- Using the third definition, for a derivation D on M at p, its image $T_pf(D)$ is a derivation on N at $f(p)$, which associates to a smooth function $g : N \to \mathbb{R}$ the real number $D(g \circ f)$. It can be shown that $T_pf(D)$ is really a derivation on N at $f(p)$.

It can be shown that the two constructions of T_pf are equivalent via the correspondence $\gamma \mapsto \partial_\gamma$. Given local charts ϕ on M around p and ρ on N around $f(p)$, the matrix of T_pf can be written in the induced basis $\partial_i^\phi(p)$ of T_pM and ∂_j^ρ of $T_{f(p)}N$. It turns out that this matrix agrees with the Jacobian $J_{\rho \circ \phi}(\phi(p))$ of $\rho \circ \phi : \mathbb{R}^m \to \mathbb{R}^n$ at $\phi(p) \in \mathbb{R}^m$.

9.5 Exercises

- perem golyokra
  - sima lekep nem fugg a charttol
  - fogantyuk es szalagok
  - gorbek ekvivalenciaja es derivalasok

10 Hopf theorem and Hopf invariant

We define homotopy invariants for smooth maps in terms of the pre-images of generic values.

10.1 Degree of a smooth map

10.2 Regular values in general

10.3 Maps from the 3-sphere to the 2-sphere

11 Poincaré–Hopf theorem

11.1 Hairy ball theorem

More precisely, hairy sphere theorem[16]. It states that there is no continuous, nowhere vanishing tangent vector field on S^2. One can give a try with the tangent vectors of the longitude curves, but in this way vortexes are created at the two poles. That is,

- either we have a continuous tangent vector field on the whole S^2 with zeros at the poles, or
- we have a continuous tangent vector field of unit length defined outside the two poles, and this vector field cannot be extended continuously to the poles.

Is it possible to give a continuous tangent vector field on S^2 with only one vortex? Yes, it can be reached by merging the two vortexes of the above vector field together. However, this new vortex is a double vortex in some sense.

In contrast, there are nowhere vanishing continuous tangent vector fields on $\mathcal{T}^2$, and one can construct such vector fields on the odd dimensional spheres S^{2n+1} as well, see exercises. What's going on here?

11.2 Tangent bundle and tangent vector fields

First of all, what does it mean that a tangent vector field is continuous? A tangent vector field is a map v defined on S^2 with values $v(p) \in T_pS^2$. Each value is in a different space – then what is continuity here? This technical problem is solved by the notion of tangent bundle.

The *tangent bundle* TM of a smooth manifold M is the disjoint union of the tangent spaces of M, that is,

$$TM = \bigsqcup_{p\in M} T_pM. \tag{11.2.1}$$

Note that the tangent spaces T_pM and T_qM for different points $p \neq q \in M$ are disjoint by definition. It is a bit confusing if you imagine the manifold in an ambient space, that is, $M \subset \mathbb{R}^N$, since in this case each tangent space T_pM can be identified with a linear subspace of $\mathbb{R}^N$, hence they intersect each other at least at the origin. But the abstract definitions of the tangent vectors (namely, equivalence classes of curves at p or directional derivations at p) carry the information of p, the point at which they are tangent vectors, making the tangent spaces corresponding to different points disjoint.

If M is a smooth manifold, then its tangent bundle TM is also a smooth manifold. Indeed, any chart $\phi : U \to V \subset \mathbb{R}^m$ of M induces a chart $\Phi : TU \to \mathbb{R}^{2m}$ of TM, indicating that $\dim(TM) = 2\dim(M)$. The chart Φ is defined as follows. Take a tangent vector $v \in T_pM$ at a point $p \in U$. Recall that ϕ induces a basis $\partial_1^{(\phi)}(p), \dots, \partial_m^{(\phi)}(p)$ in T_pM, hence,

$$v = \sum_{i=1}^{m} v_i \partial_i^{(\phi)}(p). \tag{11.2.2}$$

Then define

$$\Phi(v) = (\phi(p), v_1, \dots, v_m) \in \mathbb{R}^{2m}. \tag{11.2.3}$$

There is a natural projection map $\pi : TM \to M$, defined by $\pi(v) = p$, if $v \in T_pM$. In words, π tells the point at which v is a tangent vector. Obviously, $\pi^{-1}(p) = T_pM \subset TM$. Furthermore, over

[16] Or hedgehog theorem: 'The hedgehog cannot be combed.'

the domain U of a chart ϕ, $\pi^{-1}(U) = TU$ is a direct product

$$TU \quad \cong \quad U \times \mathbb{R}^m \tag{11.2.4}$$

$$v = \sum_{i=1}^{m} v_i \partial_i^{(\phi)}(p) \quad \mapsto \quad (p, v_1, \ldots, v_m). \tag{11.2.5}$$

Importantly, it is not true globally: in general, TM is not diffeomorphic (neither homeomorphic) to $M \times \mathbb{R}^m$, as we will see immediately.

A *tangent vector field* is a *section of the tangent bundle*, that is, a map $s : M \to TM$, such that $s(p) \in T_pM$ for all p. Or in a fancy formulation: $\pi \circ s = \mathrm{id}_M$. As a map between smooth manifolds M and TM, it makes sense that a tangent vector field s is continuous or smooth.

It is easy to see that the tangent bundle TM is *trivial*, that is, diffeomorphic to $M \times \mathbb{R}^m$ if and only if there are m linearly independent smooth sections $s_1, \ldots, s_m$, that is, the vectors $s_1(p), \ldots, s_m(p)$ form a basis in T_pM for all $p \in M$. But in general there is not even one nowhere vanishing continuous[17] tangent vector field, hence, TM is not homeomorphic to $M \times \mathbb{R}^m$ in general, for example, for S^2.

We define the index of a smooth tangent vector field s at a point p. First we do it for $M = \mathbb{R}^m$. Using the identification $T_p\mathbb{R}^m \cong \mathbb{R}^m$, the tangent vector field s can be considered as a map $s : \mathbb{R}^m \to \mathbb{R}^m$. Assume that s has an isolated zero at p. That is, $s(p) = 0$, and there is a neighborhood W of p such that $s(q) = 0$ at $q \in W$ implies that $q = p$. Take a ball B^m inside W, then s is nowhere zero on $\partial B^m = S^{m-1}$. The normalization of s induces a map

$$\widetilde{s} := \left.\frac{s}{\|s\|}\right|_{S^m} : S^{m-1} \to S^{m-1}. \tag{11.2.6}$$

The *index* of s at p is defined as the degree of $\widetilde{s}$, that is,

$$\mathrm{ind}_p(s) := \deg(\widetilde{s}). \tag{11.2.7}$$

It can be shown that the index does not depend on the choice of the sphere S^{m-1}, if p is the only zero of s inside B^m.

The general case is actually the same, using that a smooth manifold M locally looks like $\mathbb{R}^m$. In a local chart $\phi : U \to \mathbb{R}^m$ around p, the tangent vector field $s|_U$ can be considered as a map $s|_U : U \to \mathbb{R}^m$, which associates to every p the m coordinates of $s(p)$ in the basis $\partial_i^{(\phi)}(p)$. Then we can repeat the above construction, and one can show that the index $\mathrm{ind}_p(s)$ we obtain does not depend on the choice of the chart ϕ.

Of course, the index also can be defined if $s(p) \neq 0$, but in this case $\mathrm{ind}_p(s) = 0$. Moreover, the construction also works if s is not defined at p, but it is defined smoothly in $W \setminus \{p\}$, that is, p is an isolated singularity of s. The index also can be defined for continuous, but not smooth vector fields, since the notion of degree can be extended to continuous maps.

Theorem 11.2.8 (Poincaré–Hopf theorem, Part 1.)**.** *Let M be a closed, oriented, smooth manifold, and let $s : M \to TM$ be a smooth tangent vector field with isolated zeros at the points $p_1, \ldots, p_k$. Then the sum of the indices*

$$\sum_{i=1}^{k} ind_{p_i}(s) \tag{11.2.9}$$

does not depend on s, it is the same for any such tangent vector field. Hence, this sum is an invariant of M.

Sketch of proof. This is rather an intuition than a precise proof, because it uses many arguments which were not introduced precisely. The main point is that any two smooth tangent vector fields are homotopic. During a homotopy, the zeros of the vector field not only move away, but they

[17] Note that smooth or continuous is not a big difference in this situation, any continuous section can be modified by a homotopy to obtain a smooth section.

can merge together or split into more zeros. The claim is that the sum of the indices (say, *total index*) remains the same during these merging processes. The index behaves like a kind of 'charge' of the zeros, and the total charge is preserved by the 'phase transitions'. More precisely, observe the following.

The image of a smooth tangent vector field is a smooth submanifold $M \cong M_s \subset TM$. Locally, over the domain $U \subset M$ of a chart, $U_s = \mathrm{im}(s|_U) \subset TU \cong U \times \mathbb{R}^m$ is the graph of a map $U \to \mathbb{R}^m$. The image of the *zero section* $s_0(p) = 0 \in T_pM$ is another submanifold $M \cong M_0 \subset TM$. The zeros of s are the intersection points $M_s \cap M_0$.

Since M_s and M_0 are two smooth manifolds of dimension m in the manifold TM of dimension $2m$, they intersect each other in isolated points, if they are in general position. Moreover, a generic intersection point p of M_s and M_0 is *transverse*, meaning that the tangent spaces T_pM_s and T_pM_0 span the tangent space $T_p(TM)$. It can be shown that a transverse intersection point of M_s and M_0 corresponds to a zero of s with index ± 1. The sign is positive (respectively, negative), if the direct sum of the orientations of T_pM_s and T_pM_0 coincide with (respectively, is the opposite of) the orientation of $T_p(TM)$.

This argument implies that every smooth tangent vector field s can be perturbed (that is, modified by a small smooth homotopy) to a smooth tangent vector field s' with only isolated zeros with indices ± 1, such that the total index of s' equals to the total index of s. Consider two such smooth tangent vector fields s_1 and s_2 and a generic smooth homotopy $H : M \times I \to TM$ between them. Then the zeros of H form line segments in $M \times I$ joining either two zeros of the same section s_1 or s_2 with opposite indices, or one zero of s_1 with one zero of s_2 with the same index. This shows that the total indices of s_1 and s_2 are the same.

□

11.3 Euler characteristic

What is this invariant, the sum of the indices of a tangent vector field? As we saw, for S^2 it is 2. For $\mathcal{T}^2$ and for S^{2n+1} it is 0, since in these cases nowhere vanishing smooth tangent vector fields exist.

So, what is 2 for S^2? Consider a convex polyhedron, count its vertices (v), edges (e) and faces (f), and create the alternate sum $v - e + f$ of these numbers. For example:

- *Cube:* $8 - 12 + 6 = 2$
- *Regular dodecahedron:* $20 - 30 + 12 = 2$
- *Prism over an n-gon:* $2n - (3n) + (n + 2) = 2$

Of course, convexity is not important. The point is that the surface of the polyhedron (that is, the union of the closed faces) is homeomorphic to S^2.

Theorem 11.3.1 (Euler's polyhedron formula)**.** *Consider a polyhedron whose surface is homeomorphic to S^2. Then*

$$v - e + c = 2 \tag{11.3.2}$$

holds, where v, e and f are the number of vertices, edges and faces, respectively.

Sketch of proof. The polyhedron defines a special cell decomposition of S^2. We construct another cell complex of S^2 with only one face by removing one edge in each step. Starting with one face, we remove one edge for each face to unionize it with the starting face, hence the number of removed edges is $f - 1$. The graph of the remaining edges is a tree, hence, the number of these edges is $v - 1$. Therefore,

$$(f - 1) + (v - 1) = e, \tag{11.3.3}$$

proving the theorem. □

Observe that the value of the expression $v - e + f$ remains unchanged in the steps of the proof, since both the e and f decreases by 1 when one edge is removed to unionize faces. We can further simplify the cell complex by modifying the graph in the following steps:

- *(Step 1)* Deleting a degree-1 vertex together with the corresponding edge.
- *(Step 2)* Deleting a degree-2 vertex to unionize the two adjacent edges.
- *(Step 3)* Deleting an edge (which is not a loop) to unionize its two vertices.

Both steps remain $v - e + f$ unchanged. By these steps (in fact, by only the Step 1) we can reach the simplest cell complex of S^2 with one 0-cell and one 2-cell, hence, $v - e + f = 1 - 0 + 1 = 2$.

Can we adapt this method to a polyhedron whose surface is homeomorphic to $\mathcal{T}^2$? Yes, but when we reach only one face, the graph of the remaining edges is not a tree, but it has two circles, say, meridian and longitude. By the further steps 1-3 we can simplify the graph to obtain the standard cell decomposition of $\mathcal{T}^2$ with one 0-cell, two 1-cells and one 2-cell. For this, $v-e+f = 1-2+1 = 0$.

Definition 11.3.4. Let X be a cell complex of dimension n, and let c_i denote the number of i-cells of X. The *Euler-characteristic* of X is defined as

$$\chi(X) = \sum_{i=0}^{n} (-1)^i c_i. \tag{11.3.5}$$

Theorem 11.3.6. *1. If two cell complexes X and Y are homeomorphic, then $\chi(X) = \chi(Y)$.*

2. Moreover, if two cell complexes X and Y are homotopy equivalent, then $\chi(X) = \chi(Y)$.

Therefore, the Euler characteristic is a topological invariant, moreover, it is a homotopy invariant, similarly to the homotopy groups $\pi_k(X)$, but their values are groups associated to X, while the Euler characteristic is an integer.

Theorem 11.3.7. $\chi(X \times Y) = \chi(X) \cdot \chi(Y)$.

Example 11.3.8. One can directly calculate the following Euler characteristics by considering a cell complex decomposition.

- $\chi(S^1) = \chi(S^1 \times B^1) = \chi(S^1 \times B^2) = \chi(S^1 \times S^1) = 0$,
- $\chi(B^n) = 1$
- $\chi(S^n) = 1 + (-1)^n$.

To complete the Poincaré–Hopf theorem, note that every smooth manifold can be constructed as a cell complex, although, it is not trivial.

Theorem 11.3.9 (Poincaré–Hopf theorem, Part 2.)**.** *Let M be a closed, oriented, smooth manifold, and let $s : M \to TM$ be a smooth tangent vector field with isolated zeros at the points $p_1, \dots, p_k$. Then the sum of the indices is equal to the Euler characteristic, that is,*

$$\sum_{i=1}^{k} ind_{p_i}(s) = \chi(M). \tag{11.3.10}$$

In particular, the sum of the indices does not depend on s, it is the same for any such tangent vector field.

Sketch of proof. Because of Part 1 of the theorem, it is enough to construct a smooth tangent vector field with sum of the indices equals to the Euler characteristic. Consider a polyhedron homeomorphic to M. By further dividing the cells, it can be assumed that each cell is a simplex. Fix one point in the interior of each cell, considered as the center of the cell. Then one can construct a continuous tangent vector field whose zeros are at the centers of the cells, and for each of them the index is equal to $(-1)^d$, where d is the dimension of the corresponding cell. The vector field can be chosen smoothly. Clearly, the total index of this vector field is equal to $\chi(M)$. □

11.4 Exercises

12 Vector bundles

12.1 Basic definitions and examples

Locally trivial fiber bundles, or shortly *fiber bundles*, *bundles* or *fibrations* are 'locally direct product spaces'. More precisely, a fibration involves three spaces and a projection map. Intuitively, the *total space* E of a fibration is decomposed as the disjoint union $E = \bigsqcup_{p\in B} F_p$ of the *fibers* F_p, each homeomorphic to a *typical fiber* F, and the fibers are parametrized continuously with the points of the *base space* B. The formal definition is the following.

Definition 12.1.1. A locally trivial fiber bundle is a continuous map $\pi : E \to B$ between topological spaces E and B with the following properties.

1. $F_p := \pi^{-1}(p)$ is homeomorphic to F for all $p \in B$.
2. Moreover, every $p \in B$ has a neighborhood $U \subset B$ such that there is a homeomorphism
$$h : \pi^{-1}(U) \to U \times F \tag{12.1.2}$$
with
$$h|_{F_p} : F_p \to \{p\} \times F. \tag{12.1.3}$$

E is the total space, B is the base space and F is the fiber of the fibration, and π is called the projection map.

In our cases E, B and F are smooth manifolds and π is a smooth map. In this case,
$$\dim(F) + \dim(B) = \dim(E). \tag{12.1.4}$$

Notation: Sometimes we use one notation for a whole package, for example, $\xi = (E, B, F, \pi)$. In this case we write $B(\xi)$ for the base space and $E(\xi)$ for the total spacea and $\pi(\xi)$ for the projection map. However, in some cases we use the same notation for the bundle and its total space.

Definition 12.1.5. A locally trivial fiber bundle is called a real *vector bundle* of rank k, if $F \cong \mathbb{R}^k$, each fiber F_p is a real vector space of dimension k, and the local trivialization homeomorphisms $h : \pi^{-1}(U) \to U \times F$ are linear isomorphisms $h|_{F_p} : F_p \to \{p\} \times \mathbb{R}^k$ on every fiber.

We define complex vector bundles similarly, in this case $F \cong \mathbb{C}^k$, F_p is a complex vector space and $h|_{F_p}$ are linear isomorphisms over $\mathbb{C}$.

Definition 12.1.6. Two bundles ξ and η over the same base space $B = B(\xi) = B(\eta)$ are called *isomorphic* if there is a homeomrphism $h : E(\xi) \to E(\eta)$ which takes each fiber $(\pi(\xi))^{-1}(p)$ of ξ to the corresponding fiber $(\pi(\eta))^{-1}(p)$ of η. In other words, h is compatible with the projection maps:
$$\pi(\eta) \circ h = \pi(\xi). \tag{12.1.7}$$

The vector bundles are isomorphic if, in additional to the above, the restriction of h to the fibers are linear ismorphisms.

Example 12.1.8 (Tangent bundle)**.** The tangent bundle of a smooth manifold M is a vector bundle over the base space $B = M$ with total space $E = TM$. Its fibers are the tangent spaces $F_p = T_pM \cong \mathbb{R}^m$. The projection map $\pi : TM \to T_pM$ is defined as $\pi(v) = p$ if $v \in T_pM$.

Example 12.1.9 (Trivial bundle)**.** Every direct product is a fiber bundle with total space $E = X \times Y$, base space $B = X$, fiber $F = Y$, and projection map $\pi(x, y) = x$. Of course, in this case, the role of X and Y can be exchanged. These bundles are called *trivial bundles*. More precisely, a bundle is trivial if it is isomorphic to the direct product bundle of its base and fiber.

The main point in the study of bundles is to determine, how far is a bundle from being trivial. However, locally every fiber bundle is trivial by definition, so the main point is the global 'twist'. Our goal is to measure this twist.

Example 12.1.10 (Cylinder, torus, Möbius, Klein)**.** Therefore, the projection maps $S^1 \times B^1 \to B^1$, $S^1 \times B^1 \to S^1$, $S^1 \times S^1 \to S^1$ define fiber bundles, but these are trivial bundles. $S^1 \times \mathbb{R} \to S^1$ is a trivial real vector bundle of rank 1, that is, a trivial real *line bundle*.

The projection $\pi : \mathcal{M} \to S^1$ of the Möbius band onto its middle circle also defines a fiber bundle with $B = S^1$, $E = \mathcal{M}$, $F = B^1$. It is not a trivial bundle, indeed, $\mathcal{M}$ is not homeomorphic to $S^1 \times B^1$. The Möbius band is locally a a direct product, but globally not, since it is twisted. Replacing the B^1 fibers with $\mathbb{R}$ fibers results in an 'infinite Möbius band' $\breve{\mathcal{M}}$, which is a non-trivial real line bundle over S^1.

The Klein bottle $E = \mathcal{K}$ is also a fiber bundle over $B = S^1$, i.e., it admits a projection map $\pi : \mathcal{K} \to S^1$ onto its longitude circle. The fibers are $F = S^1$. This bundle is not trivial, since $\mathcal{K}$ is not homeomorphic to $S^1 \times S^1$.

Example 12.1.11 (Hopf fibration)**.** The Hopf map $\pi : S^3 \to S^2$ is the projection of a fiber bundle over $B = S^2$ with $E = S^3$, $F = S^1$. It is not a trivial bundle, since S^3 is not homeomorphic to $S^2 \times S^1$. For example, their fundamental groups are not isomorphic: $\pi_1(S^3) = 0$, $\pi_1(S^2 \times S^1) = \mathbb{Z}$.

Example 12.1.12 (Projective spaces)**.** The quotient map $\pi : S^n \to \mathbb{R}P^n$ defining the real projective space is actually the projection map of a fiber bundle over $B = \mathbb{R}P^n$ with $E = S^n$ and with $F = S^0$ consists of two points. Similarly, $\pi : S^{2n+1} \to \mathbb{C}P^n$ is a fiber bundle over $B = \mathbb{C}P^n$ with fiber $U(1) = S^1$ and total space $E = S^{2n+1}$.

Example 12.1.13 (Discrete fibers)**.** The quotient maps $\pi : \mathbb{R} \to S^1$ (realised by $\pi(t) = (\cos(t), \sin(t))$ and $\pi : S^1 \to S^1$ (realized by $\pi(z) = z^n$) are projection maps of fiber bundles with discrete fiber F. Actually, the fiber bundles of discrete fiber are exactly the covering maps, see many examples in Section 7.6.

Example 12.1.14 (Eigenvector bundles)**.** Consider the set $X \subset \mathrm{Herm}(n)$ of $n \times n$ hermitian matrices with non-degeenrate ground state, that is, $\lambda_1 < \lambda_2 \leq \lambda_3 \leq \cdots \leq \lambda_n$. As an open subset of $\mathrm{Herm}(n) \cong \mathbb{R}^{n^2}$, X is a smooth manifold of dimension n^2.

Let $F_H \subset \mathbb{C}^n$ be the ground state of $H \in X$, that is, the eigenspace corresponding to λ_1:

$$v \in F_H \Leftrightarrow Hv = \lambda_1 v. \tag{12.1.15}$$

F_H is a complex linear subspace of $\mathbb{C}^n$. The *eigenvector bundle* corresponding to λ_1 is the complex line bundle ξ with fiber F_H over H. Its base space is $B = X$, the fibers are $F_H \cong \mathbb{C}$. But what is the total space $E = E(\xi)$?

The construction of E is quite artificial. Intuitively, it the disjoint union of the fibers:

$$E = \bigsqcup_{H \in X} F_H. \tag{12.1.16}$$

However, the fibers are not disjoint, $F_H \cap F_{H'} = \{0\}$ for $H \neq H'$. How can they be considered as disjoint, if they are not? Intuitively, the origin of $\mathbb{C}^n$ should be replaced by a manifold X_0 diffeomorphic to X, serving as the set of different origins $0_H \in F_H$ of the fibers.

Precisely the total space is defined as

$$E := \{(H, v) \in X \times \mathbb{C}^n \mid Hv = \lambda_1 v\} \tag{12.1.17}$$

E is a smooth submanifold in $X \times \mathbb{C}^n$, and its first projection $\pi(H, v) = H$ defines the projection $E \to X$ of the eigenvector bundle ξ. The second projection $\pi_2(H, v) = v$ is a quotient map $\pi_2 : E \to \mathbb{C}^n$. The preimage $(\pi_2)^{-1}(0)$ of the null vector is a submanifold $X_0 \subset E$ diffeomorphic to X, indeed, $v = 0$ satisfies the equation $Hv = \lambda_1 v$ for every $H \in X$. Out of X_0, π_2 is a diffeomorphism $E \setminus M_0 \to \mathbb{C}^n \setminus \{0\}$.

If a smooth map $f : M \to X$ defined on a smooth manifold M is given, then there is an induced eigenvector bundle over M, corresponding to the ground state. This bundle is denoted by $f^*(\xi)$. Its base space is M, its fiber over $p \in M$ is $\mathbb{C} \cong F_{f(p)} \subset \mathbb{C}^n$, that is, the fiber of ξ over $f(p)$. The total space of $f^*(\xi)$ is

$$E(f^*(\xi)) = \{(p, v) \subset M \times \mathbb{C}^n \mid f(p)v = \lambda_1(v)\}. \tag{12.1.18}$$

Example 12.1.19 (Tautological line bundles)**.** There is a *tautological real line bundle* $\tau_n^{\mathbb{R}}$ over $\mathbb{R}P^n$ and a *tautological complex line bundle* $\tau_n^{\mathbb{C}}$ over $\mathbb{C}P^n$. Its construction is tautological: the fiber F_p over $p \in \mathbb{R}P^n$ is actually itself, as a 1D subspace $F_p \subset \mathbb{R}^{n+1}$. Indeed, recall that by one of the constructions of $\mathbb{R}P^n$, its points are the 1D subspaces of $\mathbb{R}^{n+1}$, hence, $p = F_p$.

The construction of the total space according to the intuitive expectation

$$E = \bigsqcup_{p \in \mathbb{R}P^n} F_p \tag{12.1.20}$$

leads to the problem that F_p are not disjoint, but the above trick works in this case too:

$$E := \{(p, v) \in \mathbb{R}P^n \times \mathbb{R}^{n+1} \mid v \in p\}. \tag{12.1.21}$$

The first projection $\pi : E \to \mathbb{R}P^n$, $\pi(p, v) = p$ defines the tautological line bundle, while the second projection $\pi_2 : E \to \mathbb{R}^{n+1}$, $\pi_2(p, v) = v$ is a quotient map. $(\pi_2)^{-1}(0) \subset E$ is diffeomorphic to $\mathbb{R}P^n$ containing the null vectors $0_p \in F_p$, otherwise π_2 is a diffeomorphism $E \setminus (\pi_2)^{-1}(0) \to \mathbb{R}^{n+1} \setminus \{0\}$.

The construction is literally the same for $\tau_n^{\mathbb{C}}$ over $\mathbb{C}P^n$, just $\mathbb{R}$ has to be replaced by $\mathbb{C}$ everywhere.

$\tau_1^{\mathbb{R}}$ over $\mathbb{R}P^1 = S^1$ is isomorphic to the Möbius bundle $\breve{\mathcal{M}} \to S^1$.

$\tau_1^{\mathbb{C}}$ over $\mathbb{C}P^1 = S^2$ is actually the Hopf fibration. More precisely, consider the set $Y \subset E$ of vectors $v \in F_p$ of unit length $\|v\| = 1$. Then $Y \cong S^3$, since the map $\pi_2 : E \to \mathbb{C}^2$ is a diffeomorphism out of the null vectors. The restricted bundle $\pi|_Y : Y \to \mathbb{C}P^1$ is isomorphic to the Hopf bundle.

12.2 Euler and Chern numbers

We generalize the Poincaré–Hopf theorem.

Definition 12.2.1. Let ξ be a (continuous or smooth) fiber bundle $\pi : E \to B$. A (continuous or smooth) section of ξ is a map $s : B \to E$ such that $s(p) \in F_p$ for all $p \in B$. In other words, $\pi \circ s = \mathrm{id}_B$.

Definition 12.2.2. A bundle is oriented, if the base B, the typical fiber F and the fibers F_p are oriented manifolds, and the local trivializations $U \subset B$, $\pi^{-1}(U) \cong U \times F$ preserve the orientation on each fiber, that is, the restrictions $F_p \to \{p\} \times F$ are orientation preserving diffeomorphisms. In this case, E is also oriented via the local trivializations.

Proposition 12.2.3. *Let ξ be a rank-k (real or complex) vector bundle. Then, ξ is trivial if and only if there are k linearly independent smooth (or continuous) sections $s_1, \ldots, s_k$, that is, $s_1(p), \ldots, s_k(p)$ is a basis in F_p for all $p \in B$.*

Proof. A trivial vector bundle obviously has k linearly independent sections. Other direction: we can define the fiber-wise linear diffeomeorphism (or homeomorphism, if the sections are only continuous)

$$\begin{aligned} B \times \mathbb{R}^k &\to E & (12.2.4)\\ (p, (a_1, \ldots, a_k)) &\mapsto \sum_{i=1}^{k} a_i s_i(p) & (12.2.5) \end{aligned}$$

□

Let ξ be an oriented real vector bundle over a smooth manifold $B = M$. Assume that the rank of the bundle equals to the dimension of the base space $\dim(M) = m$. That is, each fiber is $F_p \cong \mathbb{R}^m$. A generic section $s : M \to E$ has isolated zeros. In fact, the same argument as we explained in the proof of the Poincaré–Hopf theorem 11.2.8 works in this more general situation. The image of a section s is a submanifold $M_s \subset E$, and the image of the zero section is another submanifold, $M_0 \subset E$. Both M_s and M_0 are diffeomorphic to the base space M, in particular, their dimension is m. They intersect each other in the $2m$ dimensional total space E, hence, the intersection consists of isolated points, if M_0 and M_s are in generic position, which can be reached by modifying s with a small homotopy. The intersection points are the zeros of s.

Let $p \in B$ be an isolated zero of s. That is, $s(p) = 0$, and there is a neighborhood $U \subset M$ such that $s|_{U\setminus\{p\}}$ has no zeros, and $\pi^{-1}(U) \cong U \times \mathbb{R}^m$. Hence, on U, the section is given by a map $s|_U : U \to \mathbb{R}^m$. Then we do the same as we did for tangent vector fields. Take a ball B^m inside U, then s is nowhere zero on $\partial B^m = S^{m-1}$. The normalization of s induces a map

$$\widetilde{s} := \left.\frac{s}{\|s\|}\right|_{S^m} : S^{m-1} \to S^{m-1}. \tag{12.2.6}$$

The *index* of s at p is defined as the degree of $\widetilde{s}$, that is,

$$\mathrm{ind}_p(s) := \deg(\widetilde{s}). \tag{12.2.7}$$

It can be shown that the index does not depend on the choice of the sphere S^{m-1}, if p is the only zero of s inside B^m. Moreover, it does not depend on the choice of the trivialization $\pi^{-1}(U) \cong U \times \mathbb{R}^m$ around p.

Theorem 12.2.8 (General Poincaré–Hopf theorem)**.** *Let ξ be an oriented rank-m real vector bundle over a closed, oriented smooth manifold M of dimension m. Let $s : M \to E$ be a smooth section of ξ with isolated zeros at the points $p_1, \dots, p_k$. Then the sum of the indices*

$$\sum_{i=1}^{k} ind_{p_i}(s) \tag{12.2.9}$$

does not depend on s, it is the same for any smooth section of ξ. Hence, this sum is an invariant of the vector bundle ξ.

Sketch of the proof. The (sketch of the) proof of Poincaré–Hopf theorem 11.2.8 can be adapted. Importantly: any two sections of a vector bundle are homotopic. □

Definition 12.2.10. The invariant defined above is the *Euler number* $e(\xi)$ of the vector bundle ξ. That is, for any section s with isolated zeros,

$$e(\xi) = \sum_{i=1}^{k} \mathrm{ind}_{p_i}(s). \tag{12.2.11}$$

Example 12.2.12 (Tangent bundle)**.** The tangent bundle TM of a closed, oriented, smooth manifold M is an oriented vector bundle whose rank is equal to the dimension of the manifold. Indeed, the orientation of M also determines an orientation on each tangent space $T_pM \cong \mathbb{R}^m$. Part 2. of Poincaré–Hopf theorem 11.2.8 can be reformulated as the Euler number of the tangent bundle is equal to the Euler characteristic of the manifold, that is, $e(TM) = \chi(M)$.

Example 12.2.13 (Möbius band)**.** The bundle $\breve{\mathcal{M}} \to S^1$ is not oriented. Indeed, the total space $\breve{\mathcal{M}}$ is a non-orientable manifold. Although the fibers $F_p \cong \mathbb{R}$ are *orientable*, they are not *oriented*: their orientation couldn't be decided in a compatible way. However, since this is the simplest non-trivial vector bundle, the above phenomenon can be illustrated in this bundle.

Try to define a non-zero section! The (image of the) zero section is the middle circle. Following a circle parallel to the middle circle does not close, indeed, we arrive to the other side of the middle circle of the Möbius band. In other words, this section is not continuous. To obtain a continuous section, we have to cross the middle circle at least once. Hence, any continuous section of the Möbius band has at least one zero!

Because of the lack of the orientations, the zeros of a section do not have an index or sign. However, we observe that a generic (transverse) smooth section is non-tangential to the middle circle (zero-section). Moreover, any generic smooth section has an odd number of intersections with the middle circle, that is, odd number of zeros. The non-oriented analogue of the Euler number is a $\mathbb{Z}_2$ invariant called Stieffel–Whitney number, which is just the parity of the number of zeros of a generic smooth section. In this case the first Stieffel–Whitney number w_1 is equal to $1 \in \mathbb{Z}_2$, the non-identity element of $\mathbb{Z}_2$.

A complex vector bundle over an oriented manifold is always oriented. This is because a complex vector space, considered as a real vector space of double dimension, has a well-defined orientation: swapping two vectors of a complex basis means an even number of changes for the real basis. Also, any complex linear isomorphism is orientation preserving, considered as a real linear isomorphism (its real determinant is positive).

Therefore, we can apply the results for a rank m complex vector bundle ($F_p \cong \mathbb{C}^m$) over a closed, oriented smooth manifold M of dimension $\dim(M) = 2m$. Such bundle ξ can be considered as an oriented real vector bundle of rank $2m$. In this situation the Euler number is called the (m-th) *Chern number* of the complex bundle ξ, that is,

$$c_m(\xi) = e(\xi). \tag{12.2.14}$$

Remark 12.2.15. In this situation, Chern number sounds like just another fancy name for the Euler number. However, the full picture is the following. *Euler classes* can be defined for every oriented real vector bundles over smooth manifolds, but in general (if the rank is not equal to the dimension of the base space) the Euler class is not a number, but a so-called cohomology class: it can be evaluated (integrated) on smooth submanifolds of dimension equal to the rank of the bundle, and it tells the sum of the indices of zeros on the submanifold. A rank-k complex vector bundle over a smooth manifold has *Chern classes* $c_1, c_2, \dots, c_k$, which are cohomology classes, and the top Chern class c_k is equal to the Euler class e. We used this coincidence as a definition of the Chern class in the particular case when this is just an integer number, that is, when the dimension of the base space is $2k$.

12.3 Tautological complex line bundle

As an important example, we compute the Euler number (first Chern number) of the complex tautological line bundle over $\mathbb{C}P^1$. Let us denote this bundle by $\tau = \tau_1^{\mathbb{C}}$. We define a smooth section s with only one zero and we determine its index.

Recall that each element of $\mathbb{C}P^1$ can be denoted by homogeneous coordinates, $p = [x : z]$, where $(0,0) \neq (x,y) \in \mathbb{C}^2$. A vector $0 \neq v = (a,b) \in \mathbb{C}^2$ is in F_p if and only if $[a : b] = [x : y]$, that is, $ay = bx$.

Consider the chart

$$\begin{aligned} \phi_x : U_x = \mathbb{C}P^1 \setminus \{[0:1]\} &\to \mathbb{C}, & (12.3.1)\\ \phi_x([x:y]) &= \frac{y}{x} =: z. & (12.3.2) \end{aligned}$$

τ is trivial over U_x, a nowhere zero section over U_x can be defined, for example, as

$$s(\phi_x^{-1}(z)) = s([x:y]) = (1,z) \in F_{[x:y]} \subset \mathbb{C}^2. \tag{12.3.3}$$

Then, consider the other chart

$$\begin{aligned} \phi_y : U_y = \mathbb{C}P^1 \setminus \{[1:0]\} &\to \mathbb{C}, & (12.3.4)\\ \phi_y([x:y]) &= \frac{x}{y} =: w. & (12.3.5) \end{aligned}$$

The transition map is

$$\begin{aligned} \psi = \phi_y \circ \phi_x^{-1} : \mathbb{C} \setminus \{0\} &\to \mathbb{C} \setminus \{0\}, & (12.3.6)\\ \psi(z) &= \frac{1}{z} = w. & (12.3.7) \end{aligned}$$

The section s over the second chart reads as

$$s(\phi_y^{-1}(w)) = (1, \frac{1}{w}), \tag{12.3.8}$$

hence it cannot be continuously extended to $\phi_y^{-1}(0) = [0:1]$. This is just a technical difficulty, and it can be solved e.g. by replacing s on the disc $|w| \leq 1$ with $(1, \overline{w})$. Indeed, $\overline{w} = 1/w$ for $|w| = 1$, hence the modification is a continuous section (and it also can be smoothen) with one zero at $q = [0:1]$.

To compute the index $\mathrm{ind}_q(s)$, we need a trivialization of τ around q. It can be given by a nowhere zero section t on U_y defined as

$$t(\phi_y^{-1}(w)) = t([x:y]) = (w, 1) \in F_{[x:y]}. \tag{12.3.9}$$

Then the trivialization of τ over U_y is given as

$$U_y \times \mathbb{C} \quad \to \quad \pi^{-1}(U_y) \tag{12.3.10}$$

$$([x:y], a) \quad \mapsto \quad at([x:y]) = (a\frac{x}{y}, a), \tag{12.3.11}$$

where π is the projection map of the bundle τ. The coordinate of the section $s([x:y])$ in this trivialization is

$$a = \frac{y}{x} = z = \frac{1}{w}, \tag{12.3.12}$$

indeed, $at = s$ with this choice of a. Therefore, around q, in the local chart ϕ_y, in the local trivialization induced by t, the section s is given by the map

$$w \mapsto \frac{1}{w}, \tag{12.3.13}$$

which is not continuous, but it can be replaced by $\overline{w}$ for $|w| \leq 1$ to get a continuous map ($\overline{w}$ is a real multiple of $1/w$ and they coincide on the circle $|w| = 1$). Its index is -1, hence we computed that

$$e(\tau) = c_1(\tau) = -1. \tag{12.3.14}$$

12.4 Induced bundles

In Example 12.1.14 we saw that a map induces a bundle on the source from a bundle on the target. This is a very general phenomenon. Let $f : X \to Y$ be a continuous map between topological spaces, and let ξ be a fiber bundle over Y with projection map $\pi : E \to Y$ and fibers $F_q \cong F$ for $q \in Y$. The *induced bundle* $f^*(\xi)$ over X is the bundle whose fiber over $p \in X$ is exactly the fiber of ξ over $f(p)$. Intuitively, the fibers over the target points are copied over their preimages. The total space of $f^*(\xi)$ is defined as

$$E(f^*(\xi)) = \{(p, v) \in X \times E \mid f(p) = \pi(v)\}, \tag{12.4.1}$$

the projection map is $(p, v) \mapsto p$. Obviously, the fiber over $p \in X$ equals to the fiber $F_{f(p)}$ of ξ over $f(p)$. The induced bundle $f^*(\xi)$ is also called the pull-back of ξ by f.

Theorem 12.4.2. *If the continuous maps $f, g : X \to Y$ are homotopic, then the induced bundles $f^*(\xi)$ and $g^*(\xi)$ are isomorphic bundles over X.*

In the most important cases, ξ is a tautological bundle over a projective space. We discuss here the complex case, but the real case works in the same way. Any continuous map $f : X \to \mathbb{C}P^n$ induces a complex line bundle $f^*(\tau)$, where $\tau = \tau_n^{\mathbb{C}}$ is the tautological complex line bundle over $\mathbb{C}P^n$.

What does it mean? The fibers of τ, and also $f^*(\tau)$ are 1D linear subspaces of $\mathbb{C}^{n+1}$. Vice versa, if the fibers of a complex line bundle η over X are subspaces of $\mathbb{C}^{n+1}$, depending continuously on the points of X, then this configuration determines an inducing map $f : X \to \mathbb{C}P^n$ such that $\eta = f^*(\tau)$.

The tautological bundles τ are universal in the following sense (without proof):

Theorem 12.4.3. *1. For every complex line bundle ξ over a compact topological space X there is an integer N and a continuous map $f : X \to \mathbb{C}P^N$ such that $f^*(\tau_N^{\mathbb{C}})$ is isomorphic to ξ.*

2. *For every compact topological space X there is a (probably bigger) integer N such that for every bundle ξ over X there is a continuous map $f : X \to \mathbb{C}P^N$ such that $f^*(\tau_N^{\mathbb{C}})$ is isomorphic to ξ. Moreover, two bundles $f^*(\tau_N^{\mathbb{C}})$ and $g^*(\tau_N^{\mathbb{C}})$ are isomorphic if and only if $f, g : X \to \mathbb{C}P^N$ are homotopic maps.*

By point 2. of the theorem, for a compact topological space X, the complex line bundles over X up to isomorphism are in bijection with the homotopy classes $[X, \mathbb{C}P^N]$, for a sufficiently large integer N.

Remark 12.4.4. The analogue theorem can be stated for real line bundles as well. The compactness can be replaced by a weaker property called 'paracompact'. There is an analogue theorem for vector bundles of higher rank, using tautological rank-k vector bundles over the (real or complex) Grassmannians, which are generalizations of the projective spaces.

Example 12.4.5 (Complex line bundles over surfaces)**.** Let M be a closed, oriented manifold of dimension 2, that is, a 'sphere with handles'. By the previous theorem, the complex line bundles over M are in bijection with the homotopy classes $[M, \mathbb{C}P^N]$ for a sufficiently large N. Consider the cell complex decomposition of $\mathbb{C}P^N$ with only even dimensional cells, see Section 5.3. By Example 7.2.15, there is a bijection

$$[M, \mathbb{C}P^N] \leftrightarrow [M, \mathrm{sk}_3(\mathbb{C}P^N)], \tag{12.4.6}$$

because $\dim(M) = 2$. But $\mathrm{sk}_3(\mathbb{C}P^N) = \mathrm{sk}_2(\mathbb{C}P^N) = \mathbb{C}P^1$.

We conclude that the complex line bundles over M up to isomorhism are in bijection with

$$[M, \mathbb{C}P^1] \leftrightarrow \mathbb{Z}, \tag{12.4.7}$$

where the identification with $\mathbb{Z}$ is given by the degree $[f] \mapsto \deg(f)$, according to the Hopf theorem. Based on the fact that $e(\tau) = c_1(\tau) = -1$ for the tautological line bundle over $\mathbb{C}P^1$, one concludes that for a smooth map $f : M \to \mathbb{C}P^1$,

$$e(f^*(\tau)) = c_1(f^*(\tau)) = -\deg(f). \tag{12.4.8}$$

Indeed, any smooth section $s : \mathbb{C}P^1 \to E(\tau)$ of τ induces a smooth section $s \circ f$ of $f^*(\tau)$. We leave it as an exercise that the total index of $s \circ f$ is the product of $\deg(f)$ and the total index of s.

We conclude that the Euler number (that is, the first Chern number) completely characterizes the complex line bundles over closed, oriented 2-manifolds. Moreover, every integer can be realized as the Euler number of a complex line bundle over M.

Note that the real oriented rank-2 bundles (real plane bundles), up to isomorphism, are in bijection with the complex line bundles, up to isomorphism. Indeed, on the one hand, every complex line bundle is an oriented real plane bundles. On the other hand, on an oriented real plane bundle one can define a multiplication with the complex unit i as a rotation by $+\pi/2$, using the orientation and a metric – this is not trivial, but intuitively reasonable.

In particular, for $M = S^2$, we know two classes from the classification quite well. The tangent bundle TS^2 represents 2, since $e(TM) = 2$, while the tautological bundle τ over $\mathbb{C}P^1$ represents $e(\tau) = -1$.

12.5 Exercises

13 Physical examples

13.1 Topological invariants in field configurations

Consider one of the physical vector fields introduced earlier, describing electromagnetic fields or the order parameters in magnetic systems, liquid crystals or superconductors. These fields are sections of vector bundles discussed above, but for now they may be considered smooth mappings between two manifolds. The base space will be obtained from Euclidean space $\mathbb{R}^d$, where d is the dimension, and the mapping will go to a vector space.

13.1.1 Domain walls and spin spirals

Denote the vector field by $\boldsymbol{m}$ representing the magnetization. Consider an infinite one-dimensional magnetic material, and assume $\lim_{x\to\infty} \boldsymbol{m}(x) = \lim_{x\to-\infty} \boldsymbol{m}(x)$. In this case, the base space $\mathbb{R}$ may be compactified by adding a single point in infinity, since the limit does not depend on which way this infinite point is approached. Furthermore, assume that the length of $\boldsymbol{m}$ is fixed and it is restricted to a plane; physically, this may be achieved by strong easy-plane magnetocrystalline anisotropy. This means that the mapping may be written as

$$\boldsymbol{m} : \mathbb{R} \cup \{\infty\} \cong S^1 \to S^1. \tag{13.1.1}$$

The mappings between two circles are characterized by the degree $w \in \mathbb{Z}$, also known as the *winding number*. The winding number describes how many times and in which direction the magnetization winds around the circle when going from $-\infty$ to ∞. Assume that the magnetization has a preferred orientation $\boldsymbol{m}_0$, for example a strong external magnetic field is applied. Continuous segments where $\boldsymbol{m}(x) = \boldsymbol{m}_0$ are called domains, which are separated by 360° domain walls where the magnetization makes a full rotation around the circle. The positions of the domain walls may change, but domain walls winding in the same direction cannot annihilate each other. However, domain walls of opposite winding number may merge and vanish. This explains why only the total winding number is an invariant of the mapping.

Instead of the boundary conditions in infinity mentioned above, we may consider periodic structures $\boldsymbol{m}(x + n\lambda) = \boldsymbol{m}(x)\ \forall n \in \mathbb{Z}$, where $\lambda \in \mathbb{R}$ is the period. The base space may be considered $x \in [0, \lambda[$, which is still homeomorphic to S^1. Therefore, again the winding number may be used for the topological classification. Such magnetic structures are called spin spirals, and they typically do not contain ferromagnetic domains where the magnetization points along the same direction everywhere.

13.1.2 Vortices

Consider a two-dimensional superconducting plate with order parameter $\psi(\boldsymbol{r}), \boldsymbol{r} \in \mathbb{R}^2$. Assume that a magnetic field is applied perpendicular to the plate, which tends to destroy superconductivity. In so-called Type II superconductors, there is a range of field strengths where the magnetic field only penetrates the plate along lines perpendicular to the surface. Along these lines $\psi = 0$, but it is finite outside it. Therefore, the base space may be considered as $\mathbb{R}^2$ with the position of this line, denoted by $\mathbf{0}$, removed. The order parameter may take non-zero values here. Therefore, the mapping reads

$$\psi : \mathbb{R}^2 \setminus \{\mathbf{0}\} \simeq S^1 \to \mathbb{C} \setminus \{0\} \simeq S^1. \tag{13.1.2}$$

Since both manifolds are homotopic to a circle, the relevant topological invariant here is still the winding number w. Structures characterized by a positive winding number are called vortices. Vortices have a physical meaning: the order parameter winding around the vortex represents the supercurrent screening the magnetic field going through the centre of the vortex. Due to the order parameter being single valued, the flux going through the vortex is quantized, $\Phi = w\Phi_0 = wh/(2e)$, where h is the Planck constant and e is the elementary charge. Theoretically, vortices with higher winding numbers or antivortices with $w < 0$ also exist, but typically only vortices with $w = 1$ are formed due to energy considerations. In a certain field range, these vortices form a regular triangular lattice.

Vortices also occur in the magnetization field $\boldsymbol{m}$. The dipolar interaction prefers that the magnetization points in the plane of the plate. In the centre of a vortex, the magnetization will not vanish, it will only rotate out of the plane, which can also be achieved by applying the field perpendicular to the plate. These so-called magnetic vortices are also characterized by the winding number, but there is an additional degree of freedom: the magnetization in the centre may point either up or down with respect to the plate, leading to a *polarity* of $p = 1$ or $p = -1$, respectively. Magnetic vortices are also called merons. Some authors distinguish between merons which are supposedly localized and vortices which continue infinitely far away. However, localized structures, meaning that sufficiently far away $\boldsymbol{m}(\boldsymbol{r}) = \boldsymbol{m}_0$, may only be formed if the net winding number inside this large area vanishes, i.e., the vortices and antivortices appear in pairs.

13.1.3 Skyrmions

Instead of the magnetization being confined to a plane, assume that it may point along any direction on the unit sphere. In this case, a vortex may be turned into a localized structure by rotating the magnetization out of the plane where the winding number was observed. Such localized structures satisfy $\lim_{\boldsymbol{r}\to\infty} \boldsymbol{m}(\boldsymbol{r}) = \boldsymbol{m}_0$. This means that the base space may again be compactified by adding a single point in infinity,

$$\boldsymbol{m} : \mathbb{R}^2 \cup \{\infty\} \cong S^2 \to S^2. \tag{13.1.3}$$

The degree of the mapping between two two-dimensional spheres is chracterized by the invariant $Q \in \mathbb{Z}$, called the *topological charge*. The centre of such a localized structure corresponds to a meron characterized by the winding number w and the polarity p. If there is only a single rotation in the out-of-plane component of the magnetization between the centre of the structure and infinity (i.e., from down to up or the other way around), the topological charge may be calculated as $Q = wp$. Such magnetic structures are called skyrmions.

We may also assume that the magnetic structure is periodically repeated in two dimensions, instead of converging to a fixed direction sufficiently far away from the structure. In this case, the mapping is described by

$$\boldsymbol{m} : T^2 \to S^2. \tag{13.1.4}$$

The topological charge Q may be defined analogously to the previous case, since we have seen in an earlier chapter that mappings to S^2 from any two-dimensional manifold are described by an integer degree. Such a magnetization configuration for $Q \neq 0$ is called a skyrmion lattice. Similarly to vortex lattices in superconductors, certain magnetic materials exhibit a skyrmion lattice phase when an external magnetic field is applied perpendicular to their two-dimensional plate. However, while with increasing magnetic field the superconducting vortices become more dense until the order parameter vanishes everywhere, magnetic skyrmions move further apart such that the magnetization is aligned along $\boldsymbol{m}_0$ everywhere, which is the direction of the external field.

13.1.4 Bloch points

Now consider that the magnetization is defined almost everywhere in three-dimensional space $\mathbb{R}^3$. Assume that the topological charge is calculated for planes parallel to each other. It may occur that the topological charge is not the same for two parallel planes. This can only happen if there is a singular point between the planes where the magnetization vanishes, since it cannot be continued in a smooth way. Such a singular point is called a *Bloch point*. Typically the magnetization vectors point away from the Bloch point in every direction, or towards the Bloch point from every direction, for which reason such objects are also called *hedgehogs*. Now consider a closed surface around the Bloch point, assumed to be homeomorphic to a sphere. Restricting the magnetization to this surface, the mapping reads

$$\boldsymbol{m} : S^2 \to S^2. \tag{13.1.5}$$

This mapping is still characterized by the topological charge Q. Note that this remains true if the surface enclosing the Bloch point is any two-dimensional manifold not necessarily homeomorphic to S^2, as discussed in an earlier chapter. In this context, Q may be interpreted as the charge of the Bloch point based on the Gauss-Ostrogradsky theorem. Because of this analogy, Bloch points are also called *magnetic monopoles*. While Bloch points cannot exist in smooth vector fields, they are often encountered in real materials where the magnetization field is only defined on discrete points defined by the crystal structure.

13.1.5 Hopfions

Consider a magnetization field which is smooth everywhere in three-dimensional space. In this case, if the topological charge is non-zero along a two-dimensional cross-section, then it must take the

same value along any parallel cross-section. This results in, e.g., magnetic skyrmion strings, but it does not lead to localized magnetization configurations in three dimensions. However, there are also localized magnetic configurations, $\lim_{\boldsymbol{r}\to\infty}\boldsymbol{m}(\boldsymbol{r})=\boldsymbol{m}_0$, which are interesting from the topological point of view. After compactification, the mapping may be written as

$$\boldsymbol{m}:\mathbb{R}^3\cup\{\infty\}\cong S^3\to S^2. \tag{13.1.6}$$

Such mappings are characterized by the Hopf invariant $H\in\mathbb{Z}$, one example being the Hopf fibration discussed earlier. The topological charge Q is zero for any two-dimensional cross-section of such structures, since it vanishes if the magnetization points everywhere along $\boldsymbol{m}_0$. However, consider the preimages of two generic ($\boldsymbol{m}\neq\boldsymbol{m}_0$) points of S^2 in S^3. These are generally closed curves, and it can be calculated how many times these curves link with each other. It turns out that this linking number is the same for any pairs of preimages, and it may be identified with the Hopf invariant; the sign of the linking can be defined based on the orientation of the curves. Magnetization configurations with a non-zero Hopf invariant are called magnetic hopfions.

13.2 Topological invariants in the band structure

Here, we will describe the single-particle excitations of crystals, which may be, e.g., electrons, phonons, photons, magnons or superconducting excitations (bogoliubons). The energies of these excitations are found by diagonalizing a Hamiltonian matrix $H_{\boldsymbol{k}}$, where $\boldsymbol{k}\in T^d$ is a point in the d-dimensional Brillouin zone created by the periodic arrangement of the atoms. We will assume that the parameters in the Hamiltonian change smoothly with $\boldsymbol{k}$. We consider the simplest non-trivial case, where $H_{\boldsymbol{k}}:\mathbb{C}^2\to\mathbb{C}^2$, called a two-level system. The Hamiltonian may be parametrized as

$$H_{\boldsymbol{k}}=d_0(\boldsymbol{k})\,\sigma^0+\boldsymbol{d}(\boldsymbol{k})\,\boldsymbol{\sigma}, \tag{13.2.1}$$

with $d_0\in\mathbb{R}$, $\boldsymbol{d}\in\mathbb{R}^3$, σ_0 the identity matrix and $\boldsymbol{\sigma}$ the vector constructed from the Pauli matrices. The eigenvalues and the eigenfunctions of this Hamiltonian are given by

$$H_{\boldsymbol{k}}\left|\psi_\pm(\boldsymbol{k})\right\rangle=\left(d_0(\boldsymbol{k})\pm\left\|\boldsymbol{d}(\boldsymbol{k})\right\|\right)\left|\psi_\pm(\boldsymbol{k})\right\rangle. \tag{13.2.2}$$

We will assume that the eigenvalues are not degenerate anywhere, i.e., $\left\|\boldsymbol{d}(\boldsymbol{k})\right\|\neq 0\ \forall\boldsymbol{k}$. In this case, the ground state of the Hamiltonian may be written as the pure state $\varrho_{\boldsymbol{k}}=\left|\psi_-(\boldsymbol{k})\right\rangle\left\langle\psi_-(\boldsymbol{k})\right|\in\mathbb{C}P^1\cong S^2$. Note that there is a one-to-one correspondence between the unit vector $\boldsymbol{d}(\boldsymbol{k})/\left\|\boldsymbol{d}(\boldsymbol{k})\right\|$ and $\varrho_{\boldsymbol{k}}$, since the latter is independent of $d_0(\boldsymbol{k})$ and $\left\|\boldsymbol{d}(\boldsymbol{k})\right\|$. Therefore, we may use the ground state and the direction $\boldsymbol{d}(\boldsymbol{k})/\left\|\boldsymbol{d}(\boldsymbol{k})\right\|$ interchangeably.

13.2.1 Winding number

Assume that the crystal is one-dimensional, which also implies $k\in T^1$. In order to get a topological invariant, the possible states also have to be restricted. The simplest way to represent this using the standard representation of the Pauli matrices and setting $d_y(\boldsymbol{k})=0$ everywhere. In this case, $H_{\boldsymbol{k}}$ is real-valued, so $\left|\psi_-(\boldsymbol{k})\right\rangle$ may also chosen to be real. This restricts the space of possible density operators,

$$\varrho_{\boldsymbol{k}}:T^1\cong S^1\to\mathbb{R}P^1\cong S^1. \tag{13.2.3}$$

Note that this topological characterization remains valid under unitary transformations of the Hamiltonian, which may introduce imaginary elements. As already discussed, the topological invariant associated with such mappings is the *winding number*, which describes how many times $\boldsymbol{d}(\boldsymbol{k})/\left\|\boldsymbol{d}(\boldsymbol{k})\right\|$ winds around the unit circle as we move around the Brillouin zone. This concept we already discussed in the example of a one-dimensional topological insulator.

13.2.2 Chern insulators

Now investigate a two-dimensional lattice, i.e., $\boldsymbol{k} \in T^2$. In the topologically interesting cases, the components d_x, d_y, d_z have to change sign in order to cover many directions on the unit sphere. Consider the points $d_x(\boldsymbol{k}) = 0$, which typically forms a closed curve in two dimensions. Similarly, $d_z(\boldsymbol{k}) = 0$ defines another closed curve. These closed curves generally intersect each other in points. In order to satisfy the condition $\|\boldsymbol{d}(\boldsymbol{k})\| \neq 0$, it is also necessary to consider a d_y component which does not vanish at these points. This implies that the mapping considered in this case is

$$\varrho_{\boldsymbol{k}} : T^2 \to \mathbb{C}P^1 \cong S^2. \tag{13.2.4}$$

The degree of this map is usually called the *Chern number* C of the lowest band of the Hamiltonian. Note that from the mathematical point of view, we introduced the Chern number for complex vector bundles. However, we have also seen that there is a one-to-one correspondence between the degree from any two-dimensional manifold B to $\mathbb{C}P^1 \cong S^2$ and the Chern number of the complex line bundle over B induced from the tautological vector bundle over $\mathbb{C}P^1$. This means that it is possible to interchange the degree and the Chern number in this case. Note that the degree of a similar mapping for skyrmion lattices was called the topological charge above. Despite this analogy, the topological charge or the Chern number is rarely used interchangeably, as physically one refers to real-space configurations while the other describes reciprocal space. Systems with $C \neq 0$ are called Chern insulators, which display chiral edge states propagating along the boundary of two-dimensional islands without backscattering. A simple example of a Chern insulator is given by

$$d_x(\boldsymbol{k}) = \sin(k_x), \tag{13.2.5}$$
$$d_y(\boldsymbol{k}) = \sin(k_y), \tag{13.2.6}$$
$$d_z(\boldsymbol{k}) = m + \cos(k_x) + \cos(k_y), \tag{13.2.7}$$

on a square lattice $k_x, k_y \in [0, 2\pi[$, where $m \in]-2, 2[, m \neq 0$ is a real parameter.

13.2.3 Weyl points

In three-dimensional systems with $\boldsymbol{k} \in T^3$, we may again consider where the different components of $\boldsymbol{d}$ vanishes. For each component, this usually happens on a two-dimensional surface, and all three surfaces typically intersect in points. In this case, $\|\boldsymbol{d}(\boldsymbol{k})\| \neq 0$ cannot be satisfied everywhere, and the bands touch at these points. These crossing points are called Weyl points. Note that topological insulators without band crossings may also be defined in three dimensions by assuming additional symmetries, but we do not discuss them here in detail. Instead, we consider a closed surface around such a degeneracy point, and consider the ground states when moving around this surface,

$$\varrho_{\boldsymbol{k}} : S^2 \to \mathbb{C}P^1 \cong S^2. \tag{13.2.8}$$

The degree of this mapping is called the charge of the Weyl point, based on the analogy with the Gauss-Ostrogradsky theorem. Weyl points are analogous to Bloch points in magnetization configurations in real space. Weyl points in the Brillouin zone occur in pairs with a vanishing net charge. Systems with Weyl points are mostly investigated in the case where the lower single-particle band is completely filled. In this case, at the band-crossing points the quasiparticles may be excited with vanishing energy, unlike in insulators with a gap in the spectrum; therefore, such systems are called Weyl semimetals. On the projection of the Brillouin zone onto the surface of a Weyl semimetal, the Weyl points are connected by surface states. This connecting line is called a Fermi arc, since per the above assumption it lies at the energy of the highest occupied state in the semimetal, i.e., the Fermi level.

13.3 Berry phase

We consider a Hamiltonian $H(\boldsymbol{X})$ depending on parameters $\boldsymbol{X}$ in a manifold M. This is a generalization of the $H_{\boldsymbol{k}}$ Hamiltonians defined in the Brillouin zone previously. Assume that we move

around in the parameter space along a curve $\boldsymbol{X}(t)$, smoothly varying the Hamiltonian. The time evolution of the wave function has to follow the time-dependent Schrödinger equation,

$$\mathrm{i}\hbar\partial_t \left|\psi(t)\right\rangle = H\left(\boldsymbol{X}(t)\right)\left|\psi(t)\right\rangle. \tag{13.3.1}$$

We can also consider the eigenfunctions of the Hamiltonian for each parameter value,

$$H\left(\boldsymbol{X}(t)\right)\left|n,\boldsymbol{X}(t)\right\rangle = E_n\left(\boldsymbol{X}(t)\right)\left|n,\boldsymbol{X}(t)\right\rangle, \tag{13.3.2}$$

which form a basis of the Hilbert space in the finite-dimensional case. Therefore, at each time t the wave function of the system may be expressed as a linear combination of these eigenfunctions,

$$\left|\psi(t)\right\rangle = \sum_n c_n(t)\left|n,\boldsymbol{X}(t)\right\rangle. \tag{13.3.3}$$

We want to investigate how these expansion coefficients $c_n(t)$ vary over time. Generally, this is a complicated problem, partly because the eigenfunctions, even if considered normalized, are only defined up to a phase factor. We will assume that the system is initialized in an eigenfunction, $c_m(0) = \delta_{nm}$, and that the corresponding eigenvalue is not degenerate with any other $E_m(\boldsymbol{X}) \neq E_n(\boldsymbol{X})\ \forall \boldsymbol{X}$. We will use the adiabatic approximation, in this context meaning that we do not allow the wave function to cross into a different band, i.e., $c_n(t) = 0$ for $n \neq m$. This is conventionally justified by saying that the parameters are varied very slowly, meaning that the characteristic time scale of the change τ is much larger than $\hbar/\left|E_n - E_m\right|$, the inverse of the gap between the bands. However, this does not follow the precise time evolution given by the Schrödinger equation, where the transition probabilities between the bands may be computed numerically or as a first approximation from Fermi's golden rule. We will see later that the actual relevance of the adiabatic approximation is arriving at a description independent of the time parametrization.

In the adiabatic approximation, the time evolution may be written as

$$\mathrm{i}\hbar\left(\dot{c}_m(t)\left|m,\boldsymbol{X}(t)\right\rangle + c_m(t)\,\partial_t\left|m,\boldsymbol{X}(t)\right\rangle\right) = H\left(\boldsymbol{X}(t)\right)c_m(t)\left|m,\boldsymbol{X}(t)\right\rangle = E_m\left(\boldsymbol{X}(t)\right)c_m(t)\left|m,\boldsymbol{X}(t)\right\rangle. \tag{13.3.4}$$

Taking the scalar product with $\left\langle m,\boldsymbol{X}(t)\right|$ results in a first-order linear differential equation for $c_m(t)$, with the solution

$$c_m(t) = \mathrm{e}^{-\mathrm{i}\hbar^{-1}\int_0^t E_m\left(\boldsymbol{X}\left(t'\right)\right)\mathrm{d}t'}\mathrm{e}^{\mathrm{i}\left[\mathrm{i}\int_0^t\left\langle m,\boldsymbol{X}\left(t'\right)\right|\left(\partial_{t'}\left|m,\boldsymbol{X}\left(t'\right)\right\rangle\right)\mathrm{d}t'\right]}. \tag{13.3.5}$$

The phase factor in the first exponent is called the dynamical phase, stemming from the change in the eigenvalue over the parameter space. In the second exponent, the phase factor is called the Berry phase, which has a geometric origin. To see that it is really a phase factor, one can calculate

$$\left\langle m,\boldsymbol{X}|m,\boldsymbol{X}\left(t'\right)\right\rangle = 1 \Rightarrow \left\langle m,\boldsymbol{X}\right|\left(\partial_t\left|m,\boldsymbol{X}\right\rangle\right) + \left(\partial_t\left\langle m,\boldsymbol{X}\right|\right)\left|m,\boldsymbol{X}\right\rangle = 0, \tag{13.3.6}$$

meaning that $\left\langle m,\boldsymbol{X}\right|\left(\partial_t\left|m,\boldsymbol{X}\right\rangle\right)$ is completely imaginary. Since the time dependence of the eigenfunctions only comes implicitly from the time dependence of the parameters, meaning that the Berry phase may be rewritten as

$$\mathrm{i}\int_0^t \left\langle m,\boldsymbol{X}\left(t'\right)\right|\left(\partial_{t'}\left|m,\boldsymbol{X}\left(t'\right)\right\rangle\right)\mathrm{d}t' = \mathrm{i}\int_0^t \left\langle m,\boldsymbol{X}\left(t'\right)\right|\left(\partial_{\boldsymbol{X}}\left|m,\boldsymbol{X}\left(t'\right)\right\rangle\right)\dot{\boldsymbol{X}}\left(t'\right)\mathrm{d}t' = \mathrm{i}\int_{\boldsymbol{X}(0)}^{\boldsymbol{X}(t)} \left\langle m,\boldsymbol{X}\right|\left(\partial_{\boldsymbol{X}}\left|m,\boldsymbol{X}\right\rangle\right)\mathrm{d}\boldsymbol{X}. \tag{13.3.7}$$

As alluded to above, the integrand after this transformation, also known as the Berry connection

$$A^{(m)}(\boldsymbol{X}) = \mathrm{i}\left\langle m,\boldsymbol{X}\right|\left(\partial_{\boldsymbol{X}}\left|m,\boldsymbol{X}\right\rangle\right), \tag{13.3.8}$$

does not depend on the time parametrization of the curve. The phases of the eigenfunctions may still be defined arbitrarily at different points in the parameter space, which defines the gauge freedom

of the Berry connection. However, performing the calculation over a closed curve reaches the initial eigenstate again, and the Berry phase over this closed curve is well defined.

An alternative way of calculating the Berry phase for a closed curve is based on the Berry curvature. In local coordinates, the Berry connection may be expressed as

$$A^{(m)}\left(\boldsymbol{X}\right)=A_{\mu}^{(m)}\left(\boldsymbol{X}\right)\mathrm{d}X^{\mu}. \tag{13.3.9}$$

The Berry curvature is the exterior derivative of the Berry connection, given in local coordinates as

$$F^{(m)}\left(\boldsymbol{X}\right)=\mathrm{d}A^{(m)}\left(\boldsymbol{X}\right)=\frac{1}{2}F_{\mu\nu}^{(m)}\left(\boldsymbol{X}\right)\mathrm{d}X^{\mu}\wedge\mathrm{d}X^{\nu}, \tag{13.3.10}$$

with

$$F_{\mu\nu}^{(m)}\left(\boldsymbol{X}\right)=\partial_{\mu}A_{\nu}^{(m)}\left(\boldsymbol{X}\right)-\partial_{\nu}A_{\mu}^{(m)}\left(\boldsymbol{X}\right). \tag{13.3.11}$$

Using the definition of $A_{\mu}^{(m)}\left(\boldsymbol{X}\right)$, the components of the curvature tensor may be rewritten as

$$F_{\mu\nu}^{(m)}\left(\boldsymbol{X}\right)=\mathrm{i}\left[\left(\partial_{\mu}\left\langle m,\boldsymbol{X}\right|\right)\left(\partial_{\nu}\left|m,\boldsymbol{X}\right\rangle\right)-\left(\partial_{\nu}\left\langle m,\boldsymbol{X}\right|\right)\left(\partial_{\mu}\left|m,\boldsymbol{X}\right\rangle\right)\right]. \tag{13.3.12}$$

From the eigenvalue equation of the Hamiltonian, one arrives at

$$\left(\partial_{\mu}H\right)\left|m,\boldsymbol{X}\right\rangle+H\left(\partial_{\mu}\left|m,\boldsymbol{X}\right\rangle\right)=\left(\partial_{\mu}E_{m}\left(\boldsymbol{X}\right)\right)\left|m,\boldsymbol{X}\right\rangle+E_{m}\left(\boldsymbol{X}\right)\left(\partial_{\mu}\left|m,\boldsymbol{X}\right\rangle\right). \tag{13.3.13}$$

Taking the scalar product with $\left\langle n,\boldsymbol{X}\right|,n\neq m$ results in

$$\left\langle n,\boldsymbol{X}\right|\left(\partial_{\mu}\left|m,\boldsymbol{X}\right\rangle\right)=\frac{\left\langle n,\boldsymbol{X}\right|\partial_{\mu}H\left|m,\boldsymbol{X}\right\rangle}{E_{m}\left(\boldsymbol{X}\right)-E_{n}\left(\boldsymbol{X}\right)}. \tag{13.3.14}$$

Now, inserting $I=\sum_{n}\left|n,\boldsymbol{X}\right\rangle\left\langle n,\boldsymbol{X}\right|$ into the Berry curvature and using the above relation gives

$$F_{\mu\nu}^{(m)}\left(\boldsymbol{X}\right)=\mathrm{i}\sum_{n\neq m}\frac{\left\langle m,\boldsymbol{X}\right|\partial_{\mu}H\left|n,\boldsymbol{X}\right\rangle\left\langle n,\boldsymbol{X}\right|\partial_{\nu}H\left|m,\boldsymbol{X}\right\rangle-\left\langle m,\boldsymbol{X}\right|\partial_{\nu}H\left|n,\boldsymbol{X}\right\rangle\left\langle n,\boldsymbol{X}\right|\partial_{\mu}H\left|m,\boldsymbol{X}\right\rangle}{\left[E_{m}\left(\boldsymbol{X}\right)-E_{n}\left(\boldsymbol{X}\right)\right]^{2}}. \tag{13.3.15}$$

Note that the $n=m$ term is included in the identity operator, but it cancels in the expression of the Berry curvature due to the antisymmetrization over components.

If the curve C along which the Berry phase is calculated encloses the surface S, Stokes' theorem connects the integral of the Berry connection along the curve with the integral of the Berry curvature on the surface,

$$\oint_{C}A_{\mu}^{(m)}\left(\boldsymbol{X}\right)\mathrm{d}X^{\mu}=\int_{S}=\frac{1}{2}F_{\mu\nu}^{(m)}\left(\boldsymbol{X}\right)\mathrm{d}X^{\mu}\wedge\mathrm{d}X^{\nu}. \tag{13.3.16}$$

In certain cases, it may be advantageous to calculate the integral based on the Berry curvature, since here the derivative of the Hamiltonian is taken instead of the derivative of the wave functions, the latter introducing ambiguities due to the phase factors.

13.4 Calculation of the Berry phase for two-level systems

We assume that the parameter space is the Brillouin zone with elements $\boldsymbol{k}$, as considered earlier. The Hamiltonian of a two-level system may be parameterized as

$$H_{\boldsymbol{k}}=d_{0}\left(\boldsymbol{k}\right)\sigma^{0}+\boldsymbol{d}\left(\boldsymbol{k}\right)\boldsymbol{\sigma}. \tag{13.4.1}$$

The vector $\boldsymbol{d}\left(\boldsymbol{k}\right)$ may be expressed in spherical coordinates as

$$\boldsymbol{d}\left(\boldsymbol{k}\right)=\left\|\boldsymbol{d}\left(\boldsymbol{k}\right)\right\|\begin{bmatrix}\sin\vartheta_{\boldsymbol{k}}\cos\varphi_{\boldsymbol{k}}\\ \sin\vartheta_{\boldsymbol{k}}\sin\varphi_{\boldsymbol{k}}\\ \cos\vartheta_{\boldsymbol{k}}\end{bmatrix}=\left\|\boldsymbol{d}\left(\boldsymbol{k}\right)\right\|\hat{\boldsymbol{d}}\left(\boldsymbol{k}\right), \tag{13.4.2}$$

where $\hat{\boldsymbol{d}}(\boldsymbol{k})$ denotes its direction, and $\|\boldsymbol{d}(\boldsymbol{k})\| \neq 0$ is assumed to obtain two non-degenerate bands. The eigenvectors of the Hamiltonian

$$H_{\boldsymbol{k}} \left|\psi_{\pm}(\boldsymbol{k})\right\rangle = \left(d_0(\boldsymbol{k}) \pm \|\boldsymbol{d}(\boldsymbol{k})\|\right) \left|\psi_{\pm}(\boldsymbol{k})\right\rangle \tag{13.4.3}$$

may be given in a first choice for the gauge as

$$\left|\psi_{+}^{(1)}(\vartheta_{\boldsymbol{k}}, \varphi_{\boldsymbol{k}})\right\rangle = \begin{bmatrix} \cos\frac{\vartheta_{\boldsymbol{k}}}{2} \\ \mathrm{e}^{\mathrm{i}\varphi_{\boldsymbol{k}}} \sin\frac{\vartheta_{\boldsymbol{k}}}{2} \end{bmatrix}, \left|\psi_{-}^{(1)}(\vartheta_{\boldsymbol{k}}, \varphi_{\boldsymbol{k}})\right\rangle = \begin{bmatrix} -\mathrm{e}^{-\mathrm{i}\varphi_{\boldsymbol{k}}} \sin\frac{\vartheta_{\boldsymbol{k}}}{2} \\ \cos\frac{\vartheta_{\boldsymbol{k}}}{2} \end{bmatrix}. \tag{13.4.4}$$

13.4.1 Berry phase in one dimension

We consider the parametrization describing a one-dimensional topological insulator (Su–Schrieffer–Heeger model) with nearest-neighbour hopping parameters $v, w \in \mathbb{C}$ discussed earlier,

$$d_x(\boldsymbol{k}) = |v| \cos\varphi_v + |w| \cos(k + \varphi_w), \tag{13.4.5}$$
$$d_y(\boldsymbol{k}) = |v| \sin\varphi_v + |w| \sin(k + \varphi_w), \tag{13.4.6}$$
$$d_y(\boldsymbol{k}) = 0, \tag{13.4.7}$$

with $k \in [0, 2\pi[$, and $|v|$ and φ_v denoting the absolute value and the argument of v, respectively. Using the spherical coordinates, this corresponds to $\vartheta_k = \pi/2$, while ϕ_k depends on the hopping parameters. For $|v| < |w|$, ϕ_k describes a full circle around the equator of the unit sphere, while for $|v| > |w|$, ϕ_k does not wind around the circle.

The Berry connection for the $\psi_{+}^{(1)}$ band may be calculated as

$$A^{(+)}(k) = \mathrm{i} \left\langle \psi_{+}^{(1)}(\vartheta_{\boldsymbol{k}}, \varphi_{\boldsymbol{k}}) \right| \left(\partial_k \left|\psi_{+}^{(1)}(\vartheta_{\boldsymbol{k}}, \varphi_{\boldsymbol{k}})\right\rangle\right) = \mathrm{i} \begin{bmatrix} \cos\frac{\vartheta_{\boldsymbol{k}}}{2} \\ \mathrm{e}^{\mathrm{i}\varphi_{\boldsymbol{k}}} \sin\frac{\vartheta_{\boldsymbol{k}}}{2} \end{bmatrix}^{\dagger} \begin{bmatrix} 0 \\ \mathrm{i}\partial_k\varphi_k \mathrm{e}^{\mathrm{i}\varphi_{\boldsymbol{k}}} \sin\frac{\vartheta_{\boldsymbol{k}}}{2} \end{bmatrix} = -\partial_k\varphi_k \sin^2\frac{\vartheta_{\boldsymbol{k}}}{2} = -\frac{1}{2}\partial_k\varphi_k, \tag{13.4.8}$$

and the Berry phase over the whole one-dimensional Brillouin zone is

$$\oint -\frac{1}{2}\partial_k\varphi_k \mathrm{d}k = -\frac{1}{2}\Delta\varphi. \tag{13.4.9}$$

As mentioned above, this corresponds to $-\pi$ for $|v| < |w|$ and 0 for $|v| > |w|$, displaying the difference between the topologically non-trivial and trivial cases.

Alternatively, one could consider the Berry phase in a second gauge $\left|\psi_{+}^{(2)}(\vartheta_k, \varphi_k)\right\rangle = \mathrm{e}^{-\mathrm{i}\varphi_k/2} \left|\psi_{+}^{(1)}(\vartheta_k, \varphi_k)\right\rangle, \left|\psi_{-}^{(2)}(\vartheta_k, \varphi_k)\right\rangle = \mathrm{e}^{\mathrm{i}\varphi_k/2} \left|\psi_{-}^{(1)}(\vartheta_k, \varphi_k)\right\rangle$. It can be calculated that $A^{(+)}(k) = 0$ in this gauge, i.e., it describes parallel transport. However, the Berry phase over the whole curve may still be calculated by following the evolution of the wave function by changing φ_k by 2π,

$$\left|\psi_{+}^{(2)}(\vartheta_k, \varphi_k + 2\pi)\right\rangle = \mathrm{e}^{-\mathrm{i}\frac{\varphi_k + 2\pi}{2}} \left|\psi_{+}^{(1)}(\vartheta_k, \varphi_k + 2\pi)\right\rangle = \mathrm{e}^{-\mathrm{i}\pi} \left|\psi_{+}^{(2)}(\vartheta_k \varphi_k)\right\rangle. \tag{13.4.10}$$

Here, we used $\left|\psi_{+}^{(1)}(\vartheta_k, \varphi_k + 2\pi)\right\rangle = \left|\psi_{+}^{(1)}(\vartheta_k, \varphi_k)\right\rangle$, i.e., that the system returns to its original eigenfunction after a closed curve, which is a necessary condition for applying the integral formula, but is only satisfied in the first choice of gauge.

If further terms are included in the Hamiltonian, φ_k may wind multiple times around the unit circle. The winding number may be determined from the Berry connection in the first choice of gauge as

$$w = -\frac{1}{\pi} \oint A^{(+)}(k) \, \mathrm{d}k = \frac{1}{2\pi} \oint \partial_k\varphi_k \mathrm{d}k. \tag{13.4.11}$$

13.4.2 Chern number

Now consider the Brillouin zone of a two-dimensional system, $\boldsymbol{k} \in T^2$. Introduce the unit vectors

$$\hat{\boldsymbol{e}}(\boldsymbol{k}) = \begin{bmatrix} \cos\vartheta_{\boldsymbol{k}} \cos\varphi_{\boldsymbol{k}} \\ \cos\vartheta_{\boldsymbol{k}} \sin\varphi_{\boldsymbol{k}} \\ -\sin\vartheta_{\boldsymbol{k}} \end{bmatrix}, \ \hat{\boldsymbol{f}}(\boldsymbol{k}) = \begin{bmatrix} -\sin\varphi_{\boldsymbol{k}} \\ \cos\varphi_{\boldsymbol{k}} \\ 0 \end{bmatrix}, \tag{13.4.12}$$

with $\hat{\boldsymbol{e}} \times \hat{\boldsymbol{f}} = \hat{\boldsymbol{d}}$. Note that $\left|\psi_{\pm}^{(1)}(\boldsymbol{k})\right\rangle$ were eigenvectors of the pseudospin projected onto a specific direction, $\hat{\boldsymbol{d}}\boldsymbol{\sigma}$. Here we consider an orthonormal basis adapted to the local $\hat{\boldsymbol{d}}$ direction, and express the spin components in this basis. These spin components satisfy the usual algebra of the spin operators. In particular, we can consider the spin raising and lowering operators,

$$\left(\hat{\boldsymbol{e}}(\boldsymbol{k}) + \mathrm{i}\hat{\boldsymbol{f}}(\boldsymbol{k})\right)\left|\psi_{-}^{(1)}(\boldsymbol{k})\right\rangle = 2\left|\psi_{+}^{(1)}(\boldsymbol{k})\right\rangle, \tag{13.4.13}$$

$$\left(\hat{\boldsymbol{e}}(\boldsymbol{k}) - \mathrm{i}\hat{\boldsymbol{f}}(\boldsymbol{k})\right)\left|\psi_{+}^{(1)}(\boldsymbol{k})\right\rangle = 2\left|\psi_{-}^{(1)}(\boldsymbol{k})\right\rangle, \tag{13.4.14}$$

with all their other matrix elements vanishing. To calculate the Berry curvature, we consider the derivative of the Hamiltonian,

$$\partial_\mu H = \partial_\mu d_0(\boldsymbol{k})\,\sigma^0 + \sum_{\hat{\boldsymbol{v}}=\hat{\boldsymbol{d}},\hat{\boldsymbol{e}},\hat{\boldsymbol{f}}} \left[\partial_\mu \boldsymbol{d}(\boldsymbol{k}) \cdot \hat{\boldsymbol{v}}\right] \hat{\boldsymbol{v}} \cdot \boldsymbol{\sigma}, \tag{13.4.15}$$

expressed in the spin components in the above frame. The expressions required for the curvature read

$$\begin{aligned}
&\left\langle\psi_{+}^{(1)}(\boldsymbol{k})\right| \partial_\mu H \left|\psi_{-}^{(1)}(\boldsymbol{k})\right\rangle \left\langle\psi_{-}^{(1)}(\boldsymbol{k})\right| \partial_\nu H \left|\psi_{+}^{(1)}(\boldsymbol{k})\right\rangle - \left\langle\psi_{+}^{(1)}(\boldsymbol{k})\right| \partial_\nu H \left|\psi_{-}^{(1)}(\boldsymbol{k})\right\rangle \left\langle\psi_{-}^{(1)}(\boldsymbol{k})\right| \partial_\mu H \left|\psi_{+}^{(1)}(\boldsymbol{k})\right\rangle \\
&= \left[\partial_\mu \boldsymbol{d}(\boldsymbol{k}) \cdot \left(\hat{\boldsymbol{e}} - \mathrm{i}\hat{\boldsymbol{f}}\right)\right]\left[\partial_\nu \boldsymbol{d}(\boldsymbol{k}) \cdot \left(\hat{\boldsymbol{e}} + \mathrm{i}\hat{\boldsymbol{f}}\right)\right] - \left[\partial_\nu \boldsymbol{d}(\boldsymbol{k}) \cdot \left(\hat{\boldsymbol{e}} - \mathrm{i}\hat{\boldsymbol{f}}\right)\right]\left[\partial_\mu \boldsymbol{d}(\boldsymbol{k}) \cdot \left(\hat{\boldsymbol{e}} + \mathrm{i}\hat{\boldsymbol{f}}\right)\right] \\
&= -2\mathrm{i}\left\{\left[\partial_\mu \boldsymbol{d}(\boldsymbol{k}) \cdot \hat{\boldsymbol{f}}\right]\left[\partial_\nu \boldsymbol{d}(\boldsymbol{k}) \cdot \hat{\boldsymbol{e}}\right] - \left[\partial_\nu \boldsymbol{d}(\boldsymbol{k}) \cdot \hat{\boldsymbol{f}}\right]\left[\partial_\mu \boldsymbol{d}(\boldsymbol{k}) \cdot \hat{\boldsymbol{e}}\right]\right\} = -2\mathrm{i}\left[\partial_\mu \boldsymbol{d}(\boldsymbol{k}) \times \left(\hat{\boldsymbol{e}} \times \hat{\boldsymbol{f}}\right)\right] \cdot \partial_\mu \boldsymbol{d}(\boldsymbol{k}) \\
&= 2\mathrm{i}\hat{\boldsymbol{d}}(\boldsymbol{k}) \cdot \left(\partial_\mu \boldsymbol{d}(\boldsymbol{k}) \times \partial_\nu \boldsymbol{d}(\boldsymbol{k})\right),
\end{aligned} \tag{13.4.16}$$

and

$$\left(E_+(\boldsymbol{k}) - E_-(\boldsymbol{k})\right)^2 = 4\|\boldsymbol{d}(\boldsymbol{k})\|^2. \tag{13.4.17}$$

Combining the terms leads to the expression for the curvature in local coordinates,

$$F_{\mu\nu}^{(+)} = -\frac{1}{2}\frac{\hat{\boldsymbol{d}}(\boldsymbol{k}) \cdot \left(\partial_\mu \boldsymbol{d}(\boldsymbol{k}) \times \partial_\nu \boldsymbol{d}(\boldsymbol{k})\right)}{\|\boldsymbol{d}(\boldsymbol{k})\|^2} = -\frac{1}{2}\hat{\boldsymbol{d}}(\boldsymbol{k}) \cdot \left(\partial_\mu \hat{\boldsymbol{d}}(\boldsymbol{k}) \times \partial_\nu \hat{\boldsymbol{d}}(\boldsymbol{k})\right), \tag{13.4.18}$$

where the latter form is obtained when using the antisymmetric property of the curvature, which leads to the cancellation of terms containing $\partial_\mu \|\boldsymbol{d}(\boldsymbol{k})\|$. Consequently, the Berry phase along a curve enclosing a surface S may be calculated as

$$\int_S F_{\mu\nu}^{(+)} \mathrm{d}k^\mu \wedge \mathrm{d}k^\nu = \int_S -\frac{1}{2}\hat{\boldsymbol{d}}(\boldsymbol{k}) \cdot \left(\partial_\mu \hat{\boldsymbol{d}}(\boldsymbol{k}) \times \partial_\nu \hat{\boldsymbol{d}}(\boldsymbol{k})\right) \mathrm{d}k^\mu \mathrm{d}k^\nu \tag{13.4.19}$$

Note that for a two-dimensional parameter space, the curvature only has a single component. We can rewrite the integral in terms of the angle variables using $\partial_\mu \hat{\boldsymbol{d}}(\boldsymbol{k}) = \partial_\mu \vartheta_{\boldsymbol{k}} \hat{\boldsymbol{e}}(\boldsymbol{k}) + \sin\vartheta_{\boldsymbol{k}} \partial_\mu \varphi_{\boldsymbol{k}} \hat{\boldsymbol{f}}(\boldsymbol{k})$. This yields

$$\int_S -\frac{1}{2}\sin\vartheta_{\boldsymbol{k}} \left(\partial_\mu \vartheta_{\boldsymbol{k}} \partial_\nu \varphi_{\boldsymbol{k}} - \partial_\nu \vartheta_{\boldsymbol{k}} \partial_\mu \varphi_{\boldsymbol{k}}\right) \mathrm{d}k^\mu \mathrm{d}k^\nu = \int_S -\frac{1}{2}\sin\vartheta_{\boldsymbol{k}} \mathrm{d}\vartheta_{\boldsymbol{k}} \mathrm{d}\varphi_{\boldsymbol{k}}. \tag{13.4.20}$$

Up to a prefactor, this corresponds to the area enclosed by the curve on the parameter space $(\vartheta_{\boldsymbol{k}}, \varphi_{\boldsymbol{k}}) \in S^2$. Note that a single closed curve C splits the sphere S^2 into two surfaces S and

$S' = S^2 \setminus S$. Since the Berry phase is defined along the curve, calculating it over either of the two surfaces should not lead to a difference in the observable quantities. This results in the relation between the integrals

$$\oint A^{(+)} = \int_S F^{(+)} = \int_{S'} F^{(+)} - Q2\pi, \tag{13.4.21}$$

where $Q \in \mathbb{Z}$, leading to a trivial difference in the phase. Rearranging the equation and reversing the orientation of the surface normal on S' leads to

$$Q = -\frac{1}{2\pi} \int_{S^2} F^{(+)} = \frac{1}{4\pi} \int \hat{\boldsymbol{d}}(\boldsymbol{k}) \cdot \left(\partial_\mu \hat{\boldsymbol{d}}(\boldsymbol{k}) \times \partial_\nu \hat{\boldsymbol{d}}(\boldsymbol{k}) \right) \mathrm{d}k^\mu \mathrm{d}k^\nu. \tag{13.4.22}$$

This is the formula most commonly used for calculating the topological charge of magnetization configurations in real space $\boldsymbol{r} \in \mathbb{R}^2$, where $\hat{\boldsymbol{d}}(\boldsymbol{r})$ denotes the direction of the magnetization. In the quantum-mechanical setting in reciprocal space considered here, this quantity is known in condensed-matter physics as the Chern number of the band $|\psi_+\rangle$ instead, as discussed earlier. Mathematically, it corresponds to the degree of the mapping from a two-dimensional manifold of the parameters $\boldsymbol{k} \in M^2$ to the sphere $\hat{\boldsymbol{d}} \in S^2$, which may be equivalently written as the manifold of the ground states of the Hamiltonian.

13.4.3 Berry phase in three dimensions

We will now discuss the case of a three-dimensional parameter space $\boldsymbol{k} \in T^3$. In this case, the Berry curvature can be expressed by the formula introduced in the two-dimensional case, but it has three components instead of one. These components may be collected into a vector, also known as Berry's emergent magnetic field,

$$B_\varrho^{(+)} = \frac{1}{2} \varepsilon^{\mu\nu\varrho} F_{\mu\nu}^{(+)}. \tag{13.4.23}$$

Now consider the parametrization of the Hamiltonian

$$H_{\boldsymbol{k}} = \boldsymbol{k}\boldsymbol{\sigma}. \tag{13.4.24}$$

This Hamiltonian is degenerate in $\boldsymbol{k} = 0$, which describes a Weyl point in parameter space. Away from the Weyl point, the emergent magnetic field may be written as

$$B_\varrho^{(+)} = -\frac{1}{2} \frac{\boldsymbol{k}}{\|\boldsymbol{k}\|^3}. \tag{13.4.25}$$

This is the magnetic field generated by a monopole of strength $-1/2$ at $\boldsymbol{r} = \boldsymbol{0}$. Integrating this emergent magnetic field over a closed surface around the origin reproduces the charge of the Weyl point, as is known from electrodynamics.

Finally, consider a parametrization with $\boldsymbol{k} \in S^3$, $\|\boldsymbol{d}(\boldsymbol{k})\| \neq 0$. This is again more natural to interpret in terms of a magnetization configuration in real space, where $\hat{\boldsymbol{d}}(\boldsymbol{k})$ is the direction of the magnetization. As discussed earlier, such mappings $\hat{\boldsymbol{d}} : S^3 \to S^2$ are characterized by the Hopf invariant H. The Hopf invariant may also be expressed by the emergent magnetic field and the Berry connection using Whitehead's formula,

$$H = \frac{1}{(4\pi)^2} \int_{S^3} \boldsymbol{B}^{(+)} \cdot \boldsymbol{A}^{(+)} \tag{13.4.26}$$

It can be shown that this integral does not depend on the choice of gauge in the Berry connection.